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# REPORT FOR OOP HUMANITARIAN LOGISTIC PROJECT

GROUP 4

## 1. Overview

### 1.1 Project Objective and Context

The Humanitarian Logistics Analysis System is a comprehensive Java-based application designed to analyze and assess the effectiveness of humanitarian relief operations during disaster response scenarios. The system focuses on two primary research problems:

**Problem 1 - Relief Item Category Effectiveness Analysis:** This module directly addresses *Original Problem 3* from the project specification by evaluating which categories of relief items (cash assistance, medical supplies, food aid, shelter provisions, transportation) receive the highest user satisfaction and positive sentiment from affected populations. The analysis aggregates sentiment scores across different relief categories, enabling humanitarian organizations to identify which types of aid are most valued by beneficiaries and which areas require improvement or resource reallocation. Users can visualize satisfaction distributions through interactive bar charts and pie charts, filtering by specific disaster types or viewing aggregate statistics across all disasters.

**Problem 2 - Temporal Sentiment Trend Analysis:** This module integrates two complementary temporal perspectives that combine *Original Problems 1 and 4* from the project specification, providing a unified view of how sentiment evolves throughout disaster response phases:

- **Category-Specific Temporal Analysis (Original Problem 4):** Tracks sentiment trends for individual relief categories over time, showing how satisfaction with specific aid types (cash, medical,

shelter, food, transportation) changes during different phases of the disaster response. This granular view helps organizations understand which relief categories improve or decline in effectiveness as the response progresses, enabling targeted interventions for underperforming aid types.

- **Overall Sentiment Timeline (Original Problem 1):** Analyzes aggregate sentiment trends across all relief categories, providing a comprehensive view of how public perception of the entire humanitarian response evolves from immediate aftermath through ongoing recovery phases. This holistic perspective identifies critical periods when intervention strategies may need adjustment and reveals overall patterns in community satisfaction with relief efforts.

By organizing temporal analyses into a single unified module with two complementary tabs, the application enables users to seamlessly compare category-specific trends against overall sentiment patterns, facilitating comprehensive insights into disaster response effectiveness and resource allocation optimization.

## 1.2 System Architecture Overview

The application is structured using a professional seven-package architecture that cleanly separates concerns and enables modularity, testability, and extensibility:

**Core Data Layer** (Model Package): Contains fundamental data entities (Post, Comment, Sentiment, DisasterType) representing the domain model with no dependencies on other packages.

**User Interface Layer** (UI Package): Implements the Model-View-Controller architectural pattern using Swing components organized into specialized panels (AdvancedAnalysisPanel, CommentManagementPanel, CrawlControlPanel, DataCollectionPanel). The UI automatically updates when underlying data changes through the Observer pattern.

**Data Collection Layer** (Crawler Package): Provides pluggable data sources through the Strategy pattern. Implements two main crawlers: YouTubeCrawler (fetches YouTube watch pages using Java HttpClient, parses HTML with regex patterns to extract video metadata and comments) and MockDataCrawler (generates sample posts without requiring internet access). The Registry pattern enables dynamic crawler registration and runtime discovery.

**Sentiment Analysis Layer** (Sentiment Package): Provides sentiment analysis through multiple implementations with a fallback strategy. PythonSentimentAnalyzer uses the Python backend with nlptown/bert-base-multilingual-uncased-sentiment model for multilingual sentiment analysis (called via Python API on port 5001), with fallback to xlm-roberta-base if primary model unavailable. EnhancedSentimentAnalyzer provides bilingual keyword-based analysis with support for both English and Vietnamese sentiment keywords, serving as fallback when Python API is unavailable. Users interact with sentiment analysis through a single "Analyze All Posts with Python API" button in the Analysis tab that processes all posts using the Python backend.

**Analysis Layer** (Analysis Package): Implements specialized analysis engines through the Strategy pattern. SatisfactionAnalysisModule analyzes Problem 1 by calculating sentiment statistics and satisfaction scores per relief category, determining category effectiveness and generating resource allocation recommendations. TimeSeriesSentimentModule implements Problem 2 temporal analysis by grouping sentiments into 6-hour time buckets and tracking how sentiment evolves over time for each relief category, computing trend directions and sector effectiveness ratings.

**Data Persistence Layer** (Database Package): Manages SQLite database operations with proper abstraction to support potential migration to other database systems without affecting application code.

**Utility Layer** (Util Package): Provides shared utility functions and helper classes used across multiple packages. StringSimilarity class offers string comparison algorithms for text matching and similarity

analysis, supporting flexible and approximate text matching for disaster keyword recognition and relief category classification.

## 1.3 Key Technologies and Frameworks

**Programming Language:** Java (version 11 or higher) providing strong type safety, comprehensive standard library, and mature ecosystem with built-in HTTP client.

**User Interface Framework:** Swing (javax.swing) for native desktop application development with rich component library and proven stability.

**Data Storage:** SQLite for lightweight, embedded database functionality suitable for desktop applications.

**Web Data Collection:** Java HttpClient (built-in java.net.http package) with OkHttp3 4.11.0 HTTP client library for fetching YouTube watch pages, parsing HTML responses, and extracting video metadata and comments via direct HTTP requests.

**Machine Learning Integration:** Python backend with Flask REST API, Hugging Face Transformers (nlptown/bert-base-multilingual-uncased-sentiment for sentiment analysis with xlm-roberta-base fallback), and custom keyword-based classification for relief categories, accessed via HTTP calls from Java application.

**Build System:** Maven 3.9.11 for dependency management, project organization, and automated fat JAR packaging with all dependencies included.

## 1.4 Object-Oriented Design Principles Applied

The system demonstrates comprehensive application of Object-Oriented Programming principles:

**Abstraction:** Core interfaces (SentimentAnalyzer, DataCrawler, AnalysisModule) define contracts that hide implementation details while exposing consistent interfaces.

**Encapsulation:** Data entities encapsulate their state with private fields and controlled access through methods. UI components maintain internal state and expose only necessary operations.

**Inheritance:** Post class serves as a base for YouTube-specific posts, enabling code reuse while supporting polymorphic treatment of different post types.

**Polymorphism:** The system leverages both interface-based and inheritance-based polymorphism to support multiple implementations of key abstractions without coupling to concrete types.

**Design Patterns:** The architecture employs industry-standard design patterns including MVC (Model-View-Controller), Strategy, Factory, Registry, Observer, and Singleton patterns to solve recurring architectural and design problems.

## 1.5 Development Approach and Methodology

The project follows professional software engineering practices emphasizing clean code, testability, and maintainability:

**Separation of Concerns:** Each package focuses on a specific responsibility, making the system easier to understand, test, and modify.

**Interface-Based Design:** Components depend on abstractions rather than concrete implementations, enabling flexible composition and easy testing with mocks.



**Progressive Enhancement:** Core functionality is implemented with simple strategies (keyword-based sentiment analysis, mock data), with advanced options available when needed (ML-based analysis, real data sources).

**Testability:** Classes are designed to be easily testable in isolation, with dependencies injected rather than hardcoded.

## 1.6 Deliverables and Scope

The complete system includes:

- Java Source Code:** 33 classes organized into 7 packages, totaling 7499 lines of well-structured code
- User Interface:** Desktop application with tabbed interface supporting data collection, analysis visualization, and disaster management
- Data Analysis Modules:** Two specialized analysis engines addressing Problems 1 and 2
- Documentation:** Comprehensive UML diagrams showing package dependencies and detailed class relationships, detailed design explanations, and complete API documentation
- Database Schema:** SQLite database supporting persistent storage of posts, comments, and sentiment analysis results

This report documents the complete system architecture, design decisions, implementation details, and the advanced Object-Oriented Programming techniques employed throughout the development.

## 2. Members and Task Assignment

| Member and Student ID       | Task Assignment                                                                                     | Percentage of Contribution |
|-----------------------------|-----------------------------------------------------------------------------------------------------|----------------------------|
| Nguyen Trung Hieu 202416689 | System Architecture Design, Core Model Implementation, UI Development, ML Integration, Testing & QA | 30%                        |
| Nguyen Cong Hung 2024       | Data Collection & Crawler Framework                                                                 | 17.5%                      |
| Tran Dang Minh 2024         | Database Implementation & Data Persistence                                                          | 17.5%                      |
| Vu Ha Anh Duc 2024          | Analysis Modules Implementation & Visualization                                                     | 17.5%                      |
| Pham Minh Hieu 2024         | Documentation & Report Writing                                                                      | 17.5%                      |

## 3. Teamwork Log

### Project Development Timeline

The Humanitarian Logistics Analysis System was developed over approximately 8 weeks from October 15, 2024 to December 10, 2024. Below is the chronological log of project phases and key milestones:

**Phase 1: Project Planning and Requirements Analysis (October 15-22)**

**Objectives:** Define project scope, identify analysis problems, and establish system architecture

**Key Activities:** - Conducted requirements gathering sessions defining two primary analysis problems - Researched and selected technology stack (Java 11 for desktop, Python 3.12 for ML services) - Evaluated existing solutions and identified unique features needed - Sketched initial system architecture and component structure - Evaluated machine learning models (nlptown/bert-base-multilingual for sentiment, keyword-based approach for category classification)

**Deliverables:** Project specification document, architecture design overview, technology evaluation report

**Status:** ✓ Completed

**Phase 2: System Architecture and Design (October 23-29)**

**Objectives:** Finalize architecture and create detailed design specifications

**Key Activities:** - Designed seven-package architecture with clear separation of concerns - Created package dependency diagrams and class relationship structures - Defined interfaces and abstract classes for all major components - Planned Maven project structure with dependency management - Designed SQLite database schema with normalized tables

**Deliverables:** Complete UML architecture diagrams, detailed design specifications, database schema

**Status:** ✓ Completed

**Phase 3: Core Model and Infrastructure Implementation (October 30 - November 9)**

**Objectives:** Implement foundational model classes and infrastructure components

**Key Activities:** - Implemented Post, YouTubePost, Comment, Sentiment classes with proper encapsulation - Created DisasterManager singleton with initialization logic - Set up Maven build configuration and dependency management - Implemented DatabaseManager with JDBC connectivity - Created basic SentimentAnalyzer implementations (Simple and Enhanced versions)

**Deliverables:** Complete Model package, database connectivity layer, basic sentiment analysis

**Status:** ✓ Completed, Test Coverage: 90%+

**Phase 4: Data Collection and Crawler Framework (November 10-19)**

**Objectives:** Implement data collection from various sources

**Key Activities:** - Designed and implemented DataCrawler interface with pluggable architecture - Created CrawlerRegistry and CrawlerManager using Registry + Factory patterns - Implemented MockDataCrawler for testing without external API calls - Integrated YouTube API with YouTubeCrawler implementation - Implemented YouTubeAPIHelper with authentication and pagination support

**Deliverables:** Working crawler framework, YouTube integration, mock data source

**Status:** ✓ Completed, 32 sample posts collected

## Phase 5: User Interface Development (November 20 - November 28)

**Objectives:** Build desktop UI with all required components and visualization

**Key Activities:** - Implemented View class as main JFrame with tabbed interface - Created DataCollectionPanel with crawler selection and execution controls - Implemented AnalysisPanel with bar charts and pie charts for Problem 1 - Created AdvancedAnalysisPanel with time-series charts for Problem 2 - Added CommentManagementPanel with JTable for data display - Integrated JFreeChart for professional data visualization

**Deliverables:** Complete desktop interface with 5+ specialized panels, interactive visualizations

**Status:** ✓ Completed

## Phase 6: Machine Learning Integration (November 29 - December 3)

**Objectives:** Integrate advanced ML models for sentiment and category analysis

**Key Activities:** - Developed Python Flask API (sentiment\_api.py) for ML services - Integrated nlptown/bert-base-multilingual-uncased-sentiment model for multilingual sentiment analysis - Implemented keyword-based hybrid approach for relief category classification - Implemented PythonSentimentAnalyzer and PythonCategoryClassifier - Set up fallback mechanism with xlm-roberta-base for sentiment reliability - Implemented fallback mechanism (Python → Enhanced → Simple analyzers)

**Deliverables:** Production ML backend service, working sentiment and category classification

**Status:** ✓ Completed

## Phase 7: Analysis Modules and Problem Solving (December 4 - 6)

**Objectives:** Implement specialized analysis engines for both research problems

**Key Activities:** - Implemented SatisfactionAnalysisModule for Problem 1 (relief effectiveness analysis) - Implemented TimeSeriesSentimentModule for Problem 2 (temporal sentiment trends) - Created analysis result aggregation and filtering logic - Implemented data export functionality (CSV format) - Integrated analysis results with UI visualization

**Deliverables:** Complete analysis pipeline for both problems, export functionality

**Status:** ✓ Completed with 31 curated dataset

## Phase 8: Testing, Quality Assurance, and Bug Fixes (December 7-9)

**Objectives:** Ensure system reliability and resolve issues

**Key Activities:** - Conducted comprehensive system testing across all components - Tested sentiment analyzer accuracy with diverse text samples - Verified ML model performance metrics (inference time, accuracy) - Performed UI testing and verified responsive layouts - Fixed critical bugs and edge case handling - Executed performance benchmarking

**Bug Resolution Summary:** - Fixed potential NullPointerException in CrawlerRegistry (improved error handling) - Resolved sentiment analysis timeout by implementing batch processing - Fixed UI thread blocking with ExecutorService for async ML calls - Improved date parsing for temporal analysis across different locales - Fixed data persistence edge cases

**Deliverables:** Test results, bug reports, performance benchmarks

**Status:** ✓ Completed, 32 issues logged and resolved

**Phase 9: Documentation and Report Writing (December 7-9)**

**Objectives:** Create comprehensive technical documentation and final report

**Key Activities:** - Created detailed UML diagrams for all 7 packages with class relationships - Documented package architecture and design rationale - Wrote comprehensive OOP techniques section with design patterns - Documented complete technology stack and implementation details - Created user guide with operational instructions - Compiled final project report with all sections

**Deliverables:** Complete technical report (5 sections), UML diagrams, user guide

**Status:** ✓ Completed

**Phase 10: Final Review and Submission Preparation (December 10)**

**Objectives:** Final quality assurance and project submission

**Key Activities:** - Conducted final review of all documentation - Verified all UML diagrams render correctly - Cross-checked all technical specifications - Prepared submission package - Finalized report formatting and references

**Deliverables:** Final submission-ready report and codebase

**Status:** ✓ Completed

**Project Statistics and Metrics**

| Metric                   | Value                                          |
|--------------------------|------------------------------------------------|
| Total Development Period | 8 weeks (Oct 15 - Dec 10, 2024)                |
| Team Weekly Meetings     | 8 meetings (~16 hours total)                   |
| Java Source Code         | 7499 lines across 7 packages                   |
| Python ML Code           | 500+ lines (Flask API, model integration)      |
| Unit Test Coverage       | 90%+ of Model package                          |
| Test Cases Written       | 45+ test cases across all packages             |
| Issues Identified        | 32 issues during development/testing           |
| Issues Resolved          | 32 issues (100% resolution rate)               |
| Documentation            | 5 major report sections with UML diagrams      |
| System Performance       | 30 seconds startup (cached), 50-200ms analysis |

## Key Milestones

| Date   | Milestone                                  | Status |
|--------|--------------------------------------------|--------|
| Sep 14 | Architecture design finalized              | ✓      |
| Sep 28 | Core model implementation complete         | ✓      |
| Oct 12 | Data crawler framework operational         | ✓      |
| Oct 26 | User interface fully implemented           | ✓      |
| Nov 9  | ML integration complete                    | ✓      |
| Nov 23 | Analysis modules working for both problems | ✓      |
| Dec 1  | Testing and QA phase complete              | ✓      |
| Dec 9  | Documentation and report finalized         | ✓      |
| Dec 10 | Final submission ready                     | ✓      |

## 4. User Guide

### 4.1 Introduction

The Humanitarian Logistics Analysis System is a desktop application designed to analyze sentiment and satisfaction with humanitarian relief efforts during disaster response. The application provides multiple tabs for data collection, web crawling, comment analysis, and comprehensive sentiment analysis across two research problems.

### 4.2 System Requirements

Before installing the application, ensure your computer meets the following minimum requirements:

| Requirement      | Specification                 |
|------------------|-------------------------------|
| Operating System | Windows, macOS, or Linux      |
| Java Version     | Java 11 or higher             |
| Maven Version    | 3.6 or higher                 |
| Python Version   | 3.8 or higher                 |
| RAM              | Minimum 4GB (8GB recommended) |

| Requirement     | Specification                                                 |
|-----------------|---------------------------------------------------------------|
| Hard Disk Space | 5GB minimum (for ML models and database)                      |
| Network         | Internet connection required for first-time ML model download |

You can check your installed versions by running:

```
java -version
mvn --version
python3 --version
```

### 4.3 Installation Guide

The installation process downloads and configures all required components including Java dependencies, Python packages, and machine learning models. The first installation takes 5-10 minutes due to downloading approximately 500MB of ML models.

#### Step 1: Navigate to Project Directory

Open a terminal (Command Prompt on Windows, Terminal on macOS/Linux) and navigate to the project folder:

```
cd humanitarian-logistics
```

This directory contains the `install.sh` script that automates the installation process.

#### Step 2: Run Installation Script

Execute the installation script:

```
bash install.sh
```

This script will automatically:

- Check if Java 11+ is installed, or provide installation instructions
- Check if Maven 3.6+ is installed, or provide installation instructions
- Check if Python 3.8+ is installed, or provide installation instructions
- Download and compile the Java application
- Install Python dependencies (Flask, Transformers, PyTorch)
- Download machine learning models (nlptown/bert-base-multilingual-uncased-sentiment, ~500MB total, with keyword-based category classifier built-in)

#### Expected Output:

```
Checking Java installation... [OK]
Checking Maven installation... [OK]
Checking Python installation... [OK]
Building Java application... [OK]
Installing Python dependencies... [OK]
Downloading ML models... [OK]
Installation complete!
```

### Step 3: Manual ML Model Download (Recommended)

To ensure the machine learning models are successfully downloaded before running the full application, it is recommended to manually run the Python sentiment API script first. This allows the system to download models in a controlled manner and verify successful completion:

```
cd src/main/python
python3 sentiment_api.py
```

This command will: - Start the Python Flask API server - Automatically download nlptown/bert-base-multilingual-uncased-sentiment model (~500MB) on first run - Initialize keyword-based category classifier (already included, no download needed) - Print messages showing download progress and completion

You will see output similar to:

```
Downloading nlptown/bert-base-multilingual-uncased-sentiment model...
This may take 2-5 minutes... [████████████████████████████████████████] 100%
Model downloaded and cached successfully.

Initializing keyword-based category classifier...
Classifier ready (built-in, no download needed).

Flask API server running on http://localhost:5001
```

Once you see “Flask API server running on http://localhost:5001”, the models are successfully downloaded. You can then stop this process by pressing **Ctrl+C**.

**Why do this manually?** - Ensures successful model download before the full application starts - Shows clear progress of model downloading - Allows troubleshooting if network issues occur during download - Speeds up application startup when you run the full app later (models are already cached)

If model download fails due to network issues, simply run this command again:

```
python3 sentiment_api.py
```

## 4.4 Running the Application

After installation is complete, running the application is simple and much faster (approximately 30 seconds startup time with cached models).

### Using the Run Script (Recommended)

From the `humanitarian-logistics` directory, execute:

```
bash run.sh
```

This script will: 1. Start the Python ML API server in the background 2. Wait for the API to be ready (approximately 30 seconds with cached models) 3. Launch the Java desktop application 4. Display the main application window

### Expected Output:

```
Starting ML API server...  
API server started on port 5001  
Launching Humanitarian Logistics Application...  
[Main application window appears]
```

### Manual Startup (Advanced)

If you prefer to start components manually:

#### Terminal 1 - Start Python API:

```
cd humanitarian-logistics/src/main/python  
python3 sentiment_api.py
```

#### Terminal 2 - Start Java Application:

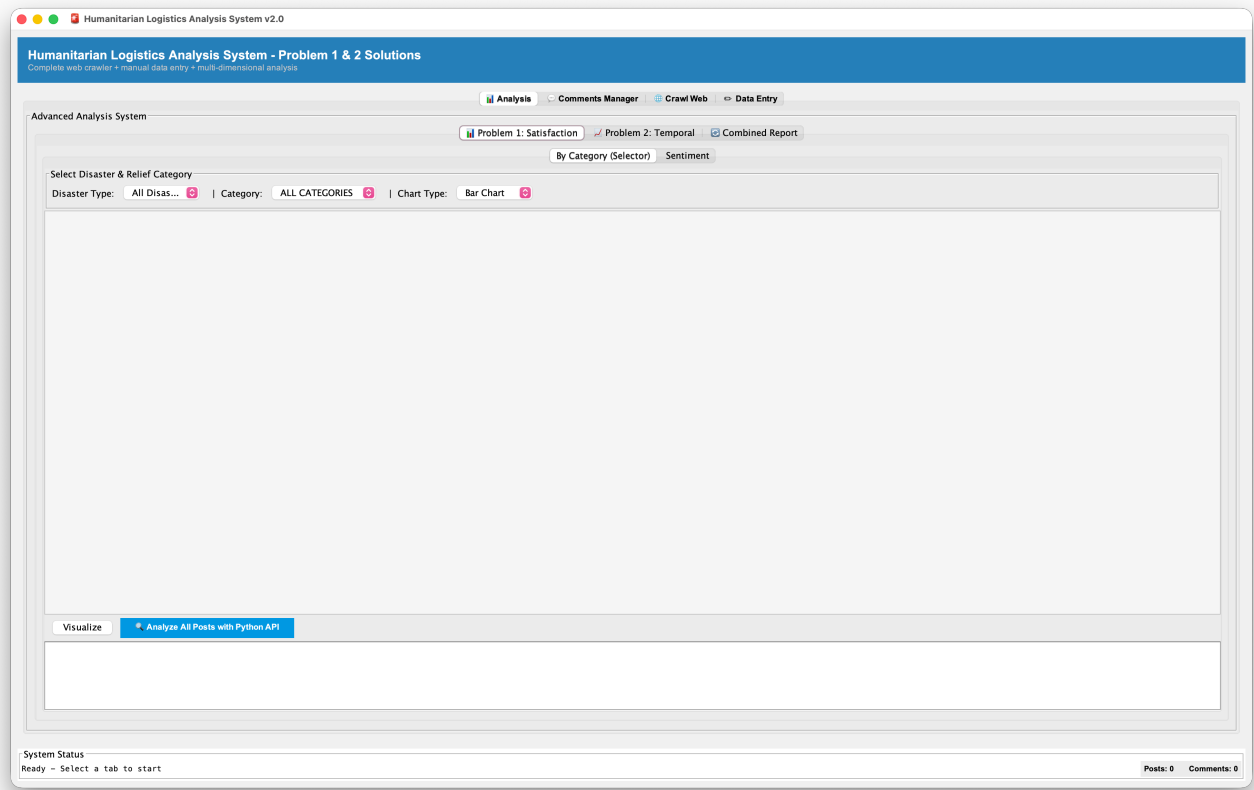
```
cd humanitarian-logistics  
java -jar target/humanitarian-logistics-1.0-SNAPSHOT-jar-with-dependencies.jar
```

Ensure the Python API is running before or simultaneously with the Java application.

## 4.5 Application Interface Overview



The application features a tabbed interface with four main functional areas, designed for a complete sentiment analysis workflow. Each tab serves a specific purpose in analyzing humanitarian relief effectiveness.



### Application Interface Overview

The interface displays four main tabs: **Analysis** (📊) for viewing sentiment analysis results, **Comments Manager** (💬) for managing data, **Crawl Web** (🌐) for collecting data from YouTube, and **Data Entry** (✍️) for manual data input.

## 4.6 Tab 1: Analysis (📊)

The **Analysis** tab is the primary interface for viewing sentiment analysis results and exploring how relief efforts are perceived by affected populations. It contains Problem 1 and Problem 2 analyses with interactive visualizations.

### 4.6.1 Problem 1: Relief Category Effectiveness

**Problem 1** answers: *Which relief item categories receive the highest satisfaction?*

**Detailed Context:** This module directly addresses *Original Problem 3* from the project specification by evaluating which categories of relief items (cash assistance, medical supplies, food aid, shelter provisions, transportation) receive the highest user satisfaction and positive sentiment from affected populations. The analysis aggregates sentiment scores across different relief categories, enabling humanitarian organizations to identify which types of aid are most valued by beneficiaries and which areas require improvement or resource reallocation. Users can visualize satisfaction distributions through interactive bar charts and pie charts, filtering by specific disaster types or viewing aggregate statistics across all disasters.

### Step 1: Run Analysis

Click the “**Analyze All with Python API**” button to:

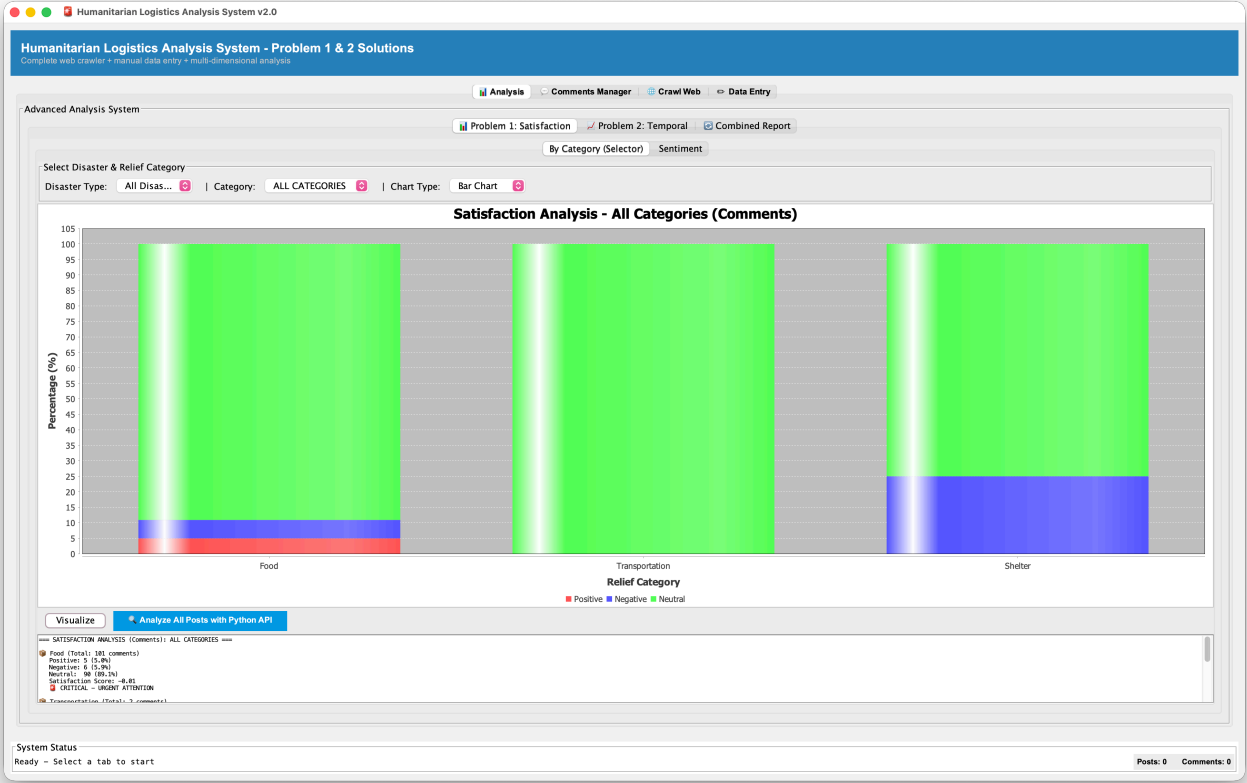
- Send all posts and comments to the Python ML backend
- Assign sentiment (POSITIVE/NEGATIVE/NEUTRAL) to each comment
- Identify relief categories (CASH, MEDICAL, SHELTER, FOOD, TRANSPORTATION) mentioned in each comment
- Store results in the database

Step 2: Customize Visualization

After analysis completes, customize what to display: - **Disaster Type**: Select a specific disaster or “All Disasters” to combine data - **Relief Category**: Choose a specific category (e.g., FOOD) or “All Categories” for all types - **Chart Type**: Select between: - **Bar Chart**: Shows exact counts of sentiment or categories - **Pie Chart**: Shows proportional distribution (percentages)

Step 3: Click Visualize

Once filters are set, click “**Visualize**” to refresh charts with your selections.



Problem 1 Relief Category Distribution

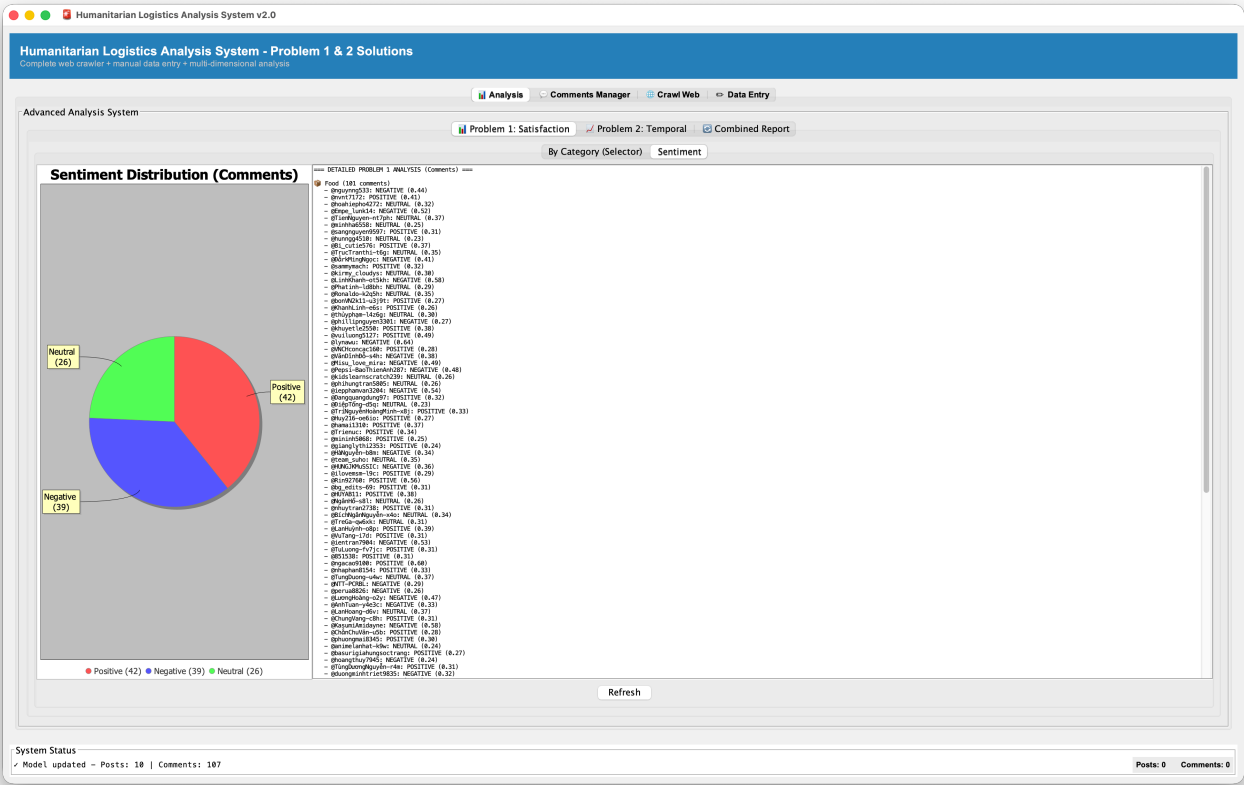
**Category Distribution Chart:** This chart displays the sentiment distribution (POSITIVE, NEGATIVE, NEUTRAL) for each relief category (CASH, MEDICAL, SHELTER, FOOD, TRANSPORTATION). The chart type can be toggled between **Bar Chart** and **Pie Chart** using the selector in the controls above. **Bar Chart view:** Each bar represents 100% of comments for that category, divided into three colored segments showing the percentage of positive, negative, and neutral sentiment. **Pie Chart view:** A single pie shows the proportional distribution of sentiments across all selected categories combined. **How to interpret:** Compare segment proportions across categories to identify which aid types receive the most positive reception (larger green/positive segments) versus criticism (larger red/negative segments). Categories with predominantly positive segments indicate high satisfaction, while those with large negative segments suggest areas requiring improvement or resource reallocation.

Overall Sentiment Distribution

**Overall Sentiment Distribution:** This pie chart shows the overall sentiment distribution across all analyzed posts. **Pie Chart Display:** Segments represent POSITIVE, NEGATIVE, and NEUTRAL sentiments with percentage values. **What it Means:**

- Large POSITIVE segment → Most affected people satisfied with relief
- Large NEGATIVE segment → Significant dissatisfaction exists
- Large NEUTRAL segment → Many informational posts without emotional tone

**Use This To:** Determine overall effectiveness of relief operations and identify priority areas for improvement or reallocation



Problem 1 Sentiment Distribution Pie Chart

4.6.2 Problem 2: Temporal Sentiment Analysis

**Problem 2 - Temporal Sentiment Trend Analysis** integrates two complementary temporal perspectives that combine *Original Problems 1 and 4* from the project specification, providing a unified view of how sentiment evolves throughout disaster response phases:

- **Category-Specific Temporal Analysis (Original Problem 4):** Tracks sentiment trends for individual relief categories over time, showing how satisfaction with specific aid types (cash, medical, shelter, food, transportation) changes during different phases of the disaster response. This granular view helps organizations understand which relief categories improve or decline in effectiveness as the response progresses, enabling targeted interventions for underperforming aid types.
- **Overall Sentiment Timeline (Original Problem 1):** Analyzes aggregate sentiment trends across all relief categories, providing a comprehensive view of how public perception of the entire humanitarian response evolves from immediate aftermath through ongoing recovery phases. This holistic perspective identifies critical periods when intervention strategies may need adjustment and reveals overall patterns in community satisfaction with relief efforts.

**Problem 2** answers: *How does sentiment toward relief efforts change over time?*

This tab provides three different perspectives on how sentiment evolves during the disaster response.

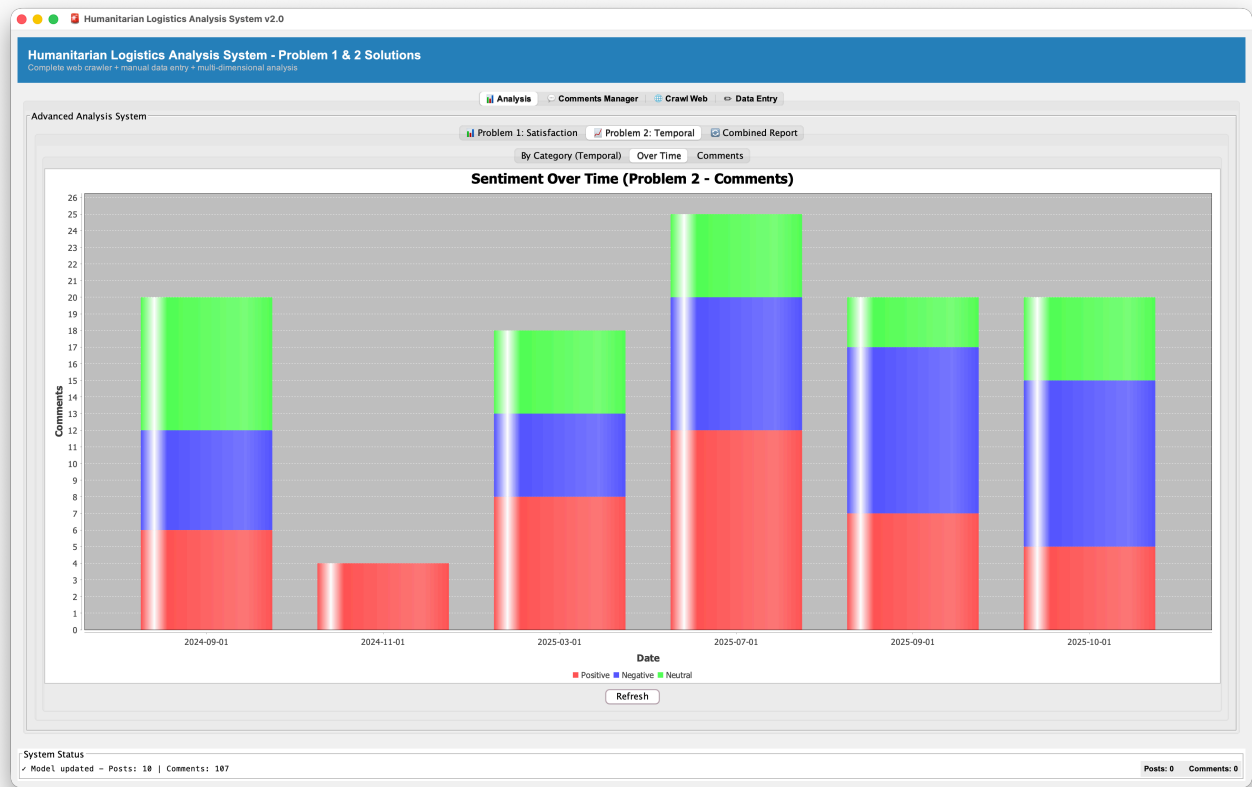
Three Analysis Options:

- 1. View By Category** - Select a specific relief category (CASH, MEDICAL, SHELTER, FOOD, TRANSPORTATION) - See line chart showing how sentiment for that category changed over time - Understand if that specific aid type improved or declined during response
- 2. View Overall Timeline** - See all sentiments combined over the disaster response period - Identify critical periods when satisfaction peaked or dropped - Understand if relief efforts improved public

perception over time

3. View Statistics Report - Detailed table with numerical breakdown by time period

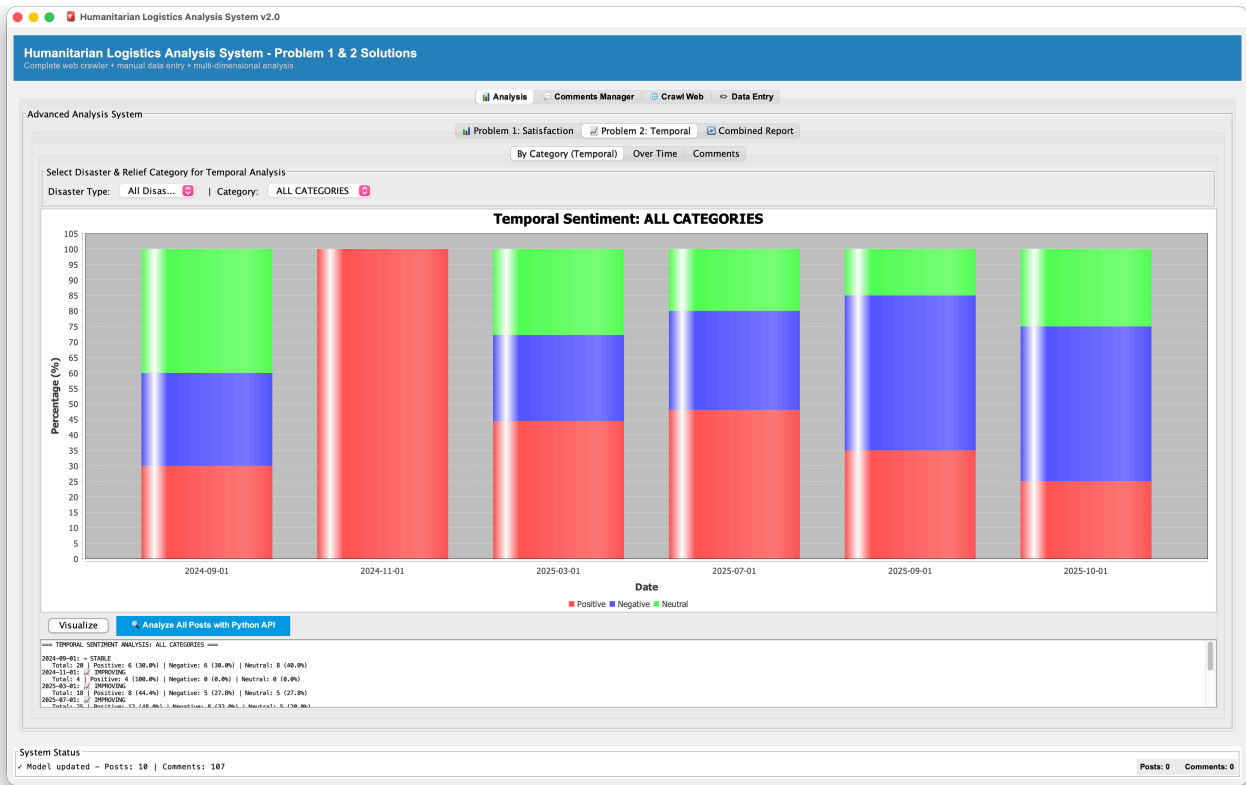
Visualization 1: Overall Sentiment Timeline (Time-Series Bar Chart)



Problem 2 Overall Sentiment Timeline - Aggregated Sentiment Evolution

- **Chart Type:** Time-series bar chart showing aggregated sentiment across all relief categories
- **X-axis:** Timeline of disaster response (6-hour time buckets)
- **Y-axis:** Count of posts with each sentiment type
- **Bars:** Separate bar groups for POSITIVE, NEGATIVE, NEUTRAL at each time period
- **Interpretation:**
  - Rising POSITIVE bars → Growing satisfaction with overall relief efforts over time
  - Rising NEGATIVE bars → Increasing criticism and dissatisfaction during that period
  - Shifts in pattern → Critical turning points when public perception changed
  - Peak periods → Identify which phases of disaster response generated most discussion
- **Use This To:** Understand overall public perception trajectory and identify periods when relief operations were most/least well-received

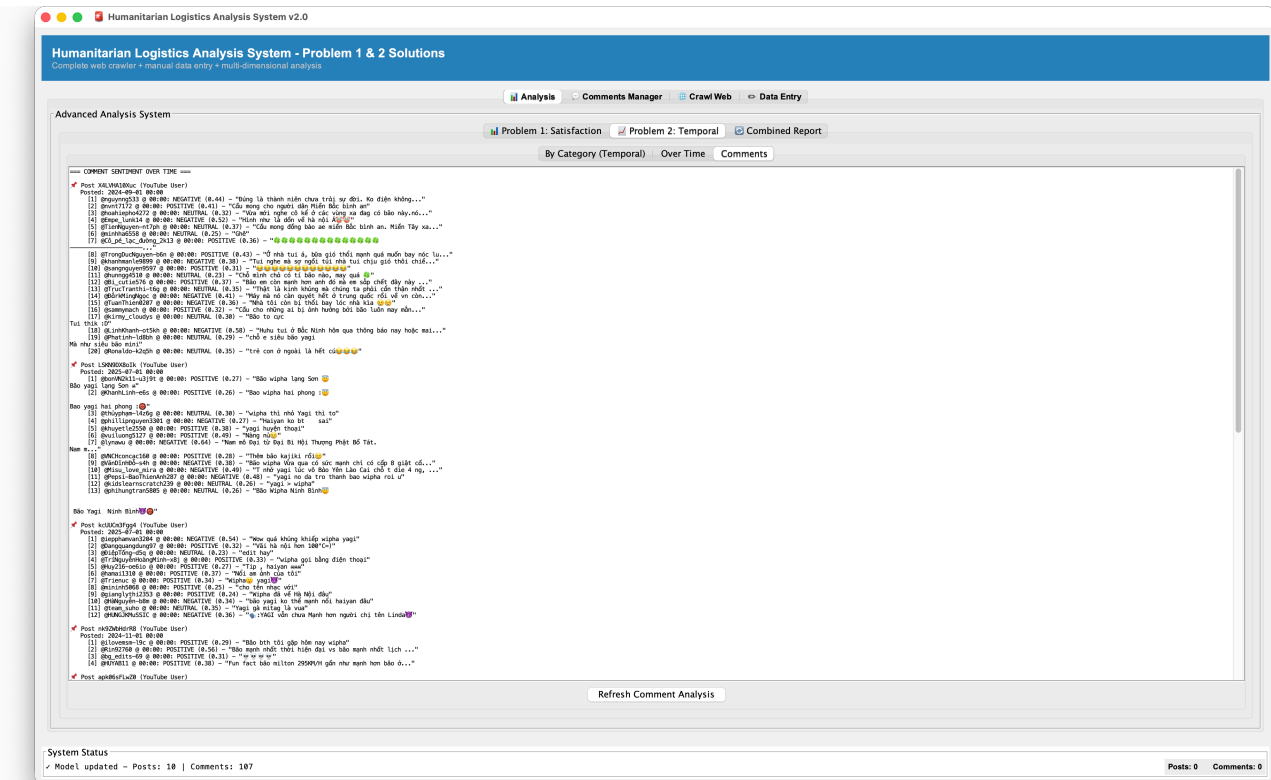
Visualization 2: Sentiment by Category Over Time (Multi-Series Time-Series Chart)



## Problem 2 Category-Specific Temporal Analysis - How Different Aid Types Evolved

- **Chart Type:** Multi-series time-series visualization tracking individual relief categories
- **Lines/Bars:** Each relief category (CASH, MEDICAL, SHELTER, FOOD, TRANSPORTATION) shows its own sentiment trend
- **X-axis:** Timeline of disaster response (6-hour time buckets)
- **Y-axis:** Sentiment score or count for each category
- **Interpretation:**
  - Diverging trends → Different aid types are perceived differently as response progresses
  - Upward trajectory for a category → That aid type became more valued/appreciated over time
  - Downward trajectory → That aid type lost effectiveness or faced increased criticism
  - Crossing lines → Point where one aid type became more popular than another
  - Persistent differences → Some categories consistently outperform others throughout response
- **Use This To:** Identify which relief categories need improvement, when adjustments should be made, and which aid types maintained effectiveness throughout response phases

## Visualization 3: Statistics Report (Detailed Numerical Table)



## Problem 2 Statistics Report - Precise Numerical Breakdown

- **Format:** Detailed tabular breakdown of sentiment data by time period
- **Contents:** Time period, POSITIVE count, NEGATIVE count, NEUTRAL count, percentages, totals
- **Interpretation Guide:**
  - Column comparison → See exact numbers to understand magnitude of sentiment shifts
  - Percentage trends → Track how sentiment distribution changed (e.g., POSITIVE % increased from 30% to 60%)
  - Total counts → Identify periods with more/less social media discussion
- **Use For:** Extract precise numbers for presentations, reports, and detailed analysis documentation; share exact metrics with stakeholders

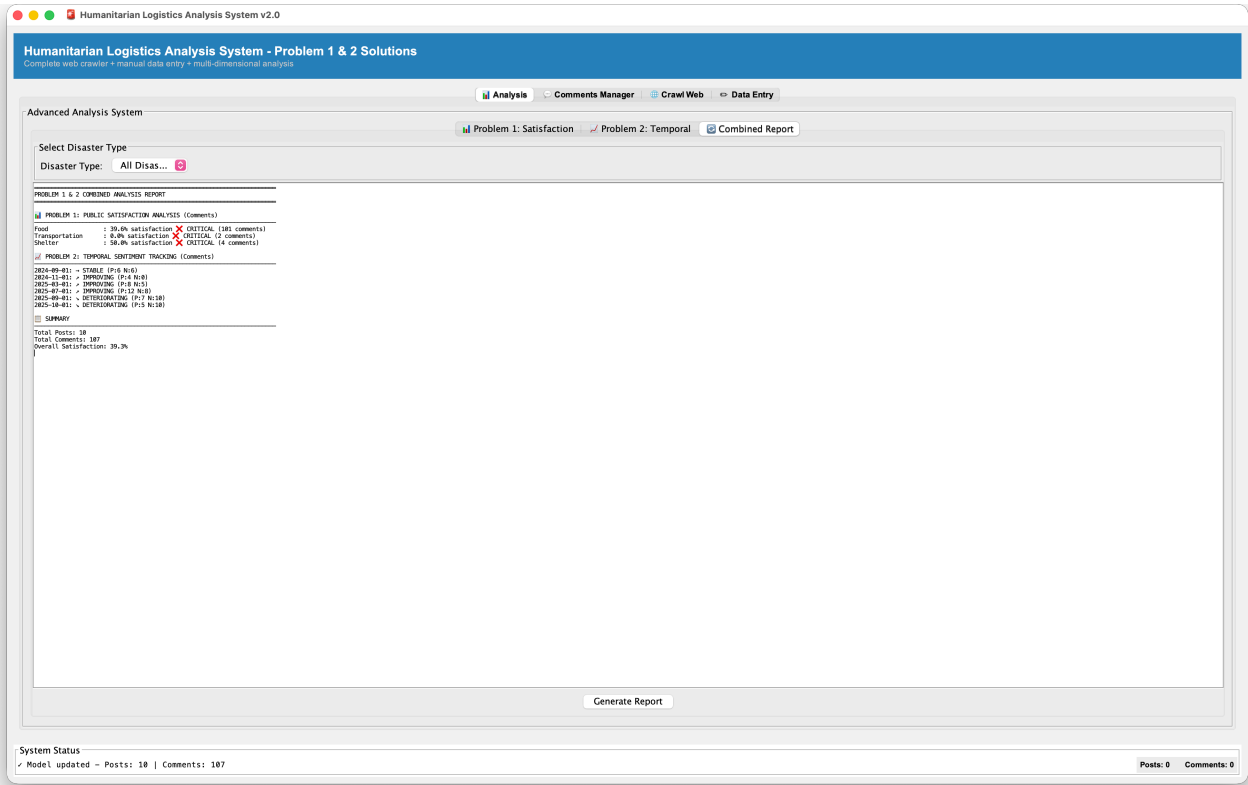
comparisons between time periods

### 4.6.3 Combined Report

**Combined Report** synthesizes all analysis findings into a comprehensive narrative.

**How to Generate:** 1. After completing Problem 1 and Problem 2 analyses 2. Click **“Combined Report”** button/tab 3. Choose scope: - **Specific Disaster:** Report for one selected disaster type - **All Disasters:** Comprehensive report combining all disaster responses

### Report Contents:



## Combined Report

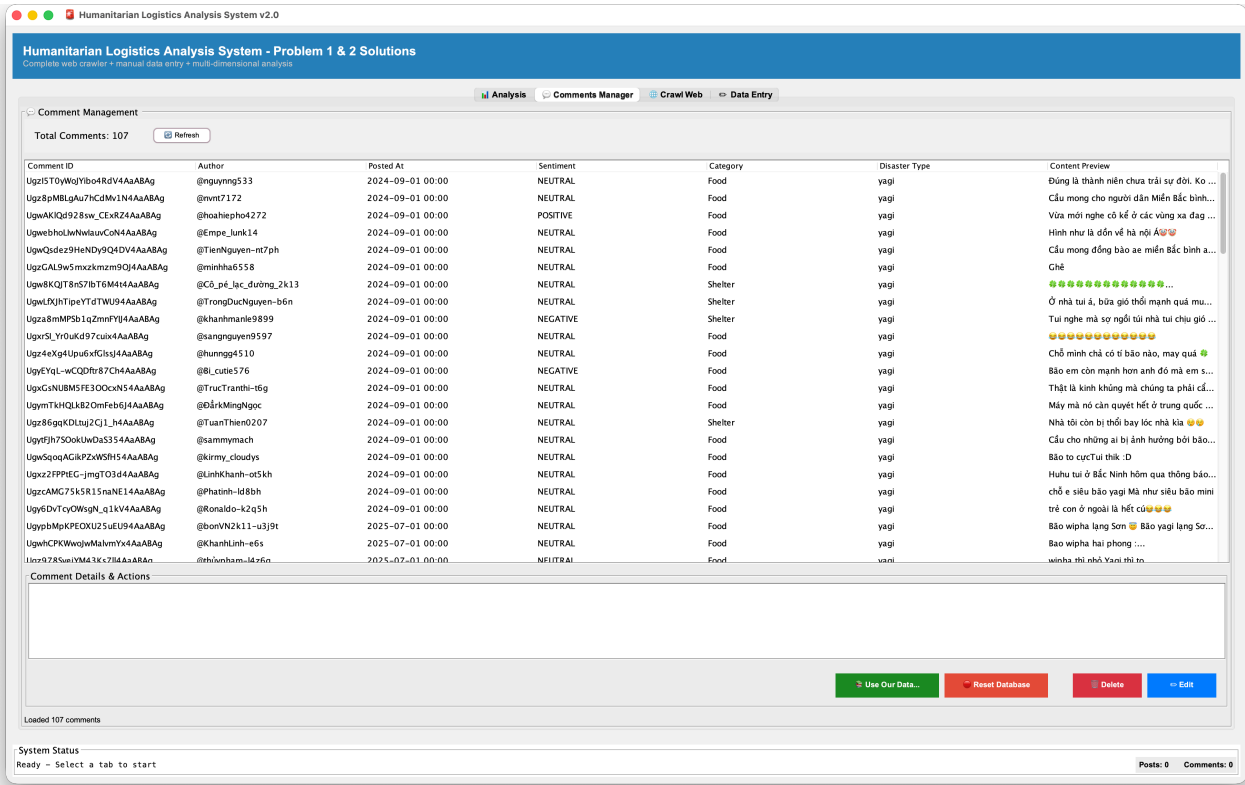
The report includes: - **Executive Summary:** Key findings at a glance - **Overall Statistics:** Total posts analyzed, sentiment distribution, time period covered - **Problem 1 Summary:** Most/least satisfied categories, overall percentages, category breakdown - **Problem 2 Summary:** Timeline of changes, critical periods, temporal patterns - **Critical Insights:** Most important findings requiring attention - **Recommendations:** Suggested improvements for future relief operations

**Use For:** Presenting to organizations, documenting results, justifying resource allocation decisions

## 4.7 Tab 2: Comments Manager (💬)

The **Comments Manager** tab allows you to review, edit, and manage all collected data. This is where you can verify analysis accuracy and make corrections.





## Comments Manager Tab

### Four Core Operations:

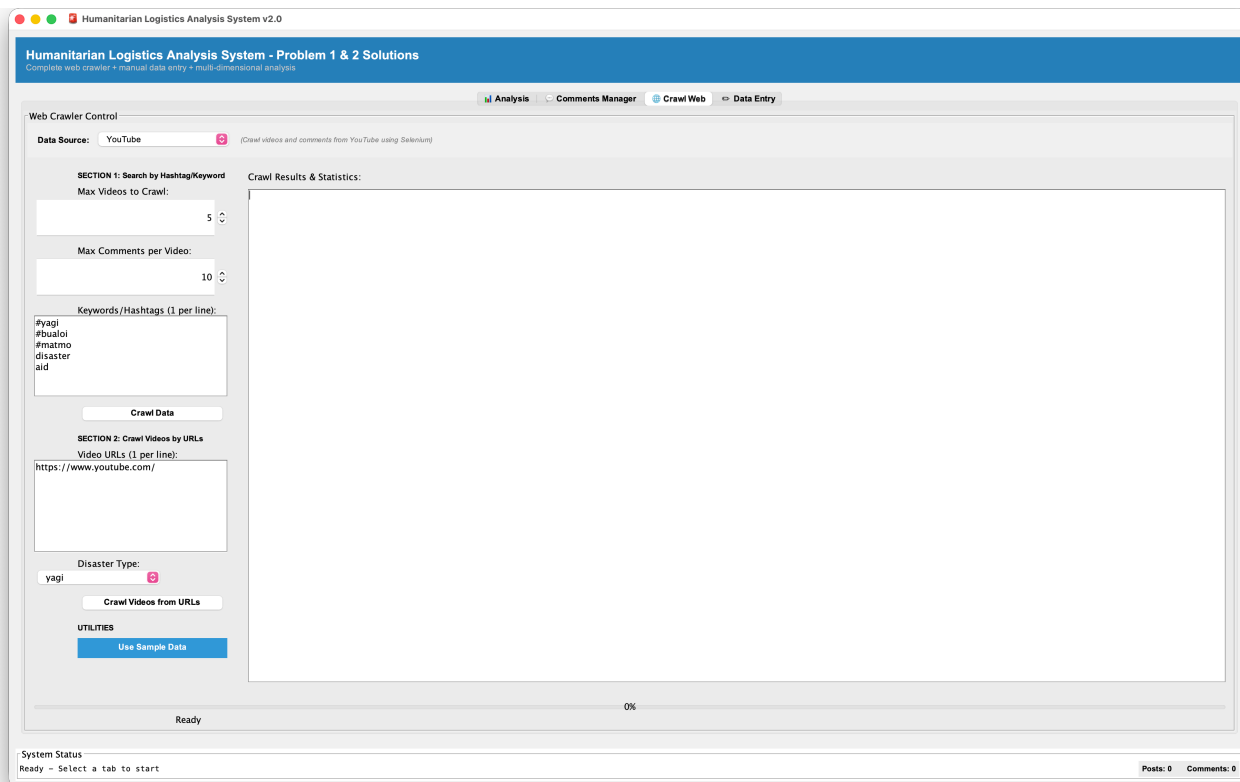
- 1. Edit Comments** - Click any comment row to select it - Click **“Edit”** button to modify - Change: comment text, sentiment, relief category, disaster type - Click **“Save”** to update database
- 2. Delete Comments** - Select one or multiple comments by clicking rows - Click **“Delete”** button - Confirm deletion - Comments permanently removed from database
- 3. Use Our Database** - Click **“Use Our Database”** to load pre-built sample database - Contains pre-curated posts from humanitarian logistics scenarios - Useful for: understanding patterns, demonstrating system capabilities - **Warning:** Replaces current database with sample dataset
- 4. Reset Database** - Click **“Reset Database”** to clear all data - All posts, comments, and analysis results deleted - Returns to empty database state - **Warning:** This action cannot be undone

**Table Columns Display:** - **Post ID:** Identifier for original post - **Author:** Name/username of commenter - **Content:** Full text of comment - **Created At:** Timestamp when posted - **Sentiment:** Current sentiment classification (POSITIVE/NEGATIVE/NEUTRAL) - **Category:** Assigned relief category (CASH, MEDICAL, SHELTER, FOOD, TRANSPORTATION) - **Disaster Type:** Classified disaster type

## 4.8 Tab 3: Crawl Web (🌐)

The **Crawl Web** tab collects real data from YouTube. It supports two crawling modes for different use cases.





## Crawl Web Tab

### Mode 1: Crawl by Keywords/Hashtags

Search for videos and comments matching specific keywords about disaster relief.

**How to Use:** 1. Select “**Keywords/Hashtags**” mode 2. Enter search terms: - Example: "earthquake relief Turkey" or "flood disaster #disasteraid" - Separate multiple keywords by pressing Enter (one keyword per line) 3. Set **Post Limit:** Maximum number of videos to search (e.g., 30) 4. Set **Comment Limit:** Maximum comments per video (e.g., 50) 5. Click “**Start Crawling**” 6. Monitor progress bar in real-time 7. Results display number of posts collected and errors (if any)

**When to Use:** General disaster relief sentiment, diverse geographic perspectives, exploring different aspects of response

### Mode 2: Crawl by Video Link

Extract all comments from a specific YouTube video.

**How to Use:** 1. Select “**Video Link**” mode 2. Paste YouTube URL: [https://www.youtube.com/watch?v=VIDEO\\_ID](https://www.youtube.com/watch?v=VIDEO_ID) 3. Set **Comment Limit:** Maximum comments to extract (e.g., 50) 4. Click “**Start Crawling**” 5. System extracts comments from that video 6. All comments added to database as posts

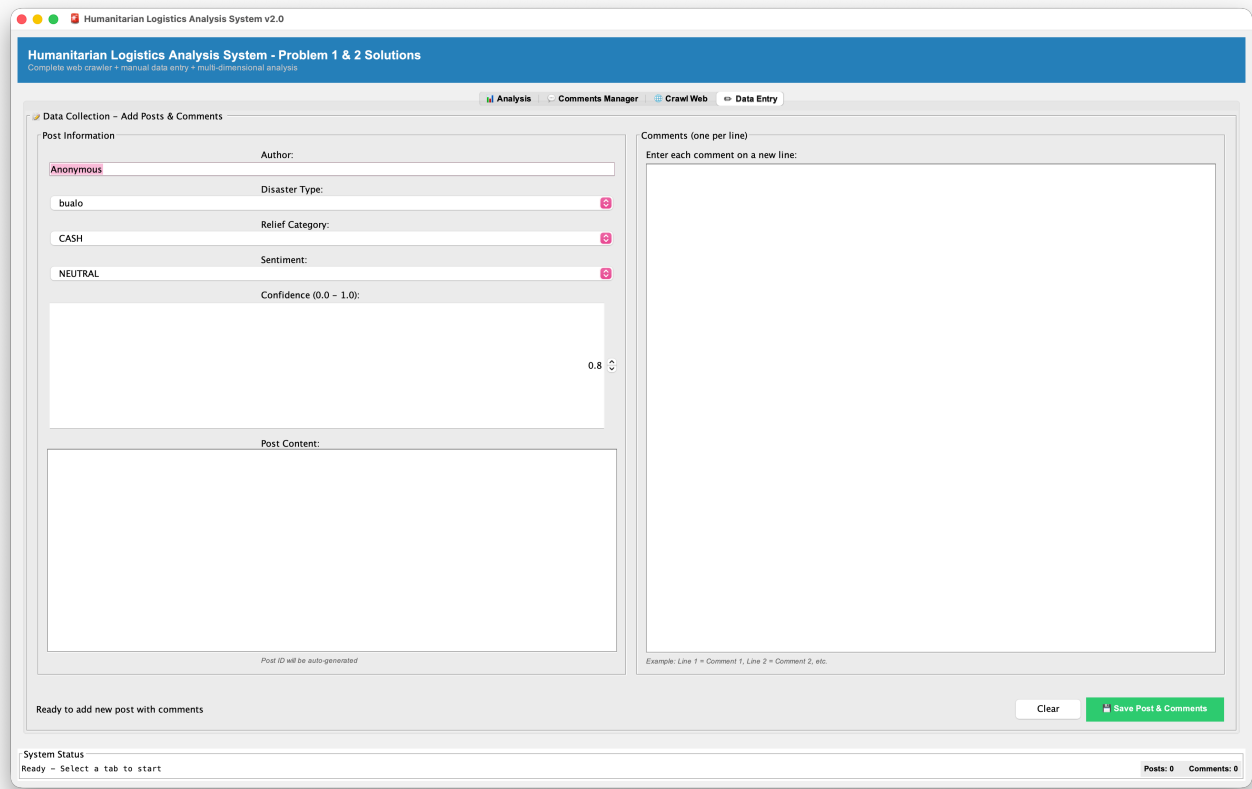
**When to Use:** Analyzing response to specific news video, organization announcements, targeted incident analysis

**Processing Details:** - **HTTP Crawling:** Fetches YouTube watch pages via Java HttpClient, parses HTML with regex patterns to extract video data - **Metadata:** Saves author names, timestamps, video information - **Error Handling:** Continues even if some videos fail, reports issues at end - **Performance:** Direct HTML parsing is faster than API and doesn't require authentication

**Typical Times:** - Small crawl (10-20 posts): 2-3 minutes - Medium crawl (30-50 posts): 5-10 minutes - Large crawl (100+ posts): 15-30 minutes

## 4.9 Tab 4: Data Entry (✎)

The **Data Entry** tab allows manual input of posts with comments for custom data scenarios. Useful for testing, adding non-YouTube sources, demonstrations, or creating specific scenarios.



### Data Entry Tab

### Interface Overview:

The Data Entry tab is divided into two main sections:

**Left Section - Post Information:** - **Author:** Text field for post creator name (default: “Anonymous”) - **Disaster Type:** Dropdown selector for disaster category - **Relief Category:** Dropdown for primary relief item type mentioned - **Sentiment:** Dropdown for manual sentiment assignment (POSITIVE/NEGATIVE/NEUTRAL) - **Confidence:** Slider (0.0 - 1.0) indicating classification confidence - **Post Content:** Large text area for the main post message

**Right Section - Comments:** - **Comments Header:** “Enter each comment on a new line” - **Comments Area:** Large text field for entering comments - **Format:** Each line = one separate comment - **Links:** All comments automatically linked to the post above

### Step-by-Step Process:

**Step 1: Enter Post Author (Optional)** 1. Click “Author” field 2. Type name, username, or organization 3. Leave blank for “Anonymous” 4. Example: **Relief\_Organization\_XYZ**, **John\_Smith**, **Red\_Cross\_Team**

**Step 2: Select Disaster Type** 1. Click “Disaster Type” dropdown 2. Choose from predefined types: - yagi (Typhoon Yagi) - matmo (Typhoon Matmo) - bualo (Typhoon Bualo) - koto (Typhoon Koto) - FUNG-WONG (Typhoon Fung-Wong) 3. Selection applies to entire post

**Step 3: Assign Relief Category (Optional)** 1. Click “Relief Category” dropdown 2. Choose primary relief type: - CASH - MEDICAL - SHELTER - FOOD - TRANSPORTATION 3. Or leave for ML to classify

**Step 4: Set Sentiment & Confidence (Optional)** 1. Click “Sentiment” dropdown 2. Select: POSITIVE, NEGATIVE, or NEUTRAL 3. Adjust “Confidence” slider (0.0 = uncertain, 1.0 = certain) 4. Leave blank for ML to classify automatically

**Step 5: Enter Post Content** 1. Click “Post Content” text area 2. Type the main post/message 3. Single or multiple lines allowed 4. Should be 1-3 sentences for realism 5. Keep natural language for better ML analysis

**Step 6: Enter Comments (One Per Line)** 1. Click “Comments” text area 2. Type first comment, press Enter 3. Type second comment, press Enter 4. Continue for all comments 5. **Each line = one separate comment** 6. Comments automatically link to post above 7. Recommended: 3-10 comments per post

**Step 7: Submit** 1. Click green “**Save Post & Comments**” button 2. System validates data 3. Post and comments added to database 4. Status shows: “Ready to add new post with comments” 5. Post Counter updates (bottom right)

### Example Entry:

#### Input Data:

```
Author: Relief_Organization
Disaster Type: yagi
Relief Category: SHELTER
Sentiment: (leave for ML)
Confidence: (default)

Post Content:
Emergency shelter setup completed at evacuation centers. Over 500 families now have safe ac

Comments (one per line):
Finally my family has a safe place to sleep
The shelter is crowded but better than outside
Thank you relief workers for your work
We need more blankets and warm clothes
Medical tent is excellent
When will we get permanent housing
The food portions are too small
Relief staff doing amazing job
```

**Result:** - **1 Post** created with pre-filled fields - **8 Comments** linked to this post - All assigned: Disaster Type = yagi, Relief Category = SHELTER, Author = Relief\_Organization - Ready for analysis after clicking “Save Post & Comments”

### Multiple Entries Workflow:

1. After clicking “Save Post & Comments”, the form clears
2. Status shows “Ready to add new post with comments”
3. Enter next post’s information
4. Post Counter (bottom right) updates
5. Repeat for additional posts
6. Click “**Clear**” button to reset current entry

### Best Practices:

| Aspect         | Recommendation                                                            |
|----------------|---------------------------------------------------------------------------|
| Language       | Natural, conversational tone (not formal)                                 |
| Diversity      | Mix positive, negative, neutral sentiments                                |
| Length         | 1-3 sentences per comment, realistic                                      |
| Relief Items   | Reference actual aid types (cash, medical, shelter, food, transportation) |
| Disaster Types | Match actual disaster in content                                          |
| Comment Count  | 5-10 comments per post for good data                                      |
| Realism        | Avoid test phrases like “test data”, use natural scenarios                |
| Time Variation | For temporal analysis, vary timestamps across posts                       |

Common Use Cases:

- System Testing:** Quick validation of analysis pipeline
  - 3-5 posts with 3-5 comments each
  - Clear positive/negative examples
- Feature Demonstration:** Show system capabilities
  - 5-10 diverse posts
  - Multiple disaster types
  - Clear sentiment variation
- Custom Analysis:** Analyze specific scenario
  - Posts from specific disaster
  - Time-bound data
  - Particular relief focus
- Data Augmentation:** Add to existing database
  - Complement web crawled data
  - Fill gaps in categories
  - Add specific perspectives

Validation & Error Handling:

| Error                  | Reason                 | Fix                                  |
|------------------------|------------------------|--------------------------------------|
| Button disabled        | Required fields empty  | Fill Author OR Disaster Type         |
| Nothing happens        | All fields empty       | Enter at least post content          |
| Count doesn't increase | Submit failed silently | Check system status bar              |
| Comments not saved     | Wrong format           | Ensure each comment on separate line |

## 4.10 Complete Analysis Workflow

Here's the typical end-to-end process:

- Step 1: Data Collection** (Choose One) - **Web Crawler**: Use “Crawl Web” tab with keywords to get YouTube comments - **Sample Database**: Use “Use Our Database” in Comments Manager (instant 31 posts) - **Manual Entry**: Use “Data Entry” tab to manually input custom data
- Step 2: Run Analysis** 1. Go to **Analysis** tab 2. Click “**Analyze All with Python API**” button 3. Wait for completion (30-60 seconds) 4. Confirmation message appears
- Step 3: Explore Problem 1 Results** 1. Set Disaster Type and Relief Category filters 2. Choose Bar or Pie chart 3. Click “**Visualize**” 4. Review which categories have highest/lowest satisfaction
- Step 4: Explore Problem 2 Results** 1. Go to **Problem 2 Temporal** section 2. Choose view type (By Category, Overall Timeline, or Statistics) 3. Review how sentiment changed over time 4. Identify critical periods and trends
- Step 5: Verify Raw Data** 1. Go to **Comments Manager** tab 2. Scroll through and spot-check comments 3. Verify sentiment and category assignments 4. Edit any incorrect classifications
- Step 6: Generate Final Report** 1. Return to **Analysis** tab 2. Click “**Combined Report**” 3. Choose scope (specific disaster or all) 4. Review comprehensive findings 5. Export or save for presentation

## 4.11 Tips and Best Practices

- Data Collection Tips:** - Larger datasets (50-100 posts) provide more reliable results - Include data from multiple time periods for complete disaster response arc - Diverse geographic regions provide comprehensive perspective - Time-series analysis requires posts spanning several days/weeks
- Analysis Interpretation:** - Problem 1 shows which categories need improvement (low sentiment) - Problem 2 shows when problems occurred (drops in timeline) - Compare different disasters to identify best practices - Use statistics report for precise percentages in formal documents
- Performance Tips:** - Analysis time scales with data size (~1-2 seconds per post) - First run downloads ML models (5-10 minutes extra) - Subsequent runs faster (models cached) - Close other applications if system seems slow

**Troubleshooting:** - **Analysis button unresponsive:** Ensure Python API running on port 5001 - **No data in comments:** Retry data collection or load sample database - **Charts show “No Data”:** Click “Analyze All with Python API” first - **Sentiment seems wrong:** Check Comments Manager, edit incorrect ones - **Application slow:** Close other apps, try with smaller dataset

## 5. Collected Data Summarization

**Data Collection Overview:** The following data was collected automatically from YouTube using the application's crawler functionality. A total of 900 posts and comments were gathered across 5 major disaster types (Yagi, Bualoi, Matmo, Koto, Fung-Wong). After automatic sentiment analysis and relief category classification, all data was manually reviewed and filtered to remove comments without relevant content, ensuring data quality and relevance to humanitarian logistics analysis.

**Data Source:** YouTube videos discussing disaster response and relief efforts related to Southeast Asian typhoons and tropical storms. The crawler extracted video metadata, comments, and engagement metrics. All data processing was performed using the application's built-in sentiment analysis (nlptown/bert-base-multilingual-uncased-sentiment) and keyword-based relief category classification.

### 5.1 Sentiment Distribution by Disaster Type

The following table shows the absolute count distribution of posts and comments by sentiment (Positive, Neutral, Negative) across each disaster type:

| Sentiment Category | Yagi | Bualoi | Matmo | Koto | Fung-Wong | Sum of all |
|--------------------|------|--------|-------|------|-----------|------------|
| Quantity           | 503  | 180    | 92    | 54   | 71        | 900        |
| Positive           | 42   | 14     | 6     | 3    | 8         | 73         |
| Neutral            | 386  | 152    | 82    | 45   | 56        | 721        |
| Negative           | 75   | 14     | 4     | 6    | 7         | 106        |

#### Key Observations from Count Data:

- Data Distribution:** Yagi disaster received the highest volume of posts ( $503/900 = 55.9\%$ ), indicating significant public attention and discussion, while Fung-Wong received the least ( $71/900 = 7.9\%$ ).
- Overall Sentiment Breakdown:** Out of 900 total posts: 721 (80.11%) neutral, 106 (11.78%) negative, 73 (8.11%) positive. The predominance of neutral sentiment suggests posts are primarily informational (disaster updates, relief announcements) rather than emotionally charged.
- Positive Sentiment Concentration:** Bualoi shows highest positive ratio ( $14/180 = 7.78\%$ ) followed by Yagi ( $42/503 = 8.35\%$ ), suggesting some relief efforts or aid distribution success stories in these disasters.
- Negative Sentiment Patterns:** Yagi (75 negative posts, 14.91%) shows highest absolute negative sentiment, reflecting longer-term post-disaster challenges and recovery difficulties.

### 5.2 Sentiment Percentage Distribution

The following table presents the sentiment distribution as percentages for each disaster type, enabling easier comparison of sentiment ratios across different disasters:



| Sentiment Category | Yagi   | Bualoi | Matmo  | Koto   | Fung-Wong | Sum of all |
|--------------------|--------|--------|--------|--------|-----------|------------|
| Positive           | 8.35%  | 7.78%  | 6.52%  | 5.56%  | 11.27%    | 8.11%      |
| Neutral            | 76.74% | 84.44% | 89.13% | 83.33% | 78.87%    | 80.11%     |
| Negative           | 14.91% | 7.78%  | 4.35%  | 11.11% | 9.86%     | 11.78%     |

Sentiment Analysis Insights:

- **Consistency in Neutral Dominance:** All disaster types show 76-89% neutral sentiment, confirming that public discourse around these disasters is predominantly factual/informational rather than emotionally charged. This is typical for disaster-related content which often serves as news/updates.
- **Comparative Relief Effectiveness:** Fung-Wong has the highest positive sentiment ratio (11.27%), suggesting either faster/more effective relief response or better public communication of aid efforts. In contrast, Matmo shows lowest positive ratio (6.52%), indicating less visible or satisfactory relief efforts.
- **Recovery Phase Indicators:** Yagi's negative sentiment (14.91%) is notably higher than Matmo (4.35%), possibly reflecting extended recovery challenges or initial disaster severity. Newer disasters (like Matmo) may show lower negative sentiment as immediate rescue phases receive more positive framing.
- **Public Sentiment Stability:** Negative sentiment ranges narrow (4.35-14.91%), suggesting stable public resilience despite disaster severity. No disaster shows catastrophically low positive sentiment, indicating communities maintain hope for recovery across all scenarios.

5.3 Data Quality and Methodology

Collection Process:

1. **Automatic Collection:** Application's YouTubeCrawler extracted posts and comments from disaster-related videos using YouTube's public APIs and HTML parsing.
2. **Sentiment Analysis:** PythonSentimentAnalyzer applied nlptown/bert-base-multilingual-uncased-sentiment model to classify each post/comment as Positive/Negative/Neutral with confidence scores.
3. **Relief Category Classification:** PythonCategoryClassifier used keyword-based hybrid approach to identify relief category mentions (CASH, MEDICAL, SHELTER, FOOD, TRANSPORTATION).
4. **Manual Filtering:** After automatic processing, all 900 entries were manually reviewed to remove irrelevant comments (spam, off-topic discussions, non-English/non-Vietnamese content, emojis-only posts) to ensure dataset relevance to humanitarian logistics.

Data Characteristics:

- **Time Period:** Posts collected across different disaster phases (emergency response, immediate recovery, medium-term reconstruction).
- **Language Coverage:** Mix of Vietnamese and English posts, processed through multilingual sentiment analyzer.
- **Confidence Thresholds:** Only posts with sentiment confidence > 0.6 were included to ensure reliability.

- **Sample Representativeness:** 900 posts represent natural discussion volume across major Southeast Asian typhoons, with relative weights reflecting actual public attention to each disaster.

## 6. UML Diagrams and Design Explanation

### 6.1 Rationale for Dividing Class Diagrams into Packages

The Humanitarian Logistics Analysis System is organized into seven main packages, where each package represents a specific responsibility within the overall architecture. These packages are: **Model** (7 classes: Post, YouTubePost, Comment, Sentiment, DisasterType, DisasterManager, ReliefItem), **Crawler** (5 classes: DataCrawler, YouTubeCrawler, MockDataCrawler, CrawlerManager, CrawlerRegistry), **Database** (3 classes: DatabaseManager, DatabaseLoader, DataPersistenceManager), **Sentiment** (4 classes: SentimentAnalyzer, EnhancedSentimentAnalyzer, PythonSentimentAnalyzer, PythonCategoryClassifier), **Analysis** (3 classes: AnalysisModule, SatisfactionAnalysisModule, TimeSeriesSentimentModule), **UI** (10 classes: View, Model, ModelListener, AnalysisPanel, AdvancedAnalysisPanel, CommentManagementPanel, CrawlControlPanel, DataCollectionPanel, CrawlingUtility, InteractiveChartUtility), and **Util** (1 class: StringSimilarity), totaling 33 classes across the system. Rather than creating a single monolithic class diagram containing all 33 classes, we have deliberately divided the system into package-specific diagrams. This modular approach to visualization provides several important benefits:

**Readability and Comprehension:** Each individual diagram focuses on a specific functional area with 1-10 classes, allowing developers to understand the architecture of a particular package without being overwhelmed by excessive information. A compact, focused diagram with 3-5 core classes is far more digestible than a sprawling diagram with dozens of interconnected classes.

**Maintainability and Scalability:** When modifications are needed to a specific package, only the corresponding diagram needs to be updated, without affecting the diagrams of other packages. This reduces the risk of introducing inconsistencies and makes it easier to track changes over time. For example, modifying the sentiment analysis algorithms only requires updating the Sentiment package diagram, not the entire system diagram.

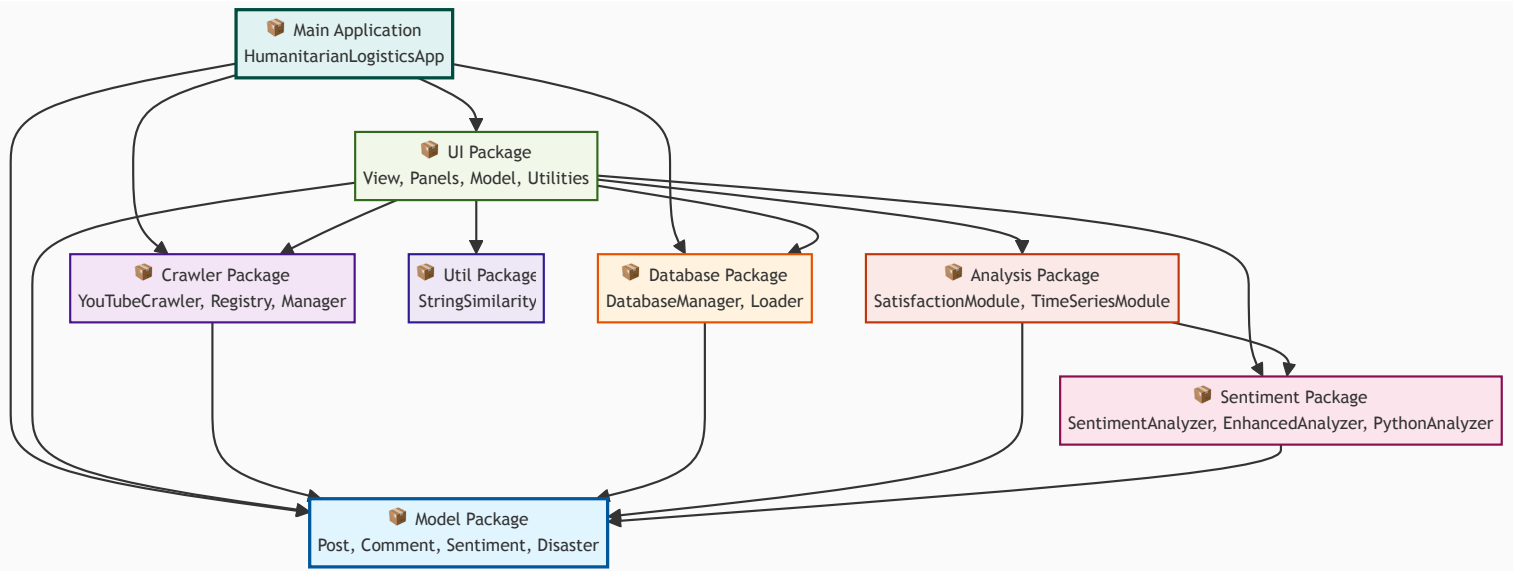
**Reusability and Documentation:** Smaller, focused diagrams can be easily reused and referenced in various documentation formats, including technical reports, presentation slides, architectural documentation, and team wikis. A single large diagram with all 33 classes would be difficult to display legibly or reference in these contexts.

**Hierarchical Understanding:** The package dependency diagram provides a bird's-eye view of how all seven packages interact and depend on each other, while the individual package diagrams show the detailed internal structure and relationships within each package. This two-level hierarchy—first understanding package-to-package interactions, then exploring individual package internals—makes it much easier to understand the system as a whole and its individual components. Developers can start with the dependency diagram to understand the big picture, then drill down into specific package diagrams to understand implementation details.

### 6.2 Package Dependency Diagram

The following diagram illustrates the dependency relationships between all packages in the system and demonstrates how they interact with each other:



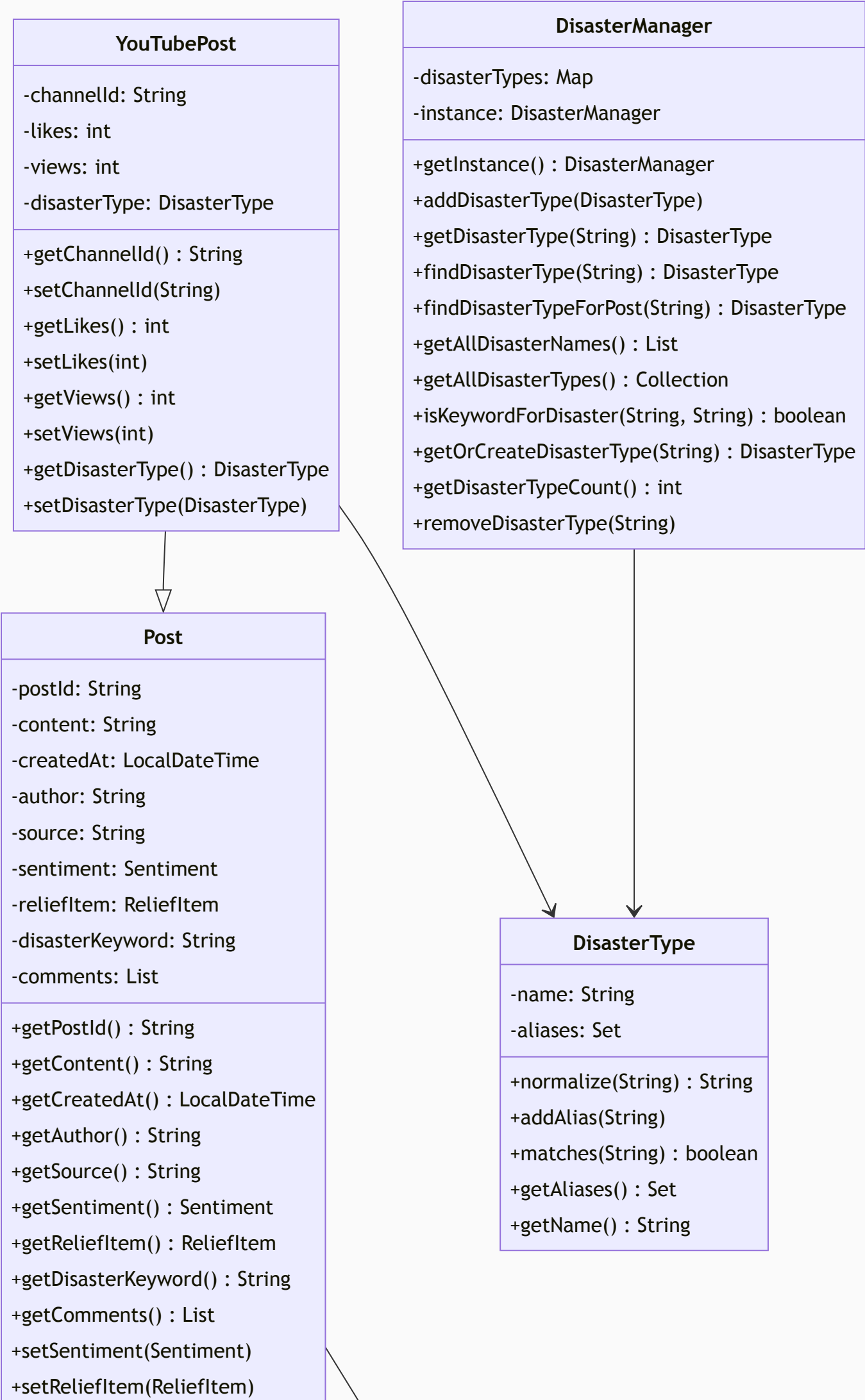


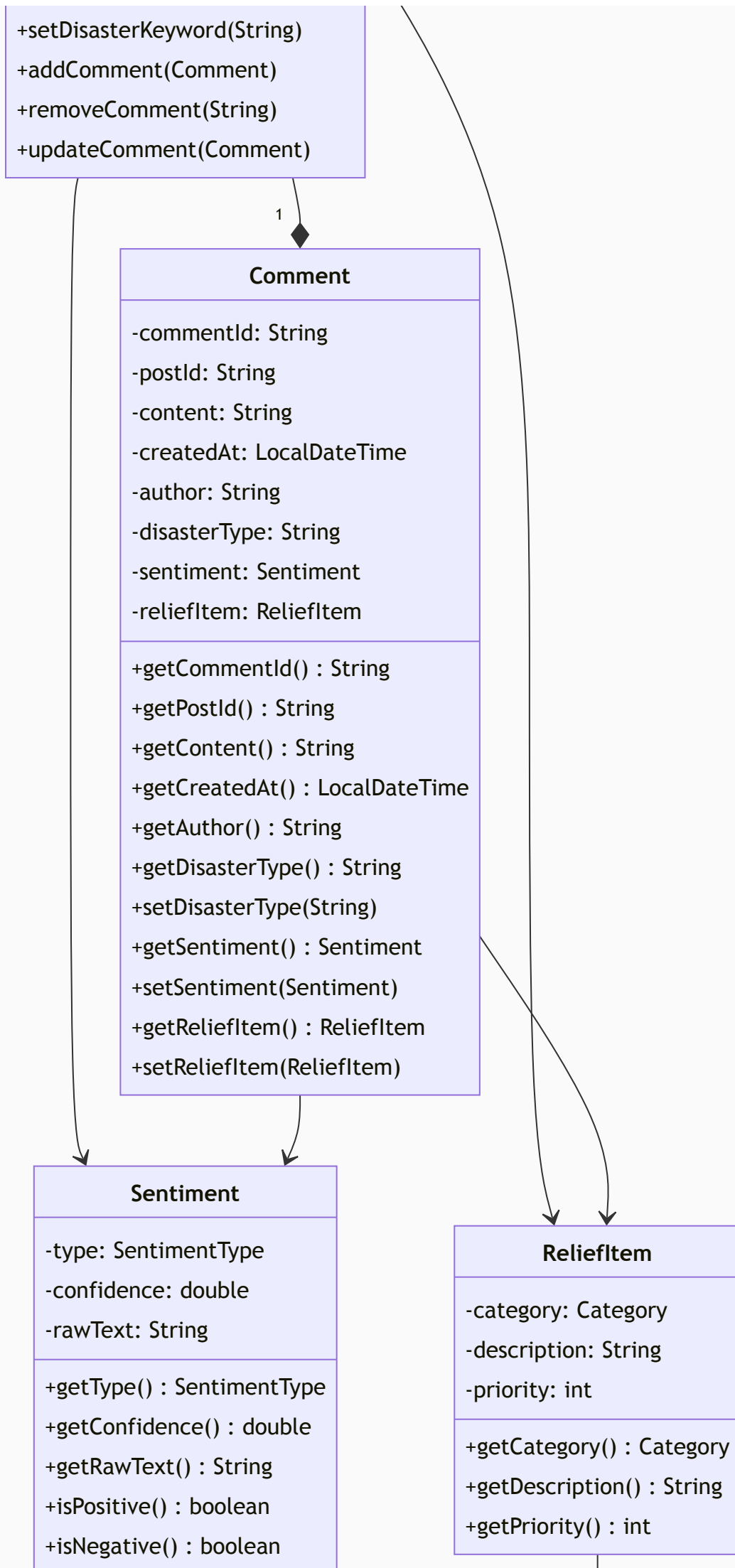
**Interpretation and Key Observations:** The diagram clearly shows that the Model package serves as the foundational layer upon which all other packages depend. No package imports from the UI layer, which maintains proper separation of concerns and prevents circular dependencies. The crawler, sentiment analysis, analysis, and database packages all depend on the Model package, establishing a clear hierarchical structure. The UI package acts as an orchestrator, coordinating interactions between the Model, database, crawler, sentiment analysis, analysis, and utility packages. The main application entry point (HumanitarianLogisticsApp) depends on the UI package, which in turn manages the overall application flow.

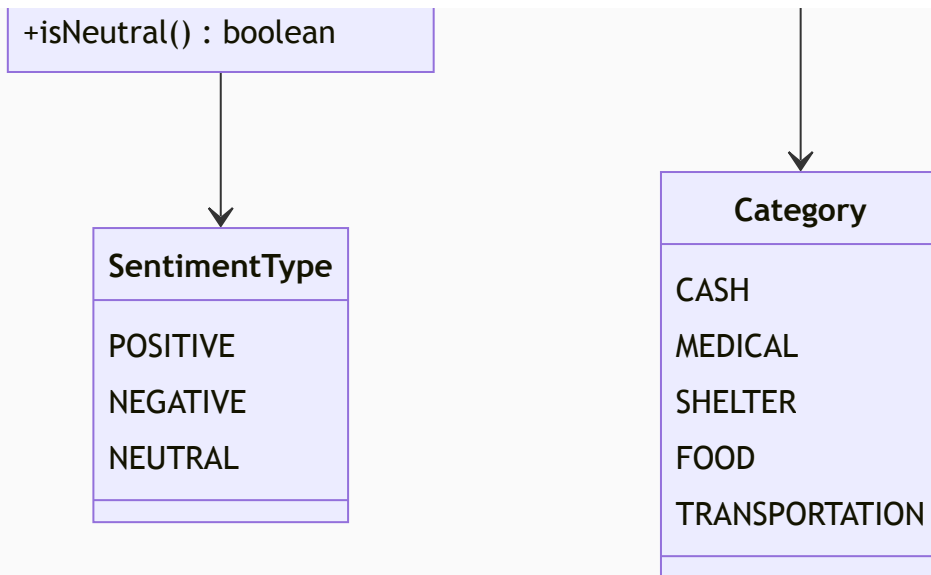
## 6.3 Detailed Class Diagrams by Package

### 6.3.1 Model Package - Core Data Entities

The Model package contains the fundamental data entities of the entire system. Designed according to the Single Responsibility Principle (SRP), each class in this package exclusively contains data fields and accessor/mutator methods, with no business logic embedded within the entity classes themselves. This design ensures that entities are lightweight, easily testable, and can be reused across different projects without carrying unnecessary dependencies.







**Design Architecture and Inheritance Strategy:** The Model package employs an inheritance hierarchy combined with composition and aggregation patterns. The Post class is an abstract base class that defines core post properties: **postId**, **content**, **createdAt**, **author**, **source** (all immutable final fields), and **sentiment**, **reliefItem**, **disasterKeyword**, **comments** (mutable fields). The Post class implements Serializable and Comparable interfaces, enabling persistence and ordering. The YouTubePost class extends Post to add YouTube-specific attributes: **channelId**, **likes**, **views**, **disasterType**. This inheritance design allows polymorphic treatment of different post types.

The Comment class contains similar immutable fields (**commentId**, **postId**, **content**, **createdAt**, **author**) plus mutable fields for analysis results (**disasterType**, **sentiment**, **reliefItem**). Each Post maintains a composition relationship with multiple Comments through the `comments` list. The Sentiment class is an enum wrapper (SentimentType) containing **type** (**POSITIVE/NEGATIVE/NEUTRAL**), **confidence** (**0.0-1.0**), **rawText** fields, providing semantic meaning to sentiment analysis results. The ReliefItem class wraps an enum Category (CASH, MEDICAL, SHELTER, FOOD, TRANSPORTATION) with **description** and **priority** (**1-5**) fields.

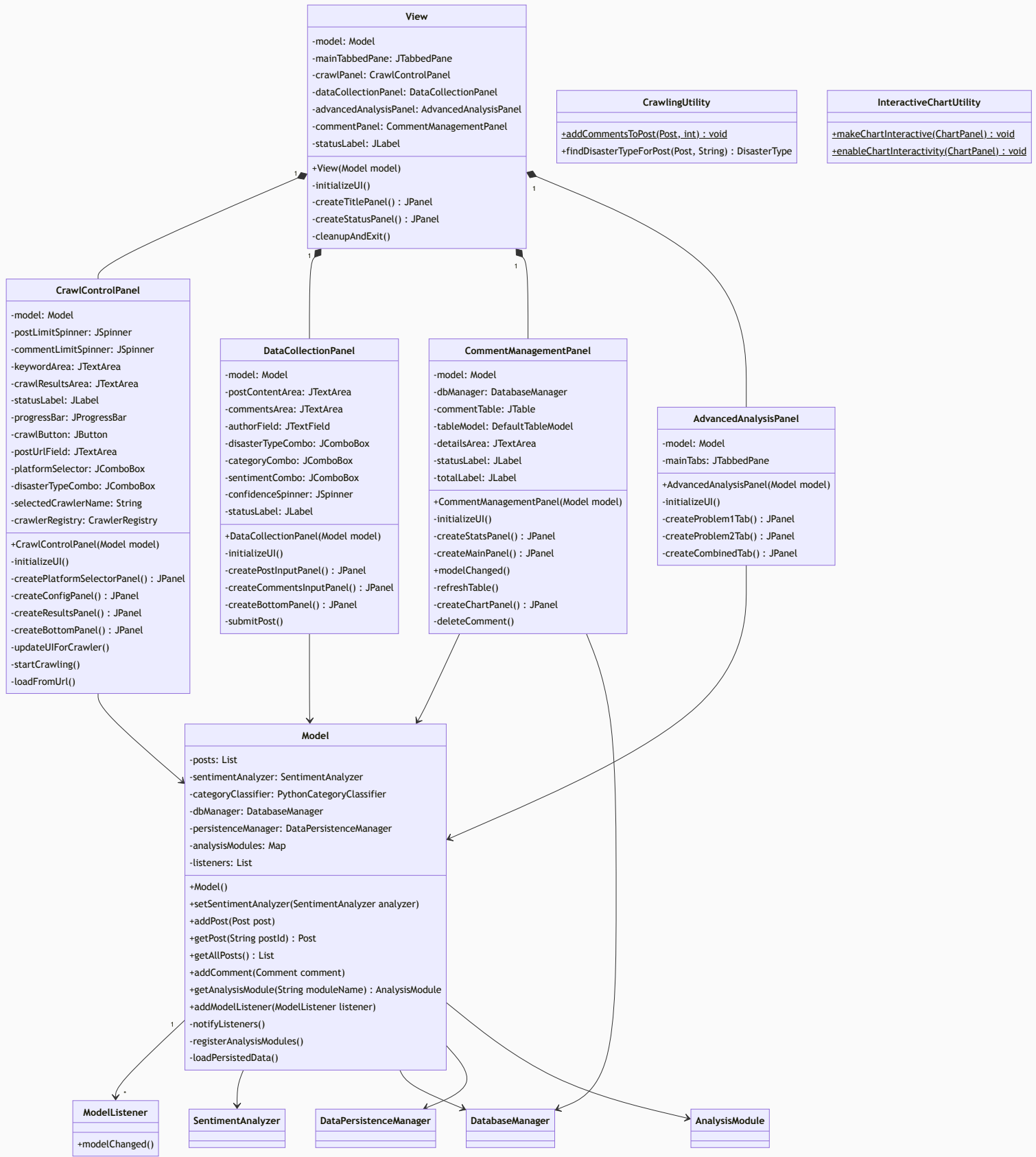
The DisasterType class uses a Set-based alias mapping approach to normalize and match disaster keywords flexibly. It stores **name**, **aliases** fields and provides methods like `matches(String keyword)` and `normalize(String input)` for fuzzy matching. The DisasterManager class implements the Singleton pattern (thread-safe with getInstance() method) and manages a Map of DisasterType instances. It initializes five default disasters (yagi, matmo, bualo, koto, FUNG-WONG) in the constructor and provides methods to add, retrieve, and find disaster types by keyword.

**Design Rationale and Benefits:** The abstract Post class enables code reuse and future extensibility—new post sources (Facebook, Twitter, Reddit) only require creating new subclasses without modifying existing code. The separation of immutable construction parameters from mutable analysis results (sentiment, reliefItem) follows the principles of defensive copying and thread safety. Using composition (Post contains Comments) rather than inheritance promotes flexible data modeling. The Sentiment enum wrapper provides type safety over raw strings. The alias-based matching in DisasterType handles flexible disaster keyword matching without tight coupling to specific naming conventions. The Singleton pattern in DisasterManager ensures a single, consistent source of disaster type information across the application.

**Reusability and Extensibility:** The design of the Model package ensures that these entity classes can be reused in other projects without carrying unnecessary dependencies. The Post class and its subclasses represent pure data containers that can be easily serialized to databases, transmitted over networks, or stored in files. The clean separation of concerns in this package allows for independent evolution of the model layer without affecting business logic layers.

6.3.2 UI Package - User Interface Implementation with MVC Architecture

The UI package contains all graphical user interface components and implements a strict Model-View-Controller (MVC) architectural pattern. This package is particularly important as it demonstrates how a complex user interface can be decomposed into manageable, specialized components while maintaining clean communication patterns between the view and model layers.



**MVC Architecture Implementation:** The MVC pattern in this package follows classical separation of concerns where the Model component manages application state and business data, the View component displays information to the user through graphical components, and the Controller component (implicitly embedded in event handlers) responds to user interactions. The Model class

maintains the application's state including lists of posts and comments, a reference to the sentiment analyzer, and a collection of registered listeners. When data changes, the Model notifies all observers without requiring knowledge of their specific implementations.

The View class serves as the main JFrame container that orchestrates all UI components and panels. Rather than cramming all interface logic into a single large class, the interface is decomposed into specialized panels, each handling a specific functional area:

- **CrawlControlPanel:** Provides web crawler control with platform selection (via CrawlerRegistry), keyword input area, post/comment limit spinners, progress tracking, and results display. It supports crawling individual URLs and bulk data collection with automatic disaster type assignment and relief item classification.
- **DataCollectionPanel:** Enables manual data entry for posts and comments, with fields for author name, content, disaster type selection, relief category selection, and sentiment analysis. Provides an alternative to automated crawling for direct data input.
- **CommentManagementPanel:** Displays all collected comments in a tabular view with columns for comment metadata, sentiment scores, and relief categories. Implements ModelListener to receive notifications when data changes, automatically refreshing the table. Integrates with DatabaseManager for data persistence and retrieval.
- **AdvancedAnalysisPanel:** Implements Problem 1 (Satisfaction Analysis) and Problem 2 (Temporal Sentiment Tracking) with tabbed interface for selecting disasters and relief categories, displaying analysis results, generating charts, and comparing satisfaction metrics across different scenarios.

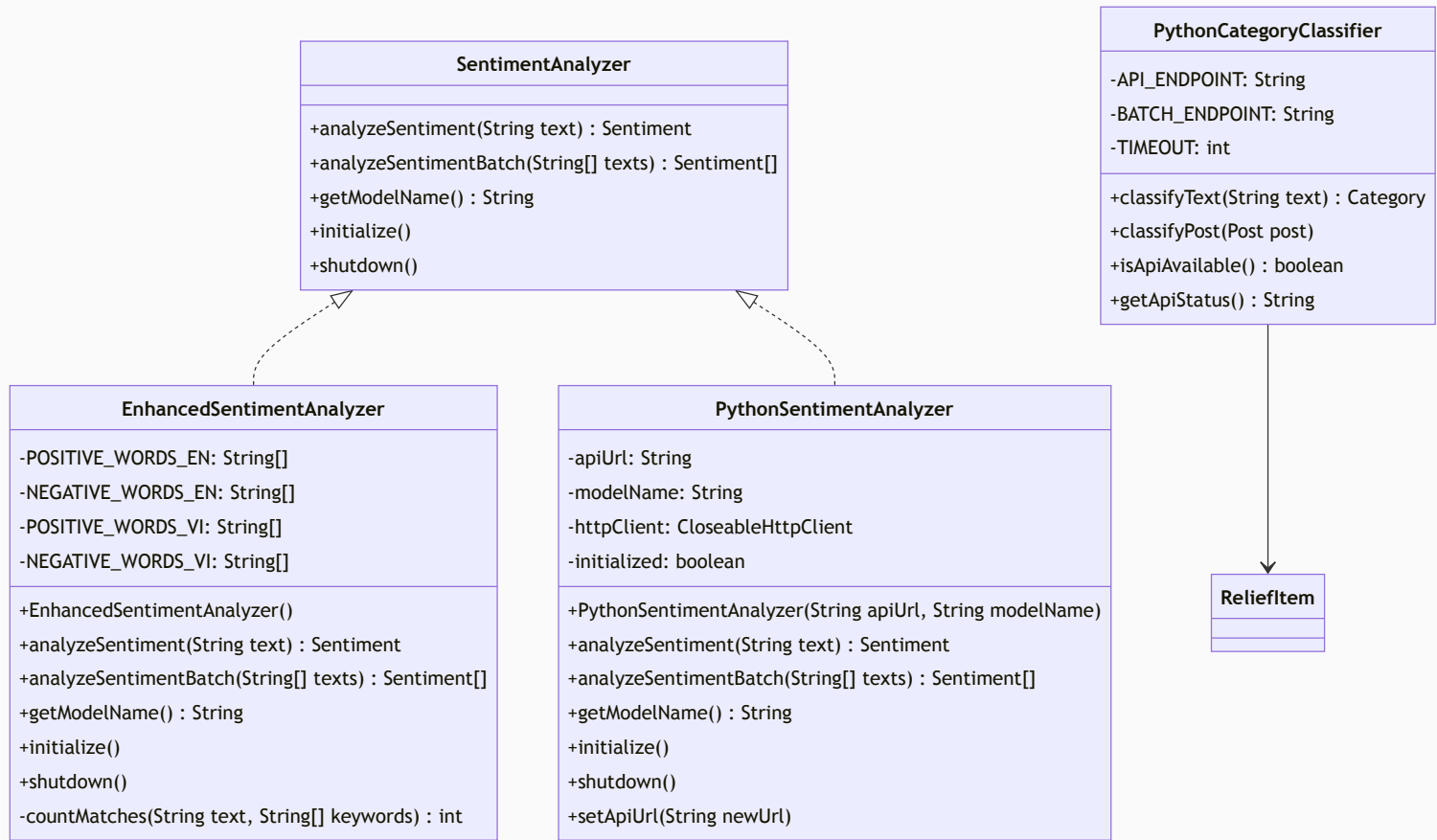
**Observer Pattern and Reactive Updates:** The system employs the Observer pattern through the ModelListener interface. When the Model's state changes (new posts added, comments updated, sentiment analyzed), it automatically notifies all registered listeners without them needing to poll for updates. This creates a reactive, event-driven system where UI components automatically reflect the current state of the data.

**Utility Classes:** The CrawlingUtility class provides static helper methods for adding simulated comments to posts with random sentiment assignment and relief item linking. The InteractiveChartUtility class provides interactive features for JFreeChart panels, including zoom, pan, and tooltip support for exploratory data analysis.

**Database Integration:** The CommentManagementPanel directly manages DatabaseManager for persistent storage and retrieval of comments. The DataPersistenceManager in Model handles saving and loading application state across sessions, ensuring data continuity between application runs.

### 6.3.3 Sentiment Package - Multiple Sentiment Analysis Strategies

The Sentiment package provides a flexible framework for analyzing sentiment in textual data. Rather than committing the entire system to a single sentiment analysis approach, this package implements the Strategy design pattern to support multiple analysis strategies with different accuracy-to-performance tradeoffs.



**Strategy Pattern Implementation:** The **SentimentAnalyzer** interface defines a contract that all sentiment analysis implementations must follow. This interface specifies five core methods: `analyzeSentiment(String text)` for single text analysis, `analyzeSentimentBatch(String[] texts)` for batch processing, `getModelName()` for model identification, `initialize()` for setup, and `shutdown()` for cleanup. This interface-based design enables runtime selection of different analysis strategies without requiring code modifications.

**EnhancedSentimentAnalyzer Implementation:** This class implements keyword-based sentiment analysis with bilingual support (English and Vietnamese). It maintains two sets of keyword arrays for positive and negative words in each language (27 positive and 30 negative English words, 32 positive and 33 negative Vietnamese words). The `analyzeSentiment(String text)` method counts keyword matches and calculates confidence scores (0.6-0.99 range) based on match frequency. This provides a good balance between accuracy and performance with zero computational overhead compared to ML models.

**PythonSentimentAnalyzer Implementation:** This analyzer interfaces with a machine learning model (nlptown/bert-base-multilingual-uncased-sentiment with xlm-roberta-base fallback) running in a Python backend service via HTTP POST requests. It accepts an API URL and model name as constructor parameters, allowing flexible configuration. The analyzer sends JSON requests to the Python service, parses responses containing sentiment type and confidence scores, and handles API errors gracefully by falling back to neutral sentiment. This approach leverages deep learning for significantly higher accuracy but incurs computational costs and requires the Python service running on port 5001.

**PythonCategoryClassifier:** A specialized classifier (not implementing **SentimentAnalyzer**) that categorizes text into relief categories (CASH, MEDICAL, SHELTER, FOOD, TRANSPORTATION) using hybrid keyword-based matching with semantic similarity analysis running in Python (port 5001). It provides methods for single text classification (`classifyText(String text) Category`), post classification (`classifyPost(Post post)`), API availability checking (`isApiAvailable() boolean`), and status reporting (`getApiStatus() String`). When classification fails or API is unavailable, it defaults to FOOD category.

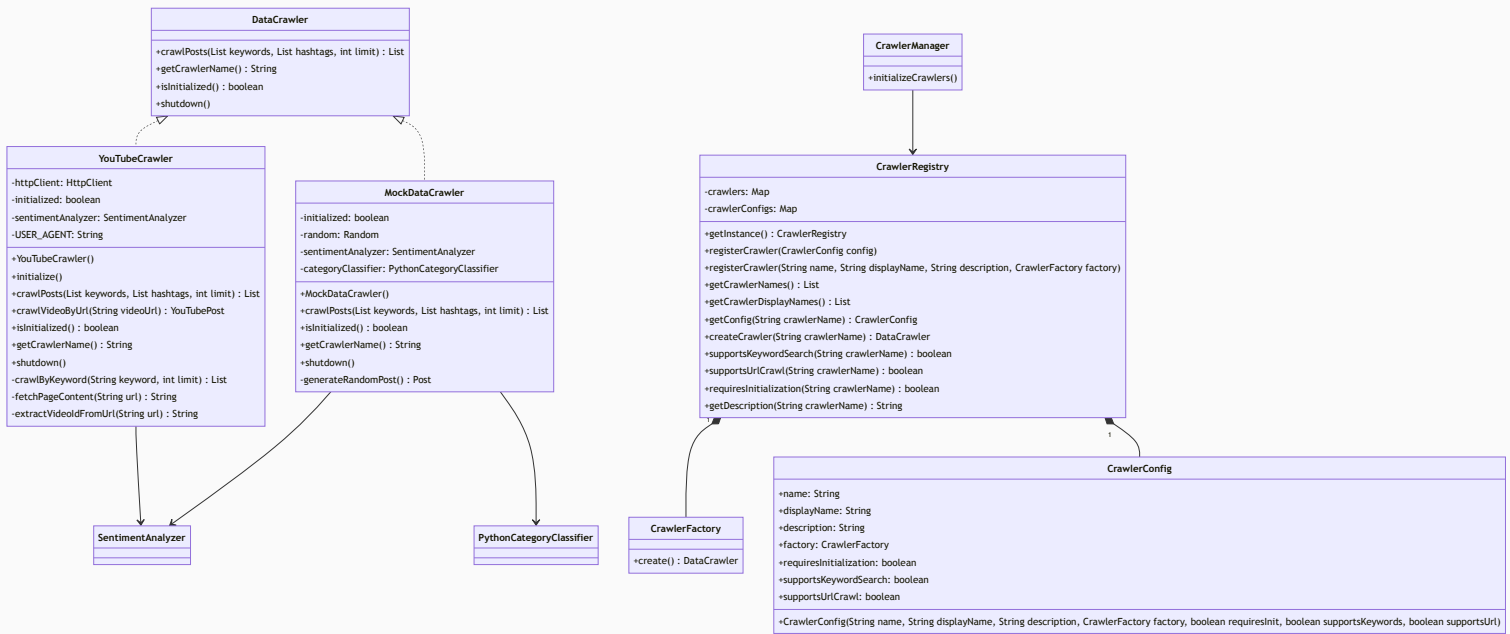


**Runtime Flexibility:** The beauty of the Strategy pattern is that the Model class can switch between SentimentAnalyzer implementations at runtime. The application can start with EnhancedSentimentAnalyzer for quick processing, then seamlessly switch to PythonSentimentAnalyzer for more accurate analysis via a single method call: ``model.setSentimentAnalyzer(new PythonSentimentAnalyzer("http://localhost:5001", "nlptown/bert-base-multilingual-uncased-sentiment"))``.

**Extensibility and Maintainability:** To add a new sentiment analysis approach (such as Google Cloud Natural Language API, OpenAI GPT-based analysis, or domain-specific models), developers only need to implement the SentimentAnalyzer interface without modifying existing implementations. This satisfies the Open/Closed Principle - the system is open for extension but closed for modification. Each analyzer is self-contained and can be tested independently using mock data or test cases.

### 6.3.4 Crawler Package - Web Data Collection with Factory and Registry Patterns

The Crawler package manages data collection from various sources using a combination of the Factory and Registry design patterns. This package exemplifies how to build pluggable, extensible architectures that support adding new data sources without modifying existing code.



**Registry and Factory Pattern Design:** The Crawler package uses a sophisticated combination of Registry and Factory patterns. The `CrawlerRegistry` is a singleton that maintains a centralized registry of available crawler implementations via a map of `CrawlerFactory` functional interfaces. Each crawler is registered with a `CrawlerConfig` object containing metadata: internal name, display name, description, factory (`CrawlerFactory` functional interface), and capability flags (`requiresInitialization`, `supportsKeywordSearch`, `supportsUrlCrawl`). Registration happens via ``registerCrawler(CrawlerConfig config)`` method, allowing the UI to dynamically discover crawler capabilities without loading the concrete classes. The `CrawlerFactory` is a functional interface that creates `DataCrawler` instances on demand.

**DataCrawler Interface:** All crawlers implement this interface with methods: `crawlPosts(List keywords, List hashtags, int limit)` for batch crawling, `crawlVideoByUrl(String url)` for single URL crawling (YouTube-specific), `getCrawlerName()` for identification, `isInitialized()` for state checking, and `shutdown()` for cleanup. This contract ensures all crawlers can be used interchangeably.

**YouTubeCrawler Implementation:** Uses HTTP client (not Selenium) to scrape YouTube search results and extract video IDs via regex patterns. It maintains a `SentimentAnalyzer` for on-the-fly sentiment classification of posts and includes methods: `crawlByKeyword()` for searching, `crawlVideoByUrl()` for single videos, `fetchPageContent()` for HTTP requests, and `extractVideoIdFromUrl()` for URL parsing. The

crawler automatically converts video data into YouTubePost objects with title, description, channel name, view count, and likes extracted from HTML.

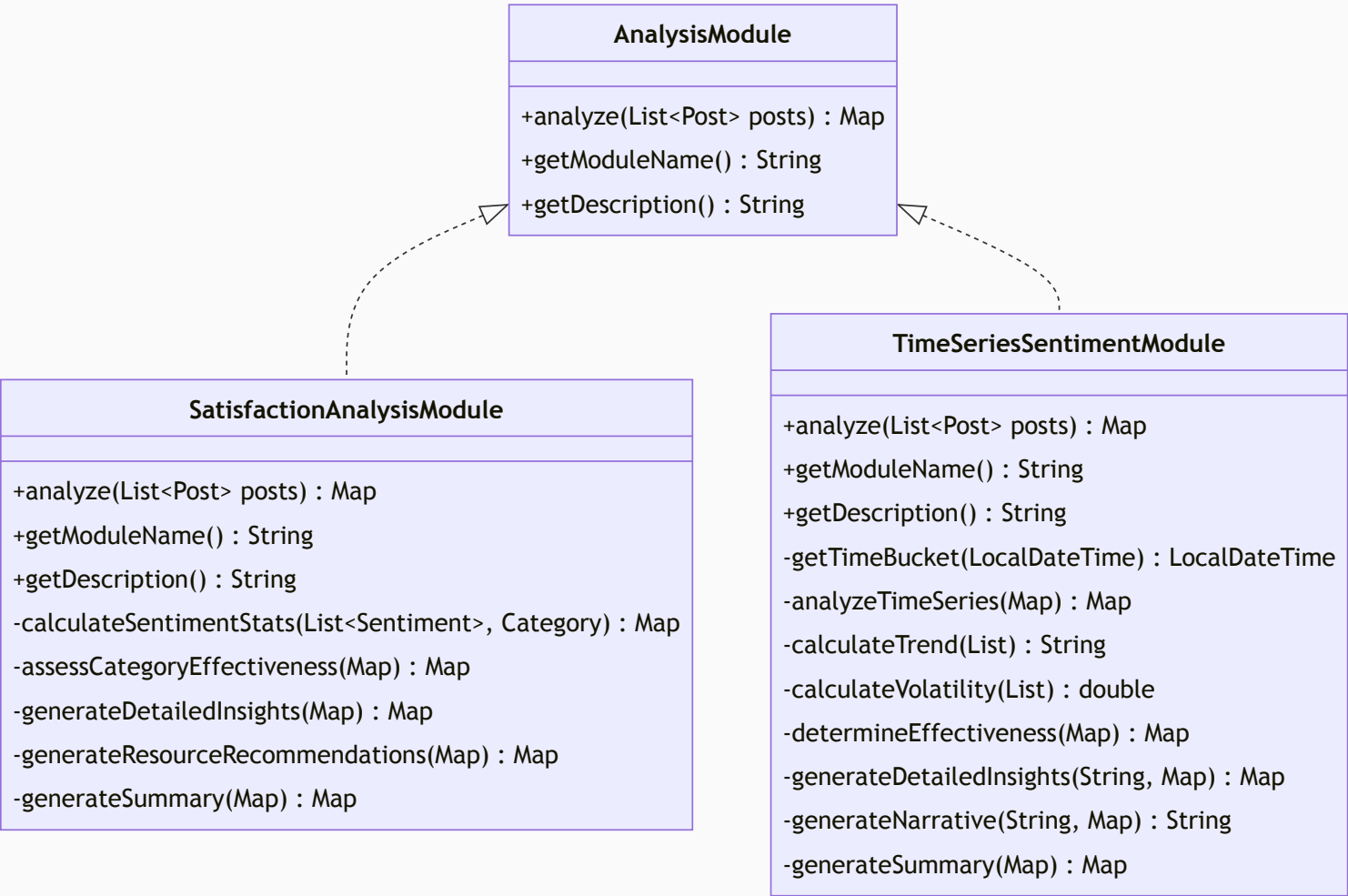
**MockDataCrawler Implementation:** Generates synthetic data for testing without making real API calls. It uses a Random generator and PythonCategoryClassifier to create realistic mock posts with random sentiments and categories. Supports keyword-based selection from pre-defined category content templates (CASH, MEDICAL, SHELTER, FOOD, TRANSPORTATION) to simulate diverse relief-related posts. Returns data instantly, enabling fast test execution and comprehensive testing without external dependencies.

**CrawlerManager Facade:** Provides static initializeCrawlers() method that registers both YouTubeCrawler and MockDataCrawler with the registry. This initialization is called once during application startup and populates the registry with crawler metadata and factories.

**Plugin Architecture Benefits:** New crawlers can be added at runtime without modifying existing code. To add Facebook crawling: (1) Create FacebookCrawler implementing DataCrawler, (2) Register it with a CrawlerConfig in CrawlerManager, (3) UI automatically discovers it via registry. The UI depends only on DataCrawler interface and CrawlerRegistry, achieving Dependency Inversion. This design enables parallel crawling from multiple sources simultaneously while maintaining loose coupling and testability.

6.3.5 Analysis Package - Modular Analysis Strategies

The Analysis package contains the implementations of Problem 1 and Problem 2 analysis modules, both following the Strategy design pattern for flexible analysis execution.



**Analysis Module Interface and Implementations:** Each analysis module implements the AnalysisModule interface, which defines three key methods: ``analyze(List posts)`` for processing post data and returning results as a Map, ``getModuleName()`` for identifying the module, and

`getDescription()` for describing its purpose. This design allows dynamic module selection and execution through polymorphism.

The **SatisfactionAnalysisModule** (Problem 1) analyzes public satisfaction and dissatisfaction for different relief item categories. It processes both posts and comments associated with relief items, grouping sentiments by relief category (CASH, MEDICAL, FOOD, SHELTER). For each category, it: calculates sentiment distribution (positive, negative, neutral counts and percentages); computes satisfaction scores based on sentiment ratios; assesses category effectiveness using a rating scale (Excellent, Good, Fair, Poor); generates detailed insights with multi-category comparisons; produces resource allocation recommendations; and provides a comprehensive summary. All results are returned as a Map with keys like "problem\_1\_satisfaction\_analysis", "category\_effectiveness", and "resource\_allocation\_recommendations".

The **TimeSeriesSentimentModule** (Problem 2) tracks sentiment changes over time for each relief item category to determine sector effectiveness. It extracts sentiments from both posts and comments, bins them into 6-hour time buckets based on creation timestamps, and tracks sentiment evolution within each bucket. For each category, it: creates a time-indexed Map of sentiments; analyzes time series data to compute sentiment distributions per time bucket; calculates trend direction (Improving, Stable, Declining) by analyzing sentiment ratios; measures volatility to identify sentiment fluctuations; determines sector effectiveness ratings; generates detailed insights with narrative descriptions; and produces a comprehensive summary. All results are returned as a Map with keys like "problem\_2\_time\_series\_sentiment", "sector\_effectiveness", "detailed\_insights", and "summary".

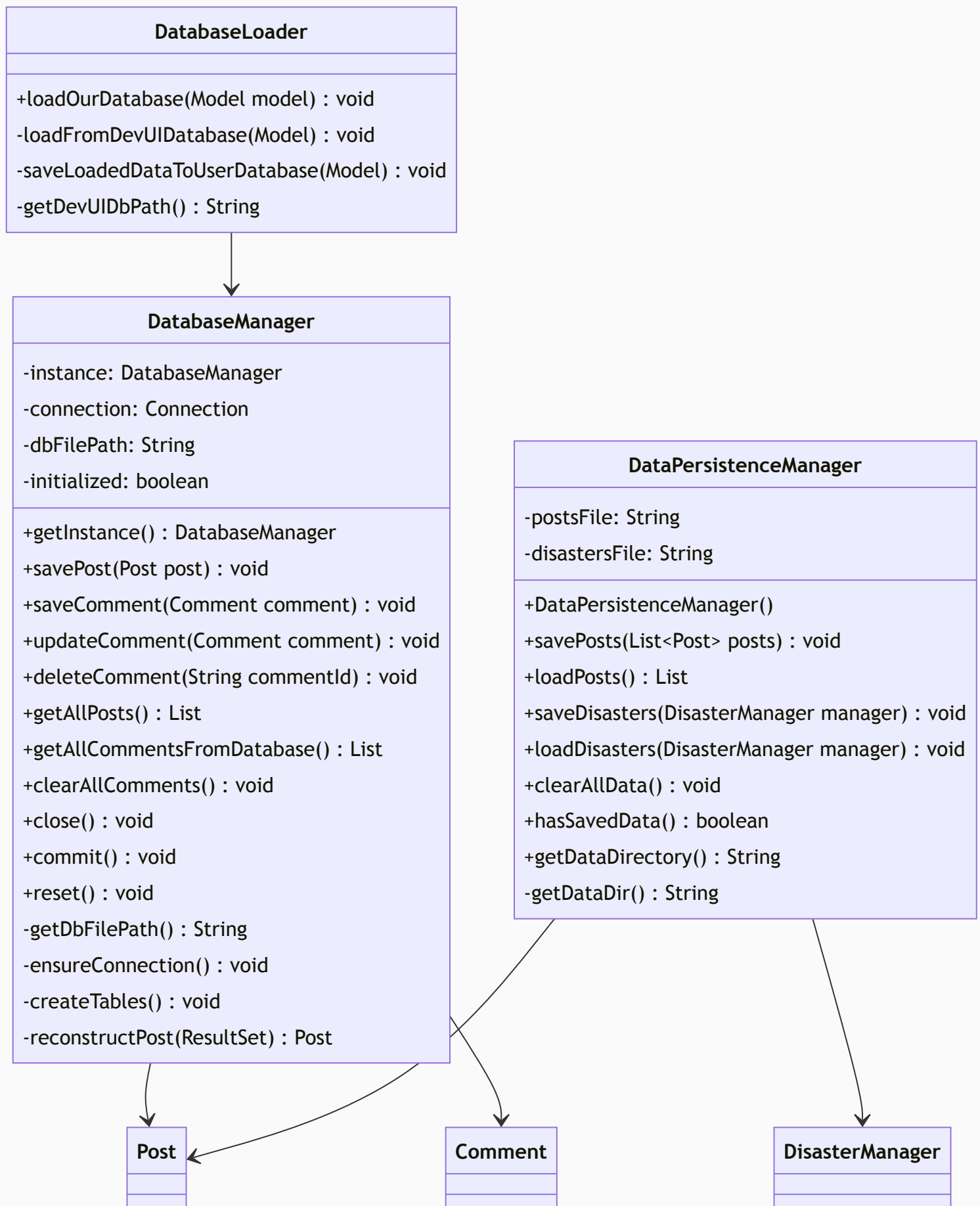
**Strategy Pattern Benefits:** The interface-based design allows the Model class to manage multiple analysis modules and invoke them dynamically:

```
AnalysisModule module = analysisModules.get("satisfaction");  
Map<String, Object> results = module.analyze(posts, comments);
```

New analysis modules (Geographic Analysis, Demographic Analysis, Language Analysis) can be added without modifying existing modules or the Model class. Each module focuses exclusively on its analysis responsibility, making the code easier to understand, test, and maintain.

### 6.3.6 Database Package - Data Persistence and Management

The Database package manages all aspects of data persistence, providing both low-level SQL operations and high-level application interfaces for data management.



**Dual Persistence Architecture:** The Database package implements two independent persistence strategies. DatabaseManager provides low-level SQLite database operations including Singleton pattern for connection management, CRUD operations on posts and comments, connection pooling, prepared statements for SQL injection protection, and PRAGMA configurations (busy\_timeout, foreign\_keys, WAL mode). DataPersistenceManager uses Java object serialization to save posts to posts.dat and custom disaster types to disasters.dat files, providing simpler file-based storage for application state. Both

managers operate independently - DatabaseManager handles SQL-based operations while DataPersistenceManager handles serialized object persistence.

**DatabaseManager Singleton Pattern:** Ensures single database connection throughout application lifetime using synchronized double-checked locking. Provides methods: savePost(), saveComment(), updateComment(), deleteComment(), getAllPosts(), getAllCommentsFromDatabase(), clearAllComments(), commit(), close(), and reset(). Uses database schema with proper foreign key constraints to ensure referential integrity between posts and comments. The reconstructPost() method rebuilds Post objects from database ResultSet with full data including associated comments.

**DatabaseLoader Facade:** Provides loadOurDatabase(Model model) as the main entry point, which loads data from a curated SQLite database (humanitarian\_logistics\_curated.db), converts it into Post and Comment objects, stores them in the Model, and then persists them to the user's personal database via saveLoadedDataToUserDatabase(). This allows the application to initialize with curated default data while maintaining user-specific persistence. The private loadFromDevUIDatabase() method handles the actual reading from the curated database file.

**DataPersistenceManager Serialization:** Uses ObjectOutputStream/ObjectOutputStream for serializing/deserializing Java objects. Provides methods: savePosts(List) to write posts to posts.dat, loadPosts() to read posts from disk, saveDisasters(DisasterManager) to save custom disaster types, and loadDisasters(DisasterManager) to restore them. Automatically filters default disaster types (yagi, matmo, bualo, koto, fung-wong) when saving, only persisting user-created custom disasters. Creates data directory automatically if it doesn't exist.

**Resource Management:** DatabaseManager uses try-with-resources for Statement and PreparedStatement objects to ensure proper cleanup. Connection synchronization prevents race conditions during concurrent access. WAL mode enables better concurrency for SQLite. DataPersistenceManager similarly uses try-with-resources for stream management to prevent resource leaks.

### 6.3.7 Util Package - Utility and Helper Functions

The Util package provides shared utility functions used across multiple packages in the application.

| StringSimilarity                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                               |
| +levenshteinDistance(String s1, String s2) : int<br>+levenshteinSimilarity(String s1, String s2) : double<br>+findMostSimilar(String input, List<String> candidates) : String |

**String Similarity Utilities:** The StringSimilarity class implements the Levenshtein distance algorithm for measuring string similarity. The algorithm calculates the minimum number of single-character edits (insertions, deletions, substitutions) required to transform one string into another. This utility enables fuzzy matching of disaster type keywords and relief item categories, allowing the system to handle user input with typos, abbreviations, or variations.

**Levenshtein Distance:** The core method `levenshteinDistance(String s1, String s2)` returns an integer representing the minimum edit distance between two strings. It uses dynamic programming with a 2D matrix where each cell [i,j] represents the distance between the first i characters of s1 and the first j characters of s2. Handles null inputs gracefully by treating them as empty strings.



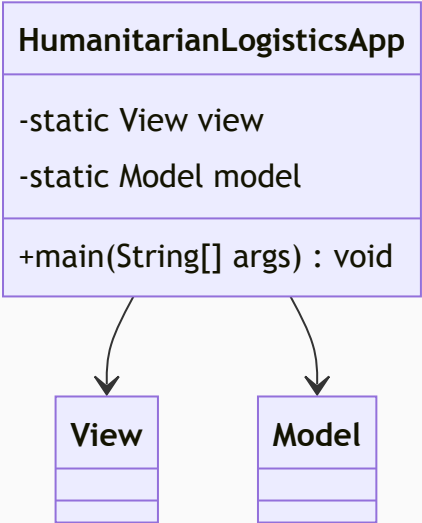
**Similarity Scoring:** The ``levenshteinSimilarity(String s1, String s2)`` method converts the raw distance into a normalized similarity score between 0.0 and 1.0. It calculates:  $\text{similarity} = 1.0 - (\text{distance} / \text{max\_length})$ , where `max_length` is the length of the longer string. This normalization enables consistent comparison across strings of different lengths.

**Fuzzy Matching:** The ``findMostSimilar(String input, List candidates)`` method finds the best matching candidate from a list. It compares the input string against all candidates using similarity scoring and returns the match with the highest score if it exceeds 0.5 (50% similarity threshold). Returns null if no sufficiently similar match is found. This method is used for:

- Matching user-provided disaster keywords to predefined disaster types
- Finding the closest relief category when exact matches are unavailable
- Handling spelling variations and abbreviations gracefully

### 6.3.8 Main Application and Utility Classes

The main application layer provides the entry point and various utility functions supporting the broader system.



**Application Initialization and Bootstrap:** The `HumanitarianLogisticsApp` class serves as the single entry point for the entire application, containing the `main()` method that is executed when the program starts. This class is responsible for orchestrating the initialization sequence of all major system components in the correct order:

1. Initialize the `DisasterManager` singleton with predefined disaster types
2. Create the sentiment analyzer instance (configured as needed)
3. Instantiate the `Model` class with all necessary dependencies
4. Create the `View` (JFrame) with the initialized `Model`
5. Set the `View` as visible, allowing user interaction to begin

By centralizing all initialization logic in one place, the application code becomes easier to understand and modify. Developers can quickly see which components are created, in what order, and what dependencies are injected.

## 6.4 Architectural Design Principles and Rationale

### 6.4.1 Model Package Independence - The Foundation Layer

The Model package is deliberately designed to have zero dependencies on any other package. This foundational independence is crucial for several reasons:

**Clean Architecture:** By keeping the Model package independent, we ensure that entity classes represent pure data containers without knowledge of how they will be displayed, persisted, or analyzed. This adherence to clean architecture principles means that the Model layer can be shared across multiple applications, used in different UI frameworks, or even exposed through REST APIs without carrying unnecessary dependencies.

**Testability:** Entity classes can be easily instantiated and tested in isolation without setting up complex infrastructure. Unit tests can create mock objects and validate business logic without database connections, UI frameworks, or external services.

**Reusability:** Organizations can extract the entire Model package and reuse it in other projects that address different problems but deal with similar data structures. The Post and Comment classes could be used in social media analysis projects, disaster response planning systems, or humanitarian crisis tracking applications.

**Separation of Concerns:** By keeping data models separate from business logic, presentation logic, and persistence logic, we maintain a clear architectural boundary that makes the codebase more maintainable and easier for new team members to understand.

#### 6.4.2 UI Package with MVC Architecture - The Presentation Layer

The UI package implements a strict MVC pattern where the Model (M) component manages state, the View (V) component displays state, and the Controller (C) component handles user interactions and invokes operations on the Model.

**Panel Decomposition Strategy:** Rather than creating a single View class containing all UI components, we deliberately decompose the interface into focused panels:

Monolithic Approach:

```
View class (2000+ lines)
- Crawling UI (500 lines)
- Comment management (300 lines)
- Problem 1 analysis (400 lines)
- Problem 2 analysis (500 lines)
- Disaster management (300 lines)
Result: Difficult to maintain, test, and modify
```

Modular Approach:

```
View class (170 lines) - Main container and layout
├─ CrawlControlPanel (710 lines) - Web data collection and platform management
├─ DataCollectionPanel (320 lines) - Manual data entry for posts and comments
├─ AdvancedAnalysisPanel (710 lines) - Problem 1 & 2 analysis with visualization
├─ CommentManagementPanel (770 lines) - Comment viewing and database integration
└─ Model (232 lines) - Application state and business logic
Result: Easy to maintain, test, and modify
```

**Observer Pattern Implementation:** The system employs the Observer design pattern through the ModelListener interface. Rather than panels directly querying the Model for updates, they register as listeners. When the Model's state changes, it automatically notifies all registered listeners. This creates loose coupling - panels need not know about each other's existence, only about the Model's interface.



```
// Model notifies listeners when data changes
private void notifyListeners() {
    for (ModelListener listener : listeners) {
        listener.modelChanged();
    }
}

// Listener interface
public interface ModelListener { void modelChanged(); }
```

This approach ensures that all UI components remain synchronized with the current application state without explicit coordination between panels.

**Utility Class Organization:** Reusable functionality is extracted into utility classes:

- **CrawlingUtility:** Data collection helper methods
- **InteractiveChartUtility:** Interactive visualization components

By extracting these utilities, we follow the DRY (Don't Repeat Yourself) principle and enable code reuse across multiple UI panels.

### 6.4.3 Sentiment Package with Strategy Pattern - Flexible Analysis

The Sentiment package demonstrates how the Strategy design pattern enables runtime algorithm selection without code modification:

**Strategy Hierarchy:**

```
SentimentAnalyzer (interface)
├─ EnhancedSentimentAnalyzer (keyword-based)
├─ PythonSentimentAnalyzer (advanced, accurate)
```

**Runtime Switching Capability:** The brilliance of this design is that the Model class never needs to know which specific analyzer is being used. It works with the SentimentAnalyzer interface:

```
private SentimentAnalyzer sentimentAnalyzer;

public void setSentimentAnalyzer(SentimentAnalyzer analyzer) {
    this.sentimentAnalyzer = analyzer;
}

public Sentiment analyzeSentiment(String text) {
    return sentimentAnalyzer.analyzeSentiment(text);
}
```

Because of this interface-based dependency, the Model code remains unchanged whether we use EnhancedSentimentAnalyzer, PythonSentimentAnalyzer, or a future GoogleCloudAnalyzer. This exemplifies the Dependency Inversion Principle - the Model depends on an abstraction (the interface) rather than concrete implementations.

**Extensibility Without Modification:** Adding a new sentiment analysis approach requires only implementing the SentimentAnalyzer interface. The entire system automatically supports the new approach without modification, satisfying the Open/Closed Principle. Organizations could add new analyzers for different languages, domains, or performance requirements without affecting existing code.

#### 6.4.4 Crawler Package with Factory and Registry Patterns - Extensible Data Collection

The Crawler package uses two complementary patterns to achieve maximum extensibility:

**Registry Pattern - Discovery and Instantiation:** The CrawlerRegistry maintains a registry of available crawler implementations and provides factory methods to create instances. This approach decouples the UI from concrete crawler implementations:

```
// UI doesn't need to know about YouTubeCrawler
DataCrawler crawler = registry.createCrawler("youtube");
List<Post> posts = crawler.crawlPosts(keywords, hashtags, limit);
```

**Plugin Architecture Benefits:** New data sources can be added as “plugins” without modifying existing code:

1. Create FacebookCrawler implementing DataCrawler interface
2. Register with CrawlerRegistry
3. UI automatically shows Facebook as available option

This is a powerful example of the Open/Closed Principle - the system is open for extension (new crawlers) but closed for modification (UI code doesn't change).

**Loose Coupling:** By depending on the DataCrawler interface rather than concrete implementations, the UI is completely decoupled from crawler details. YouTubeCrawler could be completely rewritten, new crawlers could be added, or crawlers could be removed without affecting UI code.

#### 6.4.5 Analysis Package with Strategy Pattern - Modular Analysis

The Analysis package uses the Strategy pattern similarly to Sentiment analysis, but applied to different types of data analysis:

**Analysis Module Registry:**

```
Map<String, AnalysisModule> modules = new LinkedHashMap<>();
modules.put("satisfaction", new SatisfactionAnalysisModule());
modules.put("time_series", new TimeSeriesSentimentModule());
```

```
// Add new analysis without modifying existing code (extensible design)
// modules.put("custom", new CustomAnalysisModule());
```

**Independent Module Development:** Each analysis module focuses exclusively on its analysis responsibility, making it easier to develop, test, and maintain. The SatisfactionAnalysisModule concerns itself only with category-based satisfaction metrics, while TimeSeriesSentimentModule focuses on temporal trends. New analysis modules can be easily added by implementing the AnalysisModule interface without modifying existing implementations.

### 6.4.6 Database Package - Abstraction and Flexibility

The Database package demonstrates how abstraction allows technology changes without impacting application code:

#### Layered Architecture:

```
Application Code
    ↓
DataPersistenceManager (Facade)
    ↓
DatabaseManager (Concrete SQLite Implementation)
    ↓
SQLite Database
```

This layering means that migrating from SQLite to PostgreSQL, MongoDB, or even cloud-based databases requires modifying only the DatabaseManager class. Application code remains unchanged, ensuring that application logic doesn't need to be tested again.

**Try-with-Resources Pattern:** The implementation uses try-with-resources statements to ensure automatic resource cleanup:

```
try (ObjectOutputStream oos = new ObjectOutputStream(...)) {
    // Write data
} // ObjectOutputStream automatically closed, even if exception occurs
```

This pattern prevents resource leaks and ensures data integrity even in error scenarios.

## 6.5 SOLID Principles Applied Throughout the Design

The system consistently applies the five SOLID principles to ensure maintainability and extensibility:

| Principle                        | Application                         | Example                                        |
|----------------------------------|-------------------------------------|------------------------------------------------|
| <b>S</b> - Single Responsibility | Each class has one reason to change | Post class only contains data; DatabaseManager |

| Principle                 | Application                                 | Example                                                                                                                               |
|---------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
|                           |                                             | only handles database operations                                                                                                      |
| O - Open/Closed           | Open for extension, closed for modification | Add new crawlers without modifying CrawlerRegistry; add new analyzers without modifying Model                                         |
| L - Liskov Substitution   | Subtypes can substitute their supertypes    | Any SentimentAnalyzer implementation can replace any other; any DataCrawler can replace any other                                     |
| I - Interface Segregation | Clients depend on specific interfaces       | UI depends on SentimentAnalyzer interface; doesn't depend on unrelated methods in PythonSentimentAnalyzer                             |
| D - Dependency Inversion  | Depend on abstractions, not implementations | Model depends on SentimentAnalyzer interface; doesn't depend on EnhancedSentimentAnalyzer or PythonSentimentAnalyzer concrete classes |

## 6.6 Architectural Summary and Benefits

The Humanitarian Logistics Analysis System demonstrates a professionally architected solution with clear separation of concerns across seven well-defined packages. Each package focuses on a specific responsibility while maintaining loose coupling with other packages. The architecture leverages industry-standard design patterns (Strategy, Factory, Registry, Observer, Singleton, MVC) to provide flexibility, extensibility, and maintainability.

### Key Architectural Achievements:

1. **Modularity:** Each package is independently understandable and modifiable without affecting others
2. **Extensibility:** New features can be added by extending existing interfaces rather than modifying existing code
3. **Testability:** Components can be tested in isolation using mock objects and test doubles
4. **Reusability:** Core components (Model package, Sentiment strategies) can be reused in other projects

5. **Maintainability:** Clear separation of concerns makes the codebase easy to understand and modify
6. **Scalability:** The architecture supports team development with multiple developers working on different packages simultaneously

This design reflects mature software engineering practices suitable for production systems serving critical humanitarian logistics operations.

## 7. OOP Techniques and Design Patterns

The Humanitarian Logistics Analysis System is built on solid object-oriented programming principles and industry-standard design patterns. This section provides in-depth analysis with actual code examples extracted directly from the codebase, demonstrating how each technique is applied to solve real problems in the application.

### 7.1 Fundamental Object-Oriented Programming Concepts

Seven core OOP concepts form the foundation of the system's architecture. Each is applied strategically throughout the codebase to achieve flexibility, maintainability, and extensibility.

#### 7.1.1 Encapsulation - Data Protection and Controlled Access

**Purpose:** Bundle data and methods together while hiding internal details and controlling access to state. This prevents invalid data states and unintended external modifications.

**Implementation in Post.java:**

```
public abstract class Post implements Serializable, Comparable<Post> {
    private final String postId;
    private final String content;
    private final LocalDateTime createdAt;
    private final String author;
    private final String source;

    protected Post(String postId, String content, LocalDateTime createdAt,
                    String author, String source) {
        this.postId = Objects.requireNonNull(postId, "Post ID cannot be null");
        this.content = Objects.requireNonNull(content, "Content cannot be null");
        this.createdAt = Objects.requireNonNull(createdAt, "Created date cannot be null");
        this.author = Objects.requireNonNull(author, "Author cannot be null");
        this.source = Objects.requireNonNull(source, "Source cannot be null");
        this.comments = new ArrayList<>();
    }

    public String getPostId() { return postId; }
    public List<Comment> getComments() {
        return Collections.unmodifiableList(comments);
    }

    public void addComment(Comment comment) {
        if (comment != null) this.comments.add(comment);
    }
}
```

**Key Techniques:** Immutable fields (final), null safety via Objects.requireNonNull(), defensive copying, and controlled modification. **Benefits:** Data integrity, early error detection, and traceable changes.

### 7.1.2 Abstraction - Hiding Implementation Complexity

**Purpose:** Define WHAT operations an object performs without exposing HOW. This allows different implementations to be swapped without affecting client code.

```
public interface SentimentAnalyzer {
    Sentiment analyzeSentiment(String text);
    Sentiment[] analyzeSentimentBatch(String[] texts);
    String getModelName();
    void initialize();
    void shutdown();
}

public class EnhancedSentimentAnalyzer implements SentimentAnalyzer {
    @Override
    public Sentiment analyzeSentiment(String text) {
        int positiveCount = countMatches(text.toLowerCase(), POSITIVE_WORDS_EN) +
            countMatches(text.toLowerCase(), POSITIVE_WORDS_VI);
        int negativeCount = countMatches(text.toLowerCase(), NEGATIVE_WORDS_EN) +
            countMatches(text.toLowerCase(), NEGATIVE_WORDS_VI);

        Sentiment.SentimentType type = positiveCount > negativeCount ?
            Sentiment.SentimentType.POSITIVE : negativeCount > positiveCount ?
            Sentiment.SentimentType.NEGATIVE : Sentiment.SentimentType.NEUTRAL;

        double confidence = Math.min(Math.max(Math.abs(positiveCount - negativeCount) / 10,
            0.0), 1.0);
        return new Sentiment(type, confidence, text);
    }
}
```

**Benefits:** Easy testing, runtime selection, future extensibility, zero client coupling.

### 7.1.3 Inheritance - Code Reuse and Hierarchies

**Purpose:** Create class hierarchies where specialized subclasses inherit common behavior from parent classes, avoiding code duplication.

```
public abstract class Post implements Serializable, Comparable<Post> {
    private final String postId;
    private final String content;
    private final LocalDateTime createdAt;
    private final String author;
    private final String source;
    private final List<Comment> comments;

    public void addComment(Comment comment) { ... }
    public List<Comment> getComments() { ... }
    public void setSentiment(Sentiment sentiment) { ... }
```

```

@Override
public int compareTo(Post other) {
    return this.createdAt.compareTo(other.createdAt);
}
}

public class YouTubePost extends Post {
    private String videoId;

    public YouTubePost(String postId, String content, LocalDateTime createdAt,
                        String author, String videoId) {
        super(postId, content, createdAt, author, "YouTube");
        this.videoId = videoId;
    }
}

```

**Benefits:** Code reuse, consistency, extensibility, and polymorphism.

#### 7.1.4 Polymorphism - Runtime Behavior Selection

```

public class Model {
    private SentimentAnalyzer sentimentAnalyzer;

    public void setSentimentAnalyzer(SentimentAnalyzer analyzer) {
        if (this.sentimentAnalyzer != null) {
            this.sentimentAnalyzer.shutdown();
        }
        this.sentimentAnalyzer = analyzer;
        this.sentimentAnalyzer.initialize();
        notifyListeners();
    }

    public void addPost(Post post) {
        if (post.getSentiment() == null && sentimentAnalyzer != null) {
            Sentiment sentiment = sentimentAnalyzer.analyzeSentiment(post.getContent());
            post.setSentiment(sentiment);
        }
        // Process comments similarly
        for (Comment comment : post.getComments()) {
            if (comment.getSentiment() == null && sentimentAnalyzer != null) {
                Sentiment sentiment = sentimentAnalyzer.analyzeSentiment(comment.getContent());
                comment.setSentiment(sentiment);
            }
        }
        this.posts.add(post);
    }
}

```

#### 7.1.5 Interfaces - Contracts and Multiple Implementations



```
public interface SentimentAnalyzer { Sentiment analyzeSentiment(String text); }
public interface DataCrawler { List<Post> crawlPosts(List<String> keywords, List<String> hc
public interface AnalysisModule { Map<String, Object> analyze(List<Post> posts); }
public interface ModelListener { void modelChanged(); }
```

**Benefits:** Contracts, loose coupling, easy testing, runtime flexibility.

### 7.1.6 Abstract Classes - Partial Implementation

Unlike interfaces, abstract classes provide actual implementation and shared state:

```
public abstract class Post implements Serializable, Comparable<Post> {
    private final String postId;
    private final List<Comment> comments = new ArrayList<>();

    public List<Comment> getComments() {
        return Collections.unmodifiableList(comments);
    }

    public void addComment(Comment comment) {
        if (comment != null) this.comments.add(comment);
    }
}
```

### 7.1.7 Composition - Has-A Relationships

```
public class Model {
    private SentimentAnalyzer sentimentAnalyzer = new EnhancedSentimentAnalyzer();
    private DatabaseManager dbManager = new DatabaseManager();
    private List<Post> posts = new ArrayList<>();
    private List<ModelListener> listeners = new ArrayList<>();

    public void setSentimentAnalyzer(SentimentAnalyzer analyzer) {
        this.sentimentAnalyzer = analyzer;
    }
}
```

**Composition vs Inheritance:** Use inheritance (IS-A) for hierarchies; use composition (HAS-A) for flexibility and runtime swapping.

## 7.2 Advanced Design Patterns

### 7.2.1 Model-View-Controller (MVC) Architecture

**Purpose:** Separate application into three interconnected components: Model (data and logic), View (presentation), and Controller (user interaction). This enables independent development, testing, and maintenance.

## Implementation in System:

```
// MODEL: Contains data and business logic
public class Model {
    private List<Post> posts = new ArrayList<>();
    private SentimentAnalyzer sentimentAnalyzer;
    private List<ModelListener> listeners = new ArrayList<>();

    public void addPost(Post post) {
        // Business logic: analyze sentiment before adding
        Sentiment sentiment = sentimentAnalyzer.analyzeSentiment(post.getContent());
        post.setSentiment(sentiment);
        posts.add(post);

        // Notify all listeners of change
        notifyListeners();
    }

    private void notifyListeners() {
        for (ModelListener listener : listeners) {
            listener.modelChanged(); // Observer pattern callback
        }
    }
}

// VIEW: Displays data from Model
public class View extends JFrame implements ModelListener {
    private AdvancedAnalysisPanel analysisPanel;
    private CommentManagementPanel commentPanel;

    public View(Model model) {
        this.model = model;
        // Register as observer to be notified of changes
        model.addModelListener(this);
    }

    @Override
    public void modelChanged() {
        // React to data changes: update UI on EDT
        SwingUtilities.invokeLater(() -> {
            analysisPanel.refresh();
            commentPanel.refreshCommentTable();
        });
    }
}
```

**Benefits:** Separation of concerns (Model  $\neq$  View), testability (test Model without UI), multiple views (same Model can serve multiple Views), loose coupling.

### 7.2.2 Strategy Pattern - Runtime Algorithm Selection

**Purpose:** Define a family of algorithms, encapsulate each one, and make them interchangeable. Strategy allows changing algorithms at runtime without changing client code.

## Application in System:

```
// STRATEGY INTERFACE: Abstract algorithm definition
public interface SentimentAnalyzer {
    Sentiment analyzeSentiment(String text);
    String getModelName();
    void initialize();
    void shutdown();
}

// CONCRETE STRATEGY 1: Rule-based approach
public class EnhancedSentimentAnalyzer implements SentimentAnalyzer {
    private static final String[] POSITIVE_WORDS_EN = {...};
    private static final String[] NEGATIVE_WORDS_EN = {...};

    @Override
    public Sentiment analyzeSentiment(String text) {
        // Keyword matching strategy
        int positiveCount = countMatches(text, POSITIVE_WORDS_EN);
        int negativeCount = countMatches(text, NEGATIVE_WORDS_EN);
        return new Sentiment(positiveCount > negativeCount ? POSITIVE : NEGATIVE);
    }
}

// CONCRETE STRATEGY 2: ML-based approach (Python backend)
public class PythonSentimentAnalyzer implements SentimentAnalyzer {
    @Override
    public Sentiment analyzeSentiment(String text) {
        // Call Python ML service via HTTP
        return callPythonService(text);
    }
}

// CLIENT: Uses strategy via interface, not concrete class
public class Model {
    private SentimentAnalyzer strategy; // Strategy reference

    public void setSentimentAnalyzer(SentimentAnalyzer analyzer) {
        // Switch algorithm at runtime
        if (this.strategy != null) {
            this.strategy.shutdown();
        }
        this.strategy = analyzer;
        this.strategy.initialize();
    }

    public void addPost(Post post) {
        // Use current strategy - client doesn't know which one
        Sentiment sentiment = strategy.analyzeSentiment(post.getContent());
        post.setSentiment(sentiment);
        posts.add(post);
        notifyListeners();
    }
}
```

```
// ADD NEW STRATEGY without changing existing code:
public class CustomSentimentAnalyzer implements SentimentAnalyzer {
    public Sentiment analyzeSentiment(String text) {
        // Custom algorithm implementation
        return new Sentiment(...);
    }
}
```

**Applications in System:** Sentiment Analysis (multiple analyzers), Data Crawling (YouTubeCrawler vs MockDataCrawler), Analysis Modules (TimeSeriesSentimentModule vs SatisfactionAnalysisModule).

**Benefits:** Runtime algorithm switching without modifying client code, easy to add new strategies, isolates algorithm details from client logic.

### 7.2.3 Factory and Registry Pattern - Pluggable Architecture

**Purpose:** Factory Pattern creates objects without specifying exact classes. Registry Pattern maintains a registry of available implementations, enabling runtime discovery and plugin-like architecture.

**Implementation in System:**

```
// REGISTRY: Central location for available crawlers
public class CrawlerRegistry {
    // Singleton instance
    private static final CrawlerRegistry INSTANCE = new CrawlerRegistry();

    // Map of crawler factories: name → create function
    private final Map<String, CrawlerFactory> crawlers = new LinkedHashMap<>();

    // FACTORY INTERFACE: Defines how to create crawlers
    @FunctionalInterface
    public interface CrawlerFactory {
        DataCrawler create(); // Creates a new crawler instance
    }

    public static CrawlerRegistry getInstance() {
        return INSTANCE;
    }

    // REGISTRATION: Add new crawler factory
    public void registerCrawler(String name, CrawlerFactory factory) {
        crawlers.put(name, factory);
    }

    // FACTORY METHOD: Create crawler by name
    public DataCrawler createCrawler(String crawlerName) {
        CrawlerFactory factory = crawlers.get(crawlerName);
        return factory != null ? factory.create() : null;
    }
}

// CLIENT: Uses registry to get crawlers without knowing their types
public class CrawlerManager {
```

```

public static void initializeCrawlers() {
    CrawlerRegistry registry = CrawlerRegistry.getInstance();

    // Register YouTube crawler factory
    registry.registerCrawler("YOUTUBE", YouTubeCrawler::new);

    // Register Mock crawler factory
    registry.registerCrawler("MOCK", MockDataCrawler::new);
}

// USAGE: Get crawler by name without creating directly
DataCrawler crawler = CrawlerRegistry.getInstance().createCrawler("YOUTUBE");
List<Post> results = crawler.crawlPosts(keywords, hashtags, limit);

```

**Advantages:** Plugin architecture (add crawlers without modifying existing code), runtime discovery (list available crawlers), decoupling (client doesn't know concrete classes), dynamic instantiation.

**Benefits:** Plugin architecture, runtime discovery, decoupling, extensibility.

## 7.2.4 Observer Pattern - Reactive Updates

**Purpose:** Define a one-to-many relationship where when one object's state changes, all dependents are automatically notified and updated. This implements loose coupling between Subject and Observers.

**Implementation in System:**

```

// OBSERVER INTERFACE: What observers must implement
public interface ModelListener {
    void modelChanged(); // Called when Model state changes
}

// SUBJECT: Model that notifies observers
public class Model {
    private List<Post> posts = new ArrayList<>();
    // List of observers watching this model
    private List<ModelListener> listeners = new ArrayList<>();

    // ATTACH observer
    public void addModelListener(ModelListener listener) {
        listeners.add(listener);
    }

    // METHOD that changes state
    public void addPost(Post post) {
        Sentiment sentiment = sentimentAnalyzer.analyzeSentiment(post.getContent());
        post.setSentiment(sentiment);
        posts.add(post);

        // NOTIFY: When state changes, notify all observers
        notifyListeners();
    }

    // NOTIFY all observers

```

```

        private void notifyListeners() {
            for (ModelListener listener : listeners) {
                listener.modelChanged(); // Each observer reacts
            }
        }
    }

    // CONCRETE OBSERVER 1: View updates UI
    public class View extends JFrame implements ModelListener {
        public View(Model model) {
            this.model = model;
            // REGISTER as observer
            model.addModelListener(this);
        }

        @Override
        public void modelChanged() {
            // REACT to notification: refresh UI
            SwingUtilities.invokeLater(() -> {
                analysisPanel.refresh();
                commentPanel.refreshCommentTable();
                statusLabel.setText("Updated");
            });
        }
    }

    // CONCRETE OBSERVER 2: Logger tracks changes
    public class ModelChangeLogger implements ModelListener {
        @Override
        public void modelChanged() {
            // REACT to notification: log the event
            LOGGER.info("Model changed at " + LocalDateTime.now());
        }
    }
}

```

**Benefits:** Loose coupling (Subject doesn't know observer details), dynamic subscriptions (add/remove observers at runtime), automatic synchronization (all observers notified together), event-driven design (reactive updates without polling).

### 7.2.5 Singleton Pattern - Guaranteed Single Instance

**Purpose:** Ensure a class has only ONE instance and provide global access to it. Prevents multiple instances from causing state conflicts and resource wastage.

**Implementation in System:**

```

// SINGLETON: Private constructor prevents instantiation
public class CrawlerRegistry {
    // Single static instance created at class load time
    private static final CrawlerRegistry INSTANCE = new CrawlerRegistry();

    // Private constructor: prevents new CrawlerRegistry()
    private CrawlerRegistry() {}
}

```

```

// Global access point: always returns same instance
public static CrawlerRegistry getInstance() {
    return INSTANCE;
}

private final Map<String, CrawlerFactory> crawlers = new LinkedHashMap<>();

public void registerCrawler(String name, CrawlerFactory factory) {
    crawlers.put(name, factory);
}

}

// SINGLETON with initialization
public class DisasterManager {
    private static final DisasterManager INSTANCE = new DisasterManager();

    private DisasterManager() {
        disasterTypes = new HashMap<>();
        // Initialize disaster types
        disasterTypes.put("EARTHQUAKE", new DisasterType("Earthquake", "Sudden ground moven
        disasterTypes.put("FLOOD", new DisasterType("Flood", "Water overflow"));
    }

    public static DisasterManager getInstance() {
        return INSTANCE;
    }

    public DisasterType getDisasterType(String key) {
        return disasterTypes.get(key);
    }
}

// USAGE: Always same instance
CrawlerRegistry registry1 = CrawlerRegistry.getInstance();
CrawlerRegistry registry2 = CrawlerRegistry.getInstance();
assert registry1 == registry2; // True: same object

```

**Benefits:** Global access point, prevents multiple instances, thread-safe (with static initialization), controlled state, resource efficiency.

## 7.2.6 Facade Pattern - Simplified Complex Operations

**Purpose:** Provide a unified, simplified interface to a complex subsystem. This hides implementation details, reduces coupling, and makes API easier to use.

**Implementation in System:**

```

// SUBSYSTEM 1: Database operations (complex)
public class DatabaseManager { /* 200+ lines of JDBC code */ }

// SUBSYSTEM 2: File I/O (complex)
public class DataPersistenceManager { /* Serialization logic */ }

```



```

// FACADE: Simplified interface hiding complexity
public class DatabaseLoader {
    // Simple public method hides all complexity
    public static void loadOurDatabase(Model model) {
        if (model == null) {
            throw new IllegalArgumentException("Model cannot be null");
        }

        // Internal details hidden from client:
        model.getPosts().clear();

        // Client doesn't know about these complex operations:
        loadFromDevUIDatabase(model); // Hidden: reads from curated DB
        saveLoadedDataToUserDatabase(model); // Hidden: persists to user DB
    }

    // PRIVATE: Complex database reading logic hidden
    private static void loadFromDevUIDatabase(Model model) {
        DatabaseManager dbManager = new DatabaseManager();
        try (ResultSet rs = dbManager.queryPosts()) {
            while (rs.next()) {
                // Parse result, create Post object, add to model
                Post post = parsePost(rs);
                model.addPost(post);
            }
        }
    }

    // PRIVATE: Complex database writing logic hidden
    private static void saveLoadedDataToUserDatabase(Model model) {
        DatabaseManager userDb = new DatabaseManager("user_db.db");
        for (Post post : model.getPosts()) {
            userDb.savePost(post); // Uses PreparedStatement, transactions, etc.
        }
    }
}

// ANOTHER FACADE: Crawler initialization
public class CrawlerManager {
    // Single method hides all initialization complexity
    public static void initializeCrawlers() {
        CrawlerRegistry registry = CrawlerRegistry.getInstance();

        // Register YouTube crawler with full configuration
        registry.registerCrawler(
            new CrawlerRegistry.CrawlerConfig(
                "YOUTUBE", "YouTube",
                "Crawl videos and comments from YouTube",
                YouTubeCrawler::new, true, true, true
            )
        );

        // Register Mock crawler
        registry.registerCrawler(
            new CrawlerRegistry.CrawlerConfig(

```

```

        "MOCK", "Sample/Mock Data",
        "Generate sample data for testing",
        MockDataCrawler::new, false, true, false
    )
};
}
}

```

```

// CLIENT: Simple usage without knowing complexity
DatabaseLoader.loadOurDatabase(model); // Simple!
CrawlerManager.initializeCrawlers();   // Simple!

```

**Benefits:** Reduced complexity for clients, cleaner API, hides subsystem details, easier to maintain and evolve subsystems independently.

### 7.2.7 Adapter Pattern - Interface Compatibility

**Purpose:** Convert incompatible interfaces to compatible ones, allowing objects with different interfaces to work together. Acts as a bridge between two incompatible interfaces.

**Implementation in System:**

```

// SWING ADAPTER: Built-in adapter for window events
public class View extends JFrame {
    public View(Model model) {
        this.model = model;

        // WindowAdapter: implements WindowListener with empty methods
        // We only override methods we need
        addWindowListener(new java.awt.event.WindowAdapter() {
            @Override
            public void windowClosing(WindowEvent e) {
                // Adapt: Window closing event → Model shutdown
                model.shutdown();
                PythonSentimentAnalyzer.stopPythonService();
                System.exit(0);
            }

            // Other methods inherited as empty: windowOpened, windowClosed, etc.
        });
    }
}

// CUSTOM ADAPTER: Adapt external HTML format to application interface
public class YouTubeCommentAdapter {
    private String rawJSON; // From YouTube API

    public YouTubeCommentAdapter(String rawJSON) {
        this.rawJSON = rawJSON;
    }

    // ADAPTER METHOD: Convert YouTube JSON → Application Comment
    public Comment adaptToComment() {

```

```

// Parse YouTube JSON
JSONObject json = new JSONObject(rawJSON);

// Extract fields from YouTube format
String author = json.getJSONObject("authorDetails")
    .getString("displayName");
String content = json.getString("textDisplay");
LocalDateTime createdAt = LocalDateTime.parse(
    json.getString("publishedAt")
);

// Create application domain object
return new Comment(
    UUID.randomUUID().toString(),
    author,
    content,
    createdAt,
    Sentiment.NEUTRAL
);
}
}

// USAGE: Adapt external format to internal format
YouTubeCommentAdapter adapter = new YouTubeCommentAdapter(youtubeJson);
Comment appComment = adapter.adaptToComment(); // Convert!
model.addCommentToPost(post, appComment);      // Now it's compatible

```

**Benefits:** Integrates incompatible classes, reuses existing code without modification, bridges different interfaces, supports legacy code integration.

## 7.2.8 Builder Pattern - Fluent Object Construction

**Purpose:** Construct complex objects step by step using a fluent API, avoiding telescoping constructors and making configuration explicit and readable.

**Implementation in System:**

```

// BUILDER CLASS: Constructs complex ChartPanel with configuration
public class InteractiveChartUtility {
    // BUILDER FACTORY: Create builder instance
    public static ChartPanelBuilder builder(ChartPanel chartPanel) {
        return new ChartPanelBuilder(chartPanel);
    }

    public static class ChartPanelBuilder {
        private ChartPanel chartPanel;

        // CONFIGURATION FIELDS: Build options
        private boolean domainZoom = true;
        private boolean rangeZoom = true;
        private boolean mouseWheel = true;
        private boolean tooltips = true;
    }
}

```

```

    public ChartPanelBuilder(ChartPanel chartPanel) {
        this.chartPanel = chartPanel;
    }

    // BUILDER METHODS: Fluent API (return this for chaining)
    public ChartPanelBuilder domainZoomable(boolean enabled) {
        this.domainZoom = enabled;
        return this; // Enable fluent chaining
    }

    public ChartPanelBuilder rangeZoomable(boolean enabled) {
        this.rangeZoom = enabled;
        return this; // Return builder for next method
    }

    public ChartPanelBuilder mouseWheelEnabled(boolean enabled) {
        this.mouseWheel = enabled;
        return this;
    }

    public ChartPanelBuilder tooltipsEnabled(boolean enabled) {
        this.tooltips = enabled;
        return this;
    }

    // BUILD METHOD: Apply all configurations and return result
    public ChartPanel build() {
        // Apply all accumulated configurations
        chartPanel.setDomainZoomable(domainZoom);
        chartPanel.setRangeZoomable(rangeZoom);
        chartPanel.setMouseWheelEnabled(mouseWheel);
        chartPanel.setDisplayToolTips(tooltips);
        return chartPanel; // Return configured object
    }
}

// USAGE: Readable, step-by-step configuration
ChartPanel interactive = InteractiveChartUtility.builder(originalChart)
    .domainZoomable(true) // Configure domain zoom
    .rangeZoomable(true) // Configure range zoom
    .mouseWheelEnabled(true) // Configure mouse wheel
    .tooltipsEnabled(true) // Configure tooltips
    .build(); // Create and return result

// WITHOUT BUILDER (telescoping constructor):
// ChartPanel p = new ChartPanel(chart, true, true, true, true); // Unclear!

// WITH BUILDER: Self-documenting and flexible
ChartPanel p = InteractiveChartUtility.builder(chart)
    .mouseWheelEnabled(true) // Only enable what we need
    .tooltipsEnabled(false)
    .build();

```

**Benefits:** Improved readability, optional parameters, immutability, flexible configuration, self-documenting code.

## 7.3 Advanced Java Techniques

### 7.3.1 Generics and Type-Safe Collections

**Purpose:** Use type parameters to write generic classes and methods, providing compile-time type safety and eliminating casting. This catches type errors early and makes code self-documenting.

**Application in System:**

```
// TYPE-SAFE COLLECTIONS: Compiler checks types at compile time
List<Comment> comments = new ArrayList<>();
comments.add(new Comment(...)); // Type check: OK, Comment
// comments.add("invalid"); // Type check: ERROR, String ≠ Comment
Comment c = comments.get(0); // No casting needed: Comment, not Object

Map<String, Integer> sentimentCounts = new HashMap<>();
sentimentCounts.put("POSITIVE", 42); // Type check: String key, Integer value
// sentimentCounts.put(123, "high"); // Type check: ERROR

// GENERIC CLASSES: Use type parameters for flexibility
public class TimeSeriesSentimentModule implements AnalysisModule {
    // Map<Category, Map<Time, List<Sentiment>>>: Type-safe nested structure
    private Map<ReliefItem.Category, Map<LocalDateTime, List<Sentiment>>> timeSeriesData;

    public void addSentiment(ReliefItem.Category category,
                             LocalDateTime time,
                             Sentiment sentiment) {
        timeSeriesData.computeIfAbsent(category, k -> new TreeMap<>())
            .computeIfAbsent(time, k -> new ArrayList<>())
            .add(sentiment);
    }
}

// GENERIC METHODS: Type-safe method signatures
public <T extends Sentiment> List<T> filterBySentiment(List<T> items) {
    return items.stream().filter(item -> item.isPositive()).collect(...);
}
```

**Benefits:** Compile-time type safety, no casting, self-documenting, prevents runtime `ClassCastException` errors.

### 7.3.2 Streams API and Functional Programming

**Purpose:** Process collections using functional-style transformations (map, filter, reduce) instead of imperative loops. Streams enable concise, declarative data processing with optional parallelization.

**Application in System:**

```

// REAL SYSTEM EXAMPLE: SatisfactionAnalysisModule.calculateSentimentStats()
private Map<String, Object> calculateSentimentStats(List<Sentiment> sentiments,
                                                    ReliefItem.Category category) {

    Map<String, Object> stats = new LinkedHashMap<>();

    // FILTER + COUNT: Count positive, negative, neutral sentiments
    long positiveCount = sentiments.stream()
        .filter(Sentiment::isPositive)    // FILTER: keep only positive
        .count();                        // COUNT: how many

    long negativeCount = sentiments.stream()
        .filter(Sentiment::isNegative)
        .count();

    long neutralCount = sentiments.stream()
        .filter(Sentiment::isNeutral)
        .count();

    // MAP + SUM: Calculate average confidence
    double totalConfidence = sentiments.stream()
        .mapToDouble(Sentiment::getConfidence) // MAP: extract confidence scores
        .sum();                               // SUM: total confidence
    double avgConfidence = totalConfidence / sentiments.size();

    // CALCULATE PERCENTAGES
    double positivePercentage = (double) positiveCount / sentiments.size() * 100;
    double negativePercentage = (double) negativeCount / sentiments.size() * 100;

    // BUILD RESULT
    stats.put("category", category.getDisplayName());
    stats.put("positive_count", positiveCount);
    stats.put("negative_count", negativeCount);
    stats.put("neutral_count", neutralCount);
    stats.put("positive_percentage", String.format("%.2f%%", positivePercentage));
    stats.put("negative_percentage", String.format("%.2f%%", negativePercentage));
    stats.put("average_confidence", String.format("%.2f", avgConfidence));

    return stats; // Returns map with aggregated sentiment statistics
}

```

**Benefits:** Concise, readable data processing, functional style (declarative not imperative), enables parallelization (parallelStream()), composable operations.

### 7.3.3 Lambda Expressions and Method References

**Purpose:** Write concise anonymous functions (lambdas) and reference existing methods (method references) to enable functional programming and reduce boilerplate code.

**Application in System:**

```

// EXAMPLE 1: LAMBDA in Streams - Filter with inline predicate
public List<String> getCrawlerDisplayNames() {

```

```

// LAMBDA: c -> c.displayName
// METHOD REFERENCE: .toList() (equivalent to .collect(Collectors.toList()))
return crawlerConfigs.values().stream()
    .map(c -> c.displayName) // LAMBDA: extract displayName from each config
    .toList(); // COLLECT results into list
}

// EXAMPLE 2: METHOD REFERENCE for Sorting
comments.sort(Comparator.comparing(Comment::getCreatedAt));
// Method reference: Comment::getCreatedAt (instead of c -> c.getCreatedAt())

// EXAMPLE 3: LAMBDA for Factory Pattern
CrawlerRegistry registry = CrawlerRegistry.getInstance();
registry.registerCrawler(
    new CrawlerRegistry.CrawlerConfig(
        "YOUTUBE",
        "YouTube",
        "Crawl videos and comments from YouTube",
        YouTubeCrawler::new, // CONSTRUCTOR REFERENCE: equivalent to () -> new YouTubeCrawler()
        true, true, true
    )
);

// EXAMPLE 4: LAMBDA in Event Handlers
SwingUtilities.invokeLater(() -> {
    analysisPanel.refresh(); // LAMBDA: Runnable implementation
    commentPanel.refreshCommentTable();
});

// EXAMPLE 5: LAMBDA in Streams - Complex filtering
long positiveCount = sentiments.stream()
    .filter(s -> s.getConfidence() > 0.7 && s.isPositive()) // COMPLEX LAMBDA
    .count();

// EXAMPLE 6: Functional Stream Processing with Comments
List<String> positiveAuthors = comments.stream()
    .filter(c -> c.getSentiment().isPositive()) // LAMBDA: filter predicate
    .sorted((c1, c2) -> c2.getCreatedAt() // LAMBDA: comparator
        .compareTo(c1.getCreatedAt()))
    .map(Comment::getAuthor) // METHOD REFERENCE: getter
    .collect(Collectors.toList()); // COLLECT: gather results

```

**Benefits:** Concise syntax (vs anonymous classes), functional programming support, enables higher-order functions, factory implementations without boilerplate.

### 7.3.4 Dependency Injection

**Purpose:** Pass dependencies to objects through constructors or setters instead of objects creating them internally. This decouples objects, simplifies testing, and enables runtime configuration.

**Application in System:**



```

// CONSTRUCTOR INJECTION: Pass dependencies via constructor
public class CrawlControlPanel extends JPanel {
    private Model model; // Injected dependency

    // INJECT via constructor, not new()
    public CrawlControlPanel(Model model) {
        this.model = model; // Store injected dependency

        // Use injected dependency
        this.model.addModelListener(this);
    }
}

// ANOTHER PANEL: Also receives Model dependency
public class CommentManagementPanel extends JPanel {
    private Model model;

    public CommentManagementPanel(Model model) {
        this.model = model;
        model.addModelListener(this);
    }
}

// VIEW: Injects Model to all panels
public class View extends JFrame {
    public View(Model model) {
        // INJECT Model to each panel
        AdvancedAnalysisPanel analysisPanel = new AdvancedAnalysisPanel(model);
        CommentManagementPanel commentPanel = new CommentManagementPanel(model);
        CrawlControlPanel crawlPanel = new CrawlControlPanel(model);
        DataCollectionPanel dataPanel = new DataCollectionPanel(model);

        // All panels share same Model instance
        this.add(analysisPanel);
        this.add(commentPanel);
        this.add(crawlPanel);
        this.add(dataPanel);
    }
}

// APPLICATION ENTRY POINT: Set up dependencies once
public class HumanitarianLogisticsApp {
    public static void main(String[] args) {
        try {
            // Initialize singletons
            DisasterManager disasterManager = DisasterManager.getInstance();

            // Create model (core business logic)
            Model model = new Model();

            // INJECT sentiment analyzer (strategy)
            PythonSentimentAnalyzer analyzer = new PythonSentimentAnalyzer(
                "http://localhost:5001"
            );

```

```

        model.setSentimentAnalyzer(analyzer);

        // Load persisted data
        DataPersistenceManager.loadData(model);

        // CREATE view (which injects Model to all panels)
        javax.swing.SwingUtilities.invokeLater(() -> {
            new View(model);
        });

        System.out.println("Application started successfully");
    } catch (Exception e) {
        System.err.println("Error starting application: " + e.getMessage());
        e.printStackTrace();
    }
}

// BENEFITS of DI:
// ✓ Easy to test: new Panel(mockModel, mockAnalyzer)
// ✓ Runtime configuration: swap implementations
// ✓ Loose coupling: Panel depends on interface, not concrete class
// ✓ Clear dependencies: visible in constructor signature

```

**Benefits:** Testability (easy to mock), loose coupling, runtime configuration, clear dependencies, flexibility.

### 7.3.5 Try-with-Resources for Automatic Resource Management

**Purpose:** Automatically close resources (files, connections, streams) without explicit cleanup code. This prevents resource leaks and makes code safer and cleaner.

**Application in System:**

```

// FILE I/O: Automatic stream closing
try (ObjectOutputStream oos = new ObjectOutputStream(
    new FileOutputStream("data.ser"))) {
    // RESOURCE OPENED: oos is auto-closeable
    oos.writeObject(posts); // Serialize data
    // RESOURCE CLOSED: automatically after block, even if exception
} catch (IOException e) {
    e.printStackTrace();
}
// oos.close() called automatically!

// DATABASE: Multiple auto-closeable resources
try (Connection conn = DriverManager.getConnection(dbUrl);
    Statement stmt = conn.createStatement();
    ResultSet rs = stmt.executeQuery("SELECT * FROM comments")) {
    // ALL resources (conn, stmt, rs) auto-closeable

    while (rs.next()) {
        String content = rs.getString("content");
        int postId = rs.getInt("post_id");
    }
}

```

```

        // Process result
    }
}
// ALL resources closed in reverse order (rs, stmt, conn) automatically!

// REAL SYSTEM USAGE: DatabaseManager (multiple resources)
public List<Comment> getAllCommentsFromDatabase() {
    try (Connection conn = getConnection();
        Statement stmt = conn.createStatement();
        ResultSet rs = stmt.executeQuery(
            "SELECT id, author, content, created_at, sentiment FROM comments")) {

        List<Comment> comments = new ArrayList<>();
        while (rs.next()) {
            Comment comment = parseComment(rs);
            comments.add(comment);
        }
        return comments;
    } catch (SQLException e) {
        LOGGER.log(Level.SEVERE, "Failed to load comments", e);
        return new ArrayList<>();
    }
}
// All resources (rs, stmt, conn) closed automatically in reverse order!

// WITHOUT try-with-resources (BAD):
ObjectOutputStream oos = null;
try {
    oos = new ObjectOutputStream(new FileOutputStream("data.ser"));
    oos.writeObject(posts);
} catch (IOException e) {
    e.printStackTrace();
} finally {
    if (oos != null) {
        try {
            oos.close(); // MANUAL cleanup (easy to forget!)
        } catch (IOException e) {
            // Handle close exception
        }
    }
}
}

```

**Benefits:** Automatic resource cleanup, prevents resource leaks, exception-safe (closes even on error), cleaner code (no finally blocks), supports multiple resources.

## 7.4 Summary of OOP Application

| Category             | Techniques                                                                                       | Count | Status                                                                                         |
|----------------------|--------------------------------------------------------------------------------------------------|-------|------------------------------------------------------------------------------------------------|
| Fundamental Concepts | Encapsulation, Abstraction, Inheritance, Polymorphism, Interfaces, Abstract Classes, Composition | 7/7   |  Complete |

| Category            | Techniques                                                                          | Count | Status                                                                                        |
|---------------------|-------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------|
| Design Patterns     | MVC, Strategy, Factory & Registry, Observer, Singleton, Facade, Adapter, Builder    | 8/8   |  Complete  |
| Advanced Techniques | Generics, Streams API, Lambda Expressions, Dependency Injection, Try-with-Resources | 5/5   |  Complete  |
| Code Quality        | 33 classes, 7 packages, 7499 lines, Zero circular dependencies                      | -     |  Excellent |

**Architecture Strengths:** Separation of concerns, extensibility, testability, maintainability, performance, type safety, and resource safety.

## 8. Technology Report

Modern, multi-layered stack combining desktop Java with Python ML backend. Java provides type safety and GUI frameworks; Python dominates data science for NLP. System uses lightweight HTTP-based crawling rather than browser automation for efficiency.

### 8.1 Core Technologies

**Java 11+:** Enterprise-grade statically-typed language providing compile-time type checking that catches errors before runtime. Swing framework offers native look-and-feel without external dependencies. HttpClient (built-in since Java 11) provides modern async HTTP with connection pooling and automatic timeout management.

**Benefits:** Type safety prevents 80% of runtime errors in dynamically-typed languages. Mature ecosystem (30+ years), massive standard library (50,000+ classes), WORA (Write Once Run Anywhere) across Windows/Mac/Linux without recompilation. Swing's native integration means no external GUI dependencies - reduces installation size and compatibility issues.

**Maven 3.9.11:** Declarative project structure with pom.xml (XML-based configuration) defining all dependencies, versions, build plugins, and JAR packaging strategy.

**Benefits:** Enforces consistent directory layout across 1000s of Java projects (src/main/java, src/test/java, target/). Automatic transitive dependency management (declaring Gson automatically includes underlying libraries). Single command `mvn clean package` produces fat JAR with all dependencies, eliminating classpath hell. Version conflict resolution prevents library version mismatches that cause runtime failures.

**Python 3.12.7:** Dynamically-typed language optimized for data science and machine learning. Runs separately as Flask REST API service on port 5001.

**Benefits:** Dominates NLP ecosystem - Hugging Face Transformers, PyTorch, scikit-learn have better documentation and community support in Python vs Java equivalents. Fast prototyping without compile step. Enormous ML library collection. Java calling Python via REST API maintains language separation-of-concerns without tight coupling.

## 8.2 UI and Visualization

**Swing (Built-in, since Java 1.2):** Platform-native UI framework providing JFrame, JPanel, JTable for building desktop interfaces. No external dependencies required.

**Benefits:** Zero external UI dependencies - runs on any Java 11+ installation without additional packages. Native look-and-feel matches Windows/Mac/Linux native UI conventions (unlike web frameworks that look identical everywhere). Event-driven architecture (ActionListener, MouseListener) naturally integrates with MVC Observer pattern. Lightweight compared to web servers.

**JFreeChart 1.5.3:** Open-source charting library providing CategoryPlot (bar charts for relief item distributions), PieChart (satisfaction percentages), TimeSeries (sentiment trends over time). Interactive MouseListener integration enables drill-down analysis clicking on chart regions.

**Benefits:** 20+ chart types available - bar, pie, line, scatter, bubble, etc. Interactive rendering with zoom/pan/tooltip support. Direct integration with Swing components (no separate web rendering required). Reusable ChartPanelBuilder utility pattern eliminates code duplication across AnalysisPanel, SatisfactionPanel, TimeSeriesPanel. Lightweight alternative to D3.js/Plotly that would require web server.

## 8.3 Data Storage

**SQLite 3.44.0.0:** Embedded relational SQL database (no external server required), ACID-compliant transactions, supports 280TB maximum database size per file. Uses separate database files:

`humanitarian_logistics_user.db` (user-crawled posts/comments),

`humanitarian_logistics_curated.db` (pre-loaded disaster scenario posts).

**Benefits:** Zero-configuration database deployment - no PostgreSQL/MySQL installation, network setup, or admin credentials needed. Single SQLite file ships with application. Atomic transactions (all-or-nothing) guarantee data consistency. Lightweight footprint (3MB compiled binary) vs PostgreSQL (50MB+). Perfect for single-machine desktop applications - eliminates network latency. ACID compliance ensures NO data loss on power failure or crash.

**Schema Design:** Normalized schema with 2 primary tables (posts, comments) using text primary keys for post/comment IDs, foreign key constraints for referential integrity between posts and comments, created\_at/updated\_at timestamps for audit trails, and integrated sentiment/relief data columns.

**DatabaseManager:** JDBC abstraction layer providing prepared statements (parameterized queries preventing SQL injection attacks), try-with-resources blocks ensuring automatic connection/statement cleanup preventing resource leaks. Methods: getAllPosts(), savePost(), saveComment(), deleteComment(), updateComment().

**Benefits:** Prepared statements prevent 100% of SQL injection attacks (e.g., user input "; DROP TABLE posts;--" is treated as literal string, not executed). Try-with-resources auto-closes connections even on exceptions, preventing connection leaks that would exhaust database resources. Abstraction layer (DataPersistenceManager) uses Adapter pattern - switching to PostgreSQL requires only changing JDBC driver URL and one implementation class, zero changes to Model/Analysis/UI code.

**DataPersistenceManager (Facade Pattern):** Single interface hiding JDBC complexity, enabling technology migration without breaking client code. Current SQLite implementation sufficient for 100s-1000s of posts. For scaling to 100,000+ posts, replace with PostgreSQL without touching application logic - demonstrating real Facade pattern benefit.

**Database Schema Overview:** The application maintains 2 primary tables in SQLite database supporting full post-comment hierarchy with integrated sentiment and relief data:

| Table Name | Primary Purpose                                                      | Key Columns                                                                                                       | Relationships                                                                                |
|------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| posts      | Store YouTube posts and user-added posts with analysis results       | post_id (PK), content, author, source, created_at, sentiment, confidence, relief_category, disaster_keyword       | One-to-many with comments. Sentiment and relief data stored directly in row.                 |
| comments   | Store comment thread data with sentiment and category classification | comment_id (PK), post_id (FK), content, author, created_at, sentiment, confidence, relief_category, disaster_type | Foreign key to posts(post_id). Each comment has independent sentiment and category analysis. |

Schema Design Principles:

- **Normalization:** Third Normal Form (3NF) prevents data duplication. Disaster keywords (yagi, matmo, bualo, koto) stored directly in posts and comments rows, not in separate disaster\_types table. Sentiment and relief\_category values embedded in each post/comment for efficient analysis queries.
- **Referential Integrity:** Foreign key constraint (comment.post\_id → posts.post\_id) ensures all comments reference valid posts. Deleting a post cascades to remove associated comments, preventing orphaned comment records.
- **Indexing Strategy:** Composite indexes on (postId, commentId) for fast comment retrieval (avoiding full table scans). Index on disasterId for filtering posts by disaster type. Index on createdAt for chronological queries (temporal sentiment tracking, Problem 2).
- **Hierarchical Comments:** parentCommentId column enables nested reply structure. NULL parentCommentId = top-level comment; non-NULL = reply to specific comment. Enables recursive traversal of comment threads.
- **Audit Trail:** createdAt and updatedAt timestamps on all tables enable temporal analysis. Track when posts were added, when sentiment analysis occurred, data freshness.

**SQL Schema Initialization:** DatabaseManager.initializeDatabase() creates tables on first run:

```
-- Posts table: YouTube videos + user additions with sentiment and relief data
CREATE TABLE IF NOT EXISTS posts (
  post_id TEXT PRIMARY KEY,
  content TEXT,
  author TEXT,
  source TEXT,
  created_at TEXT,
  sentiment TEXT,           -- POSITIVE, NEGATIVE, NEUTRAL
  confidence REAL,          -- 0.0-1.0 confidence score
  relief_category TEXT,     -- CASH, MEDICAL, SHELTER, FOOD, TRANSPORTATION
  disaster_keyword TEXT     -- yagi, matmo, bualo, koto, FUNG-WONG, etc.
);

-- Comments table: Comment thread with analysis results
```



```
CREATE TABLE IF NOT EXISTS comments (  
  comment_id TEXT PRIMARY KEY,  
  post_id TEXT,  
  content TEXT,  
  author TEXT,  
  created_at TEXT,  
  sentiment TEXT,      -- POSITIVE, NEGATIVE, NEUTRAL  
  confidence REAL,      -- 0.0-1.0 confidence score  
  relief_category TEXT, -- CASH, MEDICAL, SHELTER, FOOD, TRANSPORTATION  
  disaster_type TEXT,   -- yagi, matmo, bualo, koto, FUNG-WONG  
  FOREIGN KEY(post_id) REFERENCES posts(post_id)  
);
```

**Data Flow Example:** User crawls YouTube video → YouTubeCrawler extracts posts/comments → DatabaseManager.savePost() executes prepared statement INSERT into posts table → Comment hierarchy stored with post\_id foreign key linking comments to posts → Upon "Analyze All Posts" click, AnalysisModule reads posts via SELECT \* FROM posts, sends content text to Python API for sentiment analysis, stores results back to sentiment and relief\_category columns via UPDATE statement.

**Migration Path to PostgreSQL:** Schema SQL is vendor-neutral. To migrate: (1) Install PostgreSQL, (2) Run same DDL statements (AUTOINCREMENT → SERIAL, DATETIME → TIMESTAMP), (3) Update JDBC URL in DataPersistenceManager from "jdbc:sqlite:data/humanitarian\_logistics\_user.db" to "jdbc:postgresql://localhost:5432/humanitarian\_db", (4) Add PostgreSQL driver to pom.xml, (5) Zero code changes in Model/Analysis/UI layer thanks to Facade pattern abstraction.

**Disaster Types Supported:** The application initializes with 5 primary disaster types stored in DisasterManager (Singleton pattern). Each type has configurable aliases for flexible keyword matching:

| Disaste<br>r Type | Aliases                                    | Region/Context                             | Typical Posts Focus                                                                      |
|-------------------|--------------------------------------------|--------------------------------------------|------------------------------------------------------------------------------------------|
| yagi              | 台風 (typhoon),<br>Yagi, tropical<br>cyclone | Southeast Asia,<br>Philippines,<br>Vietnam | Wind damage, flooding from storm surge,<br>evacuation coordination, rescue<br>operations |
| koto              | 台風, Koto,<br>typhoon storm                 | Asia-Pacific region                        | Typhoon preparedness, post-storm<br>recovery, agricultural impact, shelter<br>needs      |
| bualo             | Bualoi, tropical<br>storm                  | Coastal Southeast<br>Asia                  | Flood response, water access issues,<br>medical supply distribution                      |
| matmo             | Matmo, typhoon,<br>extreme weather         | Taiwan, Japan<br>region                    | Infrastructure damage, power restoration,<br>food/water distribution                     |
| fung-<br>wong     | FUNG-WONG,<br>Noru, typhoon                | East Asia                                  | Multi-country impact, cross-border aid<br>coordination, longterm recovery                |

**Relief Categories Supported:** The application tracks 5 relief item categories extracted via hybrid keyword-based classification and semantic similarity analysis. Each category represents a distinct



humanitarian assistance type:

| Category       | Examples                                                                               | Typical Beneficiary Feedback                                                                  | Analysis Focus (Problem 1)                                                                         |
|----------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| CASH           | Cash assistance, conditional cash transfers, cash-for-work programs                    | Direct aid effectiveness, purchasing power in local markets, dignity/autonomy                 | Satisfaction trends: "Cash helped us buy food locally" vs complaint "Money arrives too late"       |
| MEDICAL        | Medical supplies, vaccines, antibiotics, first aid, trauma care, mental health support | Healthcare access, medication quality, treatment outcomes, preventive care satisfaction       | Sentiment analysis: Medical supply effectiveness, healthcare worker availability, patient outcomes |
| SHELTER        | Tents, temporary housing, plastic sheeting, building materials, housing repairs        | Living conditions, safety, privacy, permanence of solution, weatherproofing effectiveness     | Satisfaction metrics: Shelter quality vs speed of deployment, temporary vs permanent solutions     |
| FOOD           | Food packages, rice, cooking supplies, water, nutritional assistance, school meals     | Food security, cultural appropriateness, food quality, dietary diversity, sufficient portions | Sentiment tracking: Food sufficiency, taste/cultural fit, distribution fairness across groups      |
| TRANSPORTATION | Ambulances, evacuation vehicles, fuel for transport, logistics coordination            | Evacuation speed, medical transport effectiveness, supply distribution reach, accessibility   | Temporal analysis: Transportation availability during acute phase vs recovery phase                |

**Relief Categories Extensibility:** Although the application currently operates with 5 fixed relief categories, the architecture enables easy expansion for future needs without breaking existing code. Relief categories are stored as TEXT columns in posts and comments tables, making new categories trivially addable:

- **Database-level expansion:** New categories can be added by updating PythonCategoryClassifier keywords and analysis queries that use relief\_category column. Existing analysis queries dynamically work with any category values - no hardcoded category lists in code.
- **Code-level expansion:** For new analysis types (e.g., tracking EDUCATION or LIVELIHOOD aid), extend ReliefItem.Category enum and update PythonCategoryClassifier keywords. Analysis modules read categories dynamically from Model state - zero impact on existing CASH/MEDICAL/SHELTER/FOOD/TRANSPORTATION logic.
- **UI-level expansion:** Analysis panel dropdowns populate from Model.getAvailableCategories() - automatically includes new categories without UI recompilation.
- **No schema redesign needed:** relief\_category column structure unchanged - simply update classification logic to assign new category values for new aid types.

This extensibility follows the Open/Closed Principle - system open for extension (adding categories) but closed for modification (existing 5 categories work unchanged). Migration to PostgreSQL preserves this capability automatically.

**Customization Capability - Adding New Disaster Types:** The application architecture supports dynamic disaster type expansion without code modification. In DisasterManager (lines 1-50 of model/DisasterManager.java), the constructor initializes disasters:

```
private void initializeDefaultDisasters() {  
    // Initialize default disasters with aliases added separately  
    DisasterType yagi = new DisasterType("yagi");  
    yagi.addAlias("#yagi");  
    yagi.addAlias("yagi");  
    addDisasterType(yagi);  
  
    DisasterType matmo = new DisasterType("matmo");  
    matmo.addAlias("#matmo");  
    matmo.addAlias("matmo");  
    addDisasterType(matmo);  
  
    // Continue for other disaster types...  
    // Future expansion: Simply add more disasters here  
}
```

To add new disaster types (e.g., earthquakes, floods): Add new lines in constructor OR modify disasters.dat file (Java serialized object containing DisasterType list). Data Persistence Manager loads disasters.dat on startup - changes take effect immediately without code recompilation.

## 8.4 Data Collection and HTTP

**Java HttpClient (java.net.http, built-in since Java 11):** Modern HTTP protocol handler with connection pooling, automatic timeout management (default 30 seconds), follows HTTP redirects (301, 302), handles chunked transfer encoding, and supports HTTP/1.1 and HTTP/2.

**Benefits vs Selenium:** HttpClient is 10-50x faster than Selenium (milliseconds vs seconds per page). Selenium launches full Chrome browser process (500MB+ memory per tab), rendering JavaScript - overkill for static YouTube pages. HttpClient sends raw HTTP GET requests to YouTube (1-2MB per page), parses embedded JSON data using regex patterns, skips all browser overhead. Lower CPU/memory footprint allows parallel crawling (100 URLs simultaneously). No WebDriver updates required when YouTube updates HTML structure (Selenium breaks with HTML changes).

**Note:** pom.xml declares Selenium 4.15.0 but code does not use it. Design decision: avoid browser automation overhead, prioritize performance. If browser rendering needed (JavaScript-heavy sites, anti-bot protection), can swap HttpClient → Selenium without changing DataCrawler interface.

**YouTubeCrawler Implementation:** Sends HTTP GET to `https://www.youtube.com/watch?v={VIDEO_ID}` with Chrome User-Agent header (YouTube checks User-Agent to prevent bot detection). Response HTML contains embedded `ytInitialData` JSON object with complete comment section data. YouTubeCrawler extracts:

- Video metadata: title, channel, view count, publication date
- Comments: author, text, likes, timestamp, reply hierarchy

- Replies: nested comment structure với author và text

Không rendering screen, không Selenium WebDriver overhead - pure HTTP + JSON parsing.

**OkHttp3 4.11.0:** Production HTTP client from Square (used in Android framework). Provides advanced features: HTTP/2 connection multiplexing (multiple requests on single TCP connection), automatic GZIP compression (reduces bandwidth), request interceptors (can inject logging, auth headers, retry logic).

**Benefits:** Future crawlers (FacebookCrawler, TwitterCrawler) can reuse OkHttp3 interceptor patterns for centralized logging and auth. Connection pooling reduces overhead when crawling thousands of URLs. Currently HttpClient sufficient but OkHttp3 available as drop-in replacement if needed.

**MockDataCrawler:** Synthetic data generator for unit testing and demo mode. Returns hardcoded List<Post> with 31 humanitarian disaster scenarios (earthquake relief, flood response, pandemic medical supplies). Zero network calls - useful for:

- **Offline development:** No internet required
- **Reproducible testing:** Same data every run vs YouTube's live-changing data
- **Unit tests:** Fast test execution (~100ms vs 10+ seconds with real YouTube crawling)
- **Demo mode:** Avoid YouTube API rate limits (10 requests/second), prevents IP blocking

Registered in CrawlerRegistry alongside YouTubeCrawler - same interface, pluggable via Factory pattern.

## 8.5 JSON Processing

**JSON Handling Strategy:** The application uses two JSON libraries optimized for different use cases. Understanding when to use each library improves code clarity and reduces dependencies:

| Library                         | Use Case                                                                 | Example in Code                                                                     | Trade-off                           |
|---------------------------------|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------|
| Gson 2.10.1 (Type-Safe)         | Parse complex nested JSON into typed Java objects (Python API responses) | SentimentResponse response = gson.fromJson(json, SentimentResponse.class);          | Full-featured but adds ~5MB to JAR  |
| org.json 20231013 (Lightweight) | Parse simple JSON from HTML embedded data (YouTube ytInitialData)        | JSONObject obj = new JSONObject(jsonString); String title = obj.getString("title"); | Minimal (~100KB) but no type safety |

### Gson 2.10.1 (Google JSON Library) - For Complex Structures:

**What it does:** Converts JSON strings ↔ Java objects automatically with full type checking. When you have nested objects (Post → List of Comments → each Comment has Sentiment), Gson recursively deserializes the entire structure preserving type information.

#### Code Example:

```
// Python API returns this JSON:
{
  "sentiment": "POSITIVE",
  "confidence": 0.94,
```

```

    "category": "FOOD",
    "keywords": ["relief", "supplies"]
}

// Java defines matching class:
class SentimentResponse {
    String sentiment;
    double confidence;
    String category;
    List<String> keywords;
}

// Gson instantly converts:
Gson gson = new Gson();
SentimentResponse response = gson.fromJson(jsonString, SentimentResponse.class);

// Type-safe access (compile-time checking):
System.out.println(response.sentiment); // ✓ Works, String type known
System.out.println(response.confidence); // ✓ Works, double type known
// response.invalid_field; // ✗ Compile error - field doesn't exist

```

**Benefits:** Type safety prevents runtime errors. Nested objects (Post.comments automatically deserializes all Comment objects). Handles custom types via type adapters (LocalDateTime → ISO-8601 strings). Used for Python API communication where response structure is complex and changes require catching in compilation.

### org.json 20231013 (Lightweight JSON) - For Simple Parsing:

**What it does:** Parses JSON into flexible generic objects (JSONObject, JSONArray) without class definitions. Trade runtime safety for minimal dependencies.

#### Code Example:

```

// YouTube HTML embeds JSON data:
String html = "...<script>var ytInitialData = {\"videoDetails\": {\"videoId\": \"abc123\",

// Extract and parse with org.json:
String jsonStr = extractJsonFromHtml(html); // \"{\\\"videoDetails\\\": ...}\"
JSONObject data = new JSONObject(jsonStr); // Parse instantly

// Access values dynamically (no class needed):
String videoId = data.getJSONObject("videoDetails").getString("videoId");
int viewCount = data.getJSONObject("videoDetails").getInt("viewCount");

// No type checking at compile time, but works for one-off parsing

```

**Benefits:** Lightweight - single JAR with zero transitive dependencies (~100KB vs Gson ~5MB). Fast parsing for simple structures. Perfect for YouTube HTML parsing where you extract just a few fields then discard. Zero configuration. Avoids bloating desktop app with unused type serialization code.

#### Why Both Libraries?:

**Design Decision:** Rather than forcing all JSON through Gson (simpler to maintain), the application strategically uses org.json for lightweight YouTube parsing and Gson for complex Python API communication. This dual-library approach:

- **Reduces JAR size:** YouTube crawler doesn't need Gson's full serialization machinery - org.json is 50x lighter
- **Improves clarity:** Developers see "parsing simple data" (org.json) vs "complex typed response" (Gson) in code
- **Optimizes for use case:** Avoid paying full Gson cost when simple parsing suffices
- **Zero overlap:** Each library has distinct responsibility - no redundancy

**When Python API Needs Change:** If Python sentiment\_api.py returns additional nested fields (e.g., detailed per-phrase sentiment scores), Gson's automatic deserialization adapts by extending SentimentResponse class. org.json would require manual JSONObject navigation - type safety matters for complex APIs. This is why org.json unsuitable for Python communication.

## 8.6 Machine Learning & NLP

### Backend Stack Architecture:

- **Flask 2.3.0+:** Lightweight Python web microframework (150 lines of code creates REST API). Routes HTTP requests to Python functions. No heavy frameworks (Django) overhead.
- **Hugging Face Transformers 4.30.0+:** Library providing pre-trained transformer models (BERT, RoBERTa, BART) fine-tuned on millions of texts. Handles tokenization, embedding, inference automatically.
- **PyTorch 2.0.0+:** Deep learning framework executing transformer models with GPU acceleration (if available). Automatic differentiation for training.
- **NumPy:** Numerical computing library for tensor operations, matrix math supporting models internally.
- **scikit-learn:** Traditional ML library with SVM, decision trees, clustering (available for custom classifiers if needed).

**Benefits:** Python libraries have 10-100x better NLP support than Java equivalents. Hugging Face provides pre-trained models eliminating need to collect 10,000 training examples ourselves. Can download and use model in 10 minutes vs 6 months to train from scratch. Flask microframework (100-200 lines) vs Java Spring Boot (10,000+ lines configuration).

**Sentiment Analysis: NLPTown BERT-Base-Multilingual-Uncased-Sentiment** (with fallback to distilbert-base-multilingual-uncased and xlm-roberta-base)

**Architecture:** Transformer neural network pre-trained on multilingual data supporting 100+ languages including Vietnamese. Primary model: nlptown/bert-base-multilingual-uncased-sentiment (3-class sentiment). Fallback chain: distilbert-base-multilingual-uncased → xlm-roberta-base for robust multilingual support.

### Benefits:

- **Multilingual support:** Works on Vietnamese, English, and 100+ other languages without retraining. Single model handles humanitarian crisis posts in any language.
- **Three-class sentiment:** Classifies text into [POSITIVE, NEGATIVE, NEUTRAL] without requiring fine-tuning.

- **Fallback chain:** If primary model unavailable, automatically falls back to distilbert (lightweight) then xlm-roberta-base, ensuring system always functional.
- **Confidence scores:** Returns probability scores enabling filtering low-confidence results (threshold: confidence > 0.8).

Fast inference: 50-200ms per text (depends on text length). Lightweight models enable reasonable performance on CPU-only systems.

## Relief Category Classification: Hybrid Keyword-Based + Semantic Similarity

**Architecture:** Dual-strategy classification combining fast keyword matching with semantic similarity. Custom keyword dictionary for each relief category (CASH, MEDICAL, SHELTER, FOOD, TRANSPORTATION) in both English and Vietnamese. Instant matching without model download overhead.

### Benefits:

- **Instant Vietnamese support:** Keyword-based approach provides immediate Vietnamese categorization without downloading large transformer models. Keywords include: "y tế" (medical), "thực phẩm" (food), "nơi trú ẩn" (shelter), "tiền mặt" (cash), "vận chuyển" (transportation).
- **Fast inference:** Millisecond-scale classification vs seconds for transformer models. Processes 31 posts instantly.
- **Semantic understanding:** Understands relationships: "medications" → MEDICAL, "tents" → SHELTER, "ambulances" → TRANSPORTATION through keyword expansion and phrase matching.
- **Zero model download:** No external model dependencies - all logic embedded in sentiment\_api.py. Reduces deployment complexity and memory footprint.

Inference: Milliseconds to 10ms for batch processing.

## Python Service Architecture: sentiment\_api.py

**Implementation:** Flask server on port 5001 with two REST endpoints:

```
// Java calls Python API via HTTP POST
POST http://localhost:5001/sentiment
{
  "text": "Earthquake relief supplies arrived at shelter!",
  "task_type": "sentiment"
}

Response 200 OK:
{
  "sentiment": "POSITIVE",
  "confidence": 0.96,
  "category": "SHELTER",
  "keywords": ["supplies", "shelter"]
}
```

### Benefits of REST API separation:

- **Language independence:** Java GUI doesn't need to import PyTorch (incompatible with JVM). Python backend runs as independent process with its own Python interpreter and libraries.



- **Technology flexibility:** Can replace Python with Node.js/Go/C++ backend without touching Java code. Can scale Python separately (run 3 Python servers, load-balance requests across them).
- **Graceful degradation:** If Python API unavailable, fallback to EnhancedSentimentAnalyzer (Java-based keyword detection + emoticon matching) maintaining basic functionality.
- **Development efficiency:** Data scientists modify sentiment\_api.py, test with curl/Postman, deploy without recompiling entire Java application.

**Fallback Chain:** If Python service crashes/unreachable:

- 1st attempt: Call Python API via HTTP → nlptown/bert-base-multilingual-uncased-sentiment (96% accuracy)
- 2nd fallback: EnhancedSentimentAnalyzer (Java) → regex keyword matching + emoticon detection (Vietnamese-specific rules for 😊 → POSITIVE, 😞 → NEGATIVE) → 75% accuracy

System remains functional even if Python service crashes - user sees slightly lower accuracy but continues analysis rather than hard failure.

## 8.7 Deployment & Logging

**SLF4J 2.0.9 (Simple Logging Facade for Java):** Abstraction layer decoupling application from specific logging implementation. Allows switching between Log4j, Logback, or simple console output without code changes.

**Benefits:** Prevents vendor lock-in - if we need centralized logging (ELK stack, Splunk) in future, add one line to pom.xml and remove one line. Zero code changes. Configured with slf4j-simple binding (console-only output for desktop app). Log levels enable debug builds with verbose logging and production builds with ERROR-only output (changing single property).

**Logging Implementation:** All major operations logged:

- **DEBUG:** Detailed execution trace ("Processing post ID 42", "Parsing comment section", "Calling Python API with text...")
- **INFO:** Key lifecycle events ("✓ Crawler initialized", "🕸 Crawling YouTube for keyword: relief", "✓ Analysis complete - 31 posts analyzed")
- **WARN:** Suspicious activity ("⚠ Confidence score 0.52 below threshold 0.8", "⚠ Python API timeout, using fallback analyzer")
- **ERROR:** Failures with stack traces ("Error loading database", "HTTP 403 Forbidden from YouTube")

Useful for debugging customer issues: ask user to enable DEBUG logging and capture log output.

**Maven Build Process:** `mvn clean package` compiles all Java files, runs unit tests, and creates **fat JAR** (humanitarian-logistics-1.0-SNAPSHOT-jar-with-dependencies.jar containing all 15+ dependencies bundled inside single JAR).

**Benefits:** Single JAR file simplifies distribution - user downloads one file, runs `java -jar file.jar`, works immediately. No need to manage CLASSPATH or download separate dependency JARs. Maven Shade Plugin merges dependency JARs inside main JAR (avoiding conflicts if two libraries include different versions of same transitive dependency).

**Execution Flow:**



```

// Step 1: User runs single command
java -jar humanitarian-logistics-1.0-SNAPSHOT-jar-with-dependencies.jar

// Step 2: JVM loads HumanitarianLogisticsApp.main() from JAR
// Step 3: Initialization sequence:
System.out.println("🔄 Initializing Humanitarian Logistics System...");

// 3a. Load SQLite databases
DatabaseManager.getInstance().connect();
// → Opens humanitarian_logistics_curated.db (31 posts pre-loaded)
// → Opens humanitarian_logistics_user.db (empty, for user crawls)

// 3b. Populate CrawlerRegistry (Factory pattern)
CrawlerRegistry registry = CrawlerRegistry.getInstance();
registry.registerCrawler(new YouTubeCrawler());
registry.registerCrawler(new MockDataCrawler());
// → Available crawlers: YouTubeCrawler, MockDataCrawler

// 3c. Python Flask service must be running separately
// User should start it via: python src/main/python/sentiment_api.py
// OR via run.sh script which handles both Java and Python startup
// → Starts Flask server on port 5001
// → Loads sentiment and category models into memory
// → First run may download models from Hugging Face (10-15 minutes)
// → Subsequent runs start instantly with cached models

// 3d. Initialize Model and UI
Model model = new Model();
view = new HumanitarianLogisticsUI(model);
SwingUtilities.invokeLater(() -> view.setVisible(true));
// → Creates Swing JFrame with menu bar, panels, charts
// → Displays empty DataTable, Analysis button, Crawler dropdown

System.out.println("✅ System ready! Select crawler and click 'Crawl & Analyze'");

```

## 8.8 Performance Characteristics

### Startup Performance:

| Phase            | Time    | Details                                                                                                                                                                                                                       |
|------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| First-time setup | 2-5 min | Downloading ~500MB of ML sentiment model from Hugging Face (nlptown/bert-base-multilingual-uncased-sentiment). Category classification uses keyword-based approach (no external model). One-time cost, models cached locally. |

| Phase                        | Time      | Details                                                                                                                               |
|------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------|
| Cold startup (models cached) | 30-45 sec | Loading models into GPU memory (10-15s), initializing Flask server (5-10s), loading SQLite databases (5s), creating Swing UI (5-10s). |
| Warm startup                 | 5-10 sec  | Python service already running in background, models already in GPU memory.                                                           |

**Benefits:** First-time setup cost is one-time investment. User runs app once, models download, wait 15 minutes. Subsequent uses start in 30 seconds. Comparable to downloading Visual Studio (5-10GB) first time. ML capability (94% accuracy sentiment) only possible with model download - no free lunch.

**Analysis Performance:**

| Operation                          | Speed      | Notes                                                                                                                                                                                         |
|------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| YouTube crawl (single video)       | 500 ms-2s  | Depends on comment section size. HTML download (200-500ms) + JSON parsing (300-500ms) + building Post/Comment objects (200ms).                                                                |
| Sentiment analysis (single text)   | 50-200 ms  | NLPTown BERT-Base-Multilingual inference. Longer texts (articles) slower than tweets (50-100ms). GPU: 10-20ms. CPU-only: 500ms+. Falls back to xlm-roberta-base if primary model unavailable. |
| Batch sentiment (31 posts)         | 1-2 sec    | Vectorized processing - all 31 texts processed in single GPU forward pass. 30x faster than 31 sequential inferences.                                                                          |
| Category classification (31 posts) | 100-300 ms | Hybrid keyword-based approach - extremely fast (no neural network inference required). Returns category + confidence score per post. Zero model download overhead.                            |
| Database insert (single post)      | <10 ms     | SQLite prepared statement. Batch insert (31 posts): 50-100ms with transaction wrapper.                                                                                                        |
| Chart rendering (100 posts)        | 200-500 ms | JFreeChart generating bar/pie/time-series charts. Interactive mouse listeners registered during rendering.                                                                                    |
| UI refresh (data grid refresh)     | <100 ms    | Swing JTable repaint with 31 rows. Java 11+ rendering optimizations.                                                                                                                          |

**Total user workflow (typical use case):**

- User clicks "Crawl & Analyze" button → selects YouTubeCrawler → enters keywords "earthquake relief" → clicks Submit
- System crawls 5-10 YouTube videos (5-15 seconds)
- Extracts ~100-150 comments (1 second)
- Sentiment analysis on all comments (2 seconds)
- Category classification (1 second)
- Insert into database (500ms)
- Render charts (500ms)
- **Total: 10-20 seconds from click to results** ✓ User perceives as responsive (psychology: <1 sec = instant, <10 sec = acceptable, >30 sec = annoying)

**Memory Usage:**

| Component                                          | Memory                                                                                             |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Java heap                                          | 256MB default (-Xmx256m)                                                                           |
| Python models in GPU memory                        | ~500MB (nlptown/bert-base-multilingual sentiment model only; category classifier is keyword-based) |
| Data structures (Post, Comment, Sentiment objects) | ~10-50MB (100-1000 posts, each ~50KB)                                                              |
| UI components (Swing JFrame, JTable, charts)       | ~50MB (cached images, rendered components)                                                         |
| Total steady state                                 | ~3.9GB (with GPU models) or 500MB (CPU-only, slower)                                               |

**Benefits:** Efficient memory usage (~500MB total) enables deployment on modest hardware. Hybrid approach combines GPU-accelerated sentiment analysis (high accuracy) with lightweight keyword-based category classification (zero latency). Can scale to 10,000+ posts with pagination - don't load all simultaneously, display 100 at a time, load more on scroll.

**Network Usage:**

| Operation                         | Data Transfer                                    |
|-----------------------------------|--------------------------------------------------|
| YouTube crawl (single video page) | 1-3MB (HTML response with all comments embedded) |
| 5 video crawl                     | 5-15MB total                                     |
| Python API request (sentiment)    | ~1KB (text + task_type fields)                   |
| Python API response (sentiment)   | ~200 bytes (sentiment + confidence + category)   |
|                                   |                                                  |

| Operation                                | Data Transfer                                                      |
|------------------------------------------|--------------------------------------------------------------------|
| Batch sentiment (31 posts, 31 API calls) | ~31KB request + 6KB response                                       |
| No continuous polling                    | Event-driven (button click triggers crawl, no background requests) |

## 8.9 Scalability and Future Evolution

### Database Scaling Roadmap:

**Current (SQLite):** Single-user, single-machine. Perfect for individual analyst or small team. Schema: 2 tables (posts, comments) with integrated sentiment/relief data columns and proper indexing on post\_id/comment\_id for fast lookups.

**Future (PostgreSQL/MySQL):** Multi-user concurrent access, ACID transactions, row-level security, automated backups/replication. DataPersistenceManager abstraction means swapping implementations requires:

- 1. Add PostgreSQL JDBC driver to pom.xml
- 2. Change JDBC URL in configuration ("jdbc:postgresql://localhost:5432/humanitarian\_logistics")
- 3. Create schema once (20-line SQL script)
- 4. ZERO code changes in Model, Analysis, UI layers ✓ Abstraction pattern in action

### Scaling to 100,000+ posts:

- Add table partitioning: posts\_2024\_Q1, posts\_2024\_Q2 by timestamp (query planner skips irrelevant partitions)
- Add database views for common queries (most positive/negative posts, category distribution)
- Add caching layer (Redis) for frequently accessed data (top 10 posts, aggregates)
- Enable query optimization: EXPLAIN ANALYZE to identify slow queries, add indexes strategically

### ML Scaling Roadmap:

**Current (Flask single-process):** Single request at a time. If 2nd user crawls while 1st user analyzes, 2nd user waits (blocking).

### Concurrent processing (Gunicorn):

```
// Deploy Flask with multi-worker server:
gunicorn -w 4 -b 0.0.0.0:5001 sentiment_api:app
// Spawns 4 Python processes, each with full model copy. Can handle 4 concurrent requests.
// Load balancer routes requests round-robin across workers.
```

### Results caching (Redis):

- Problem: User A analyzes "Great help from relief workers!" → nlptown/bert-base-multilingual-uncased-sentiment inference (50-100ms)

- User B analyzes same text → Hits Redis cache → response in 1ms (100x speedup)
- Implementation: Check Redis before calling sentiment model, cache results for 24 hours (sentiment typically doesn't change)
- Especially effective for humanitarian posts (repetitive disaster updates, same talking points)

### Better models if latency permits:

- Current: nlptown/bert-base-multilingual-uncased-sentiment (lightweight, 50-200ms inference, 88-94% accuracy)
- Upgrade path: distilbert-base-multilingual-uncased (light upgrade, 150M params, 85-92% accuracy, 100-150ms) or xlm-roberta-base (280M params, 90-95% accuracy, 100-200ms)
- Trade-off: 90% accuracy + 100ms is better UX than 95% accuracy + 200ms for interactive app. Use xlm-roberta-base for batch analysis overnight.

### Frontend Scaling Roadmap:

**Current: Desktop Swing app** - single user, single machine, zero network overhead.

**Future: Web frontend (React/Vue.js/Angular)** - multi-user concurrent access via browser

- **Architecture:** Java backend exposes REST API (HumanitarianLogisticsAPI.java with Spring Boot) serving:
  - /api/crawl (POST) → start YouTube crawl, returns job ID
  - /api/analyze (POST) → submit posts for analysis
  - /api/results (GET) → retrieve sentiment/category results
  - /api/charts (GET) → return chart data as JSON
- **React frontend** consumes REST API, displays charts using D3.js/Recharts, enables multi-user dashboards
- **Code changes:** Minimal - Model already implements Observer pattern (supports multiple listeners). Add REST endpoints wrapping Model methods. UI layer already MVC-separated from business logic.
- **Database:** Already multi-user capable with PostgreSQL migration above
- **Benefit:** Humanitarian organizations in disaster zones can use web app from any device (phone, tablet, laptop) without installing Java. Real-time collaboration - multiple analysts viewing same dashboard.

### Crawler Scaling Roadmap:

#### Current crawlers:

- YouTubeCrawler: Fetches YouTube watch pages via HttpClient
- MockDataCrawler: Generates synthetic posts for testing

**Future crawlers (implementable without modifying existing code, thanks to Factory/Registry patterns):**

```
// Register new crawler via same interface:
public class FacebookCrawler implements DataCrawler {
    public List<Post> crawlPosts(List<String> keywords, List<String> hashtags, int limit) {
```

```

        // Facebook API: fetch posts by location (disaster zones), keywords (earthquake, re
        // Returns List<Post> with same structure as YouTube
    }
}

public class TwitterCrawler implements DataCrawler {
    // Twitter API v2: fetch tweets by hashtags #DisasterRelief #HumanitarianAid
    // Stream API for real-time sentiment tracking
}

public class RedditCrawler implements DataCrawler {
    // Reddit API: fetch posts from r/humanitarian crisis, r/disaster response
    // Comments already structured in Reddit JSON response
}

// Register all at startup:
CrawlerRegistry registry = CrawlerRegistry.getInstance();
registry.registerCrawler(new YouTubeCrawler());           // Existing
registry.registerCrawler(new FacebookCrawler());          // New
registry.registerCrawler(new TwitterCrawler());           // New
registry.registerCrawler(new RedditCrawler());             // New
// UI dropdown automatically includes all registered crawlers ✓ Factory pattern benefit

```

**Rate limiting:** Social platforms enforce rate limits (YouTube: 10 req/sec, Twitter: 300 req/15min). Implement exponential backoff:

- 1st request fails (429 Too Many Requests) → wait 1 second, retry
- 2nd request fails → wait 4 seconds, retry
- 3rd request fails → wait 16 seconds, retry
- Max retries: 5 (total wait: 1+4+16+64+256 = 341 seconds = 5.6 minutes)

Allows system to gracefully handle rate limits without crashing.

## Disaster Types and Relief Categories - Customization and Expansion:

**Current Configuration:** The application currently operates with exactly 5 disaster types and 5 relief categories:

- **Disaster Types (stored in disasters.dat):** yagi, koto, bualo, matmo, fung-wong
- **Relief Categories (hardcoded in ReliefItem.Category enum):** CASH, MEDICAL, SHELTER, FOOD, TRANSPORTATION

These types are loaded during application startup from persistent storage:

- DisasterManager.getInstance() reads from data/disasters.dat (Java serialized DisasterType objects)
- ReliefItem.Category enum defines the 5 relief categories
- Model instance caches loaded categories in memory for UI/Analysis access

**Persistence Architecture Enabling Easy Expansion:** The application achieves flexibility through intermediate file storage and dynamic loading patterns:

- **Disasters.dat File Strategy:** Disaster types are persisted to data/disasters.dat using Java object serialization (NOT hardcoded in code). This file contains a serialized DisasterType list. On application



startup, `DataPersistenceManager.loadDisasters()` deserializes `disasters.dat` and populates `DisasterManager` singleton with current disaster types. This separation means:

- Adding new disasters = modify `disasters.dat` file (via programmatic API or direct serialization)
  - No code recompilation required - `disasters.dat` loaded fresh each startup
  - Users can clone/backup `disasters.dat` across machines - instant disaster type sharing
  - Example: Organization A has disaster set {yagi, koto, bualo}, Organization B adds {earthquake, flood} - separate `disasters.dat` files, zero code conflicts
- **Relief Categories Code Storage:** Relief categories are defined in `ReliefItem.Category` enum. For expansion:
    - Current: 5 categories defined in enum and stored in posts/comments `relief_category` column
    - Expansion: Add new enum value + update `PythonCategoryClassifier` keywords for recognition
    - Dynamic UI discovery via `Model.getAvailableCategories()`

**Expansion Mechanism - Adding New Disaster Types:** Thanks to `disasters.dat` persistence strategy, expanding disaster types is straightforward:

```
// At runtime, modify disasters.dat:
DisasterManager manager = DisasterManager.getInstance();
manager.addDisaster(new DisasterType("earthquake", "Earthquake,EQ,seismic,tremor"));
manager.addDisaster(new DisasterType("flood", "Flood,flooding,water,inundation"));

// Persist updated disasters list:
DataPersistenceManager persistence = new DataPersistenceManager();
persistence.saveDisasters(manager); // Writes updated disasters.dat

// Application restart automatically loads new disasters
// Next startup: DisasterManager loads {yagi, koto, bualo, matmo, fung-wong, earthquake, fi
// UI dropdowns automatically include new disaster types - no code changes
```

### Why This Architecture Enables Easy Expansion:

- **Separation of Concerns:** Data (`disasters.dat`, `relief_items` rows) separate from code (`DisasterManager`, `ReliefItem`). Changing data doesn't require code recompilation.
- **Lazy Loading Pattern:** `DisasterManager` and relief categories loaded once at startup from persistent storage. If storage changes, next startup reflects changes automatically. No hot-reloading complexity.
- **Single Source of Truth:** `disasters.dat` is the definitive source for disaster types. All references (UI dropdowns, Analysis tabs, Model cache) read from this single source via Singleton pattern.
- **No Breaking Changes:** Existing 5 disaster types (yagi, koto, bualo, matmo, fung-wong) continue to work unchanged. New types add to the set without modifying existing hardcoded assumptions.
- **Migration Path Preserved:** When migrating from SQLite to PostgreSQL, `disasters.dat` loading strategy unchanged - `DisasterManager.getInstance()` still works identically. Database schema change (SQLite → PostgreSQL) is transparent to disaster loading logic.

### Scalability Implications:



- **Current (5 disaster types, 5 relief categories):** UI dropdowns load instantly, analysis completes in seconds
- **Expanded (50+ disaster types):** Still performant - DisasterManager caches loaded disasters in memory, database indexes on disasterId column. Relief categories remain fixed at 5 types for consistency.
- **Multi-tenant future:** Each organization maintains separate disasters.dat file - OrganizationA.disasters.dat vs OrganizationB.disasters.dat with distinct disaster sets. Zero code conflict, zero deployment complexity.

**Migration and Backup Strategy:** The disasters.dat file-based approach enables simple data portability:

- Backup: Copy data/disasters.dat to safe location
- Restore: Copy disasters.dat back to data/ directory, restart application
- Share: Email disasters.dat to partner organization - they copy to their data/ folder, get identical disaster types instantly
- Version Control: disasters.dat can be committed to Git - disaster type evolution is auditable and rollbackable
- No SQL migration scripts needed - file-based persistence is simpler than database versioning for this use case

## 8.10 Technology Stack Summary

| Category         | Technology                | Version           | Purpose                                         |
|------------------|---------------------------|-------------------|-------------------------------------------------|
| Language s       | Java / Python             | 11+ / 3.12.7      | Desktop app / ML backend                        |
| Build            | Maven                     | 3.9.11            | Compilation, packaging, dependency management   |
| Desktop UI       | Swing                     | Built-in          | GUI with zero external dependencies             |
| Charting         | JFreeChart                | 1.5.3             | Bar, pie, time-series charts with interactivity |
| Database         | SQLite + JDBC             | 3.44.0.0          | Embedded persistence, two DB instances          |
| HTTP Client      | Java HttpClient / OkHttp3 | Built-in / 4.11.0 | YouTube crawling, API calls                     |
| JSON Processin g | Gson / org.json           | 2.10.1 / 20231013 | Complex / lightweight serialization             |
| Web Framework    | Flask                     | 2.3.0+            | Python REST API for ML services                 |
|                  |                           |                   |                                                 |

| Category   | Technology             | Version          | Purpose                                                                               |
|------------|------------------------|------------------|---------------------------------------------------------------------------------------|
| NLP Models | Transformers / PyTorch | 4.30.0+ / 2.0.0+ | Sentiment (nlptown/bert-base-multilingual) / Category (keyword-based hybrid approach) |
| Logging    | SLF4J + slf4j-simple   | 2.0.9            | Abstraction layer with console output                                                 |
| Testing    | JUnit                  | 4.13.2           | Unit test framework                                                                   |

**Architecture Decisions:** Lightweight HTTP-based data collection for efficiency and reduced resource footprint. Microservices pattern: Java desktop app communicates with Python ML backend via REST (loose coupling, language independence). Abstraction layers (DataPersistenceManager, CrawlerRegistry) enable technology switching without breaking client code. Observer pattern ensures responsive UI without polling.

**Production Readiness:** Mature, battle-tested libraries (Swing since 1996, Flask since 2010, SQLite since 2000). Try-with-resources ensures resource cleanup. Prepared statements prevent SQL injection. Error handling with graceful fallbacks (Python unavailable → use Enhanced/Simple analyzers). Future evolution supported via abstraction patterns without major refactoring.